



**COMPONENT MAINTENANCE MANUAL
WITH ILLUSTRATED PARTS LIST**

**DC STARTER-GENERATOR
23046 SERIES I**

**INCLUDES MODELS:
23046-001, -007, -007M, -009,
-017, -019, -020, -028**

Original issue, February 1, 1965

Revision 7, March 23, 2010

These commodities, technology or software were exported from the United States in accordance with the Export Administration Regulations. Diversion contrary to U.S. law is prohibited. ECCN: 9E991

© Copyright 2010 Goodrich Corporation - Copyright in this manual and associated documents and drawings belong to Goodrich Corporation. All rights reserved. No reproduction of all or part of this manual shall be made without prior written consent of Goodrich Corporation. This document contains information that may be confidential. Disclosure to others of information contained herein requires the written consent of Goodrich Corporation.

TITLE PAGE



Component Maintenance Manual with Illustrated Parts List
DC Starter-Generator, 23046 Series I

EXCLUSIVE DISTRIBUTORS

<u>Component Parts and AOG</u>		<u>Repair and Overhaul Services</u>	
Genuine Goodrich Corporation component parts are to be purchased from the following Exclusive Distributors.		Goodrich operates three major repair and overhaul service centers for electrical Power Systems Products. Each center can service the majority of Goodrich products.	
<u>ADDRESS</u>	<u>COMMUNICATIONS</u>	<u>ADDRESS</u>	<u>COMMUNICATIONS</u>
WORLDWIDE: AOG Desk (24/7 Service)	PHONE: (44) 121 451 5904 FAX: (44) 121 451 6133 EMAIL: 24-7@goodrich.com	Goodrich Corporation Maintenance Repair and Overhaul 4115 Corporate Center Drive Monroe, NC 28110-1313 USA	PHONE: (1) 704 282 2500 FAX: (1) 704 282 2560
AMERICAS: Goodrich Corporation 4115 Corporate Center Drive Monroe, NC 28110-1313 USA	PHONE: (1) 704 282 2500 AOG: (1) 800 735 7899 FAX: (1) 704 282 2560	Goodrich Corporation Maintenance Repair and Overhaul Westfield Road Pitstone Green Buckinghamshire LU7 9GT UK	PHONE: (44) 1296 663000 FAX: (44) 1296 663326 EMAIL: pitstone.repairs@goodrich.com
EUROPE/MIDEAST /AFRICA: Goodrich Corporation Stratford Road Solihull B90 4LA UK	PHONE: (44) 121 451 5788 FAX: (44) 121 451 6133	Goodrich Control Systems, Australia 84-92 Epsom Road Zetland, Sydney NSW 2017 Australia	PHONE: (61) 2 9315 3700 FAX: (61) 2 9313 7676
ASIA/PACIFIC: Goodrich Aerospace Pte Ltd 39 Changi North Crescent Republic of Singapore 499642	PHONE: (65) 6545 9975 FAX: (65) 6543 0419	In addition, Goodrich has a worldwide network of Goodrich Authorized Repair Centers (ARC) that can be found at this link: Click here	

<u>Technical Publications</u>		<u>Technical Support Information</u>	
Technical publications, distributions and subscriptions should be directed to the distribution administrators listed below:		Technical questions should be directed to the location listed below:	
<u>ADDRESS</u>	<u>COMMUNICATIONS</u>	<u>ADDRESS</u>	<u>COMMUNICATIONS</u>
Goodrich Corporation Twinsburg Technical Publications Distribution 4115 Corporate Center Drive Monroe, NC 28110-1313 USA	PHONE: (1) 704 282 2624 FAX: (1) 704 290 1906 EMAIL: techpubs.monroe@goodrich.com	Goodrich Corporation Technical Services 8380 Darrow Road Twinsburg, OH 44087 USA	PHONE: (1) 330 487 2000 FAX: (1) 330 487 1902 EMAIL: twinsburg.techpubs@goodrich.com
Goodrich Corporation Power Systems Division Westfield Road Pitstone Green Buckinghamshire LU7 9GT UK	PHONE: (44) 1296 663000 x3149 FAX: (44) 1296 663334 EMAIL: techpubs.pitstone@goodrich.com	Goodrich Corporation Power Systems Division Westfield Road Pitstone Green Buckinghamshire LU7 9GT UK	PHONE: (44) 1296 663000 FAX: (44) 1296 663334

EXCLUSIVE DISTRIBUTORS



Component Maintenance Manual with Illustrated Parts List
DC Starter-Generator, 23046 Series I

HIGHLIGHTS

TO: Holders of Component Maintenance Manual with IPL for DC Starter-Generator, 23046 Series I.

Attached to this transmittal letter is Revision No. 7 of the Component Maintenance Manual with IPL (original issue dated February 1, 1965).

REVISION 7, DATED MARCH 23/10

Remove the complete existing manual and replace with this full revision of the manual. Retain the Highlights page(s) in the front of the manual for future reference.

This Component Maintenance Manual has been fully revised per standards set by the Air Transport Association of America Specifications ATA-100 and ATA-2200 to include the latest engineering information and also to include Service Bulletin(s) and Service Information Letter(s) as listed below.

Key items included in this revision:

- Incorporated SB 23046-0XX-24-05, Rev 1.
- Incorporated SB 23046-0XX-24-06, Rev 0.
- Incorporated SB 23046-0XX-24-07, Rev 1.
- Incorporated GSIL 2006-01, Rev 0.
- Incorporated GSIL 2006-02, Rev 0.
- Changed the IPL to numerical order of 10s instead of 5s. This was reflected throughout the manual.

The technical changes in this revision are individually identified with revision bars.



Component Maintenance Manual with Illustrated Parts List
DC Starter-Generator, 23046 Series I

THIS PAGE INTENTIONALLY LEFT BLANK



Component Maintenance Manual with Illustrated Parts List

DC Starter-Generator, 23046 Series I

RECORD OF REVISIONS

[illegible]

RECORD OF REVISIONS



Component Maintenance Manual with Illustrated Parts List
DC Starter-Generator, 23046 Series I

THIS PAGE INTENTIONALLY LEFT BLANK

RECORD OF REVISIONS



Component Maintenance Manual with Illustrated Parts List
DC Starter-Generator, 23046 Series I

RECORD OF TEMPORARY REVISIONS

Temporary Revision Number	Page Number	Date Issued	Date Inserted	Initials	Date Removed	Initials
23046-0XX-05	2 pages	Jan 25/00	Jan 25/00	GR	Jun 27/01	GR
23046-0XX-06	1 page	Jan 25/00	Jan 25/00	GR	Jun 27/01	GR
23046-0XX-07	2 pages	Feb 3/00	Feb 3/00	GR	Jun 27/01	GR
24-08	6 pages	Feb 13/09	Feb 13/09	GR	Mar 23/10	GR
Retain this record in the front of the CMM. On receipt of temporary revisions, insert affected pages into the manual. Record the temporary revision number, page number, date issued, date inserted into the CMM, and initials on this page. When temporary revisions are removed from the CMM, record the date removed and initials.						

RECORD OF TEMPORARY REVISIONS



Component Maintenance Manual with Illustrated Parts List
DC Starter-Generator, 23046 Series I

. THIS PAGE INTENTIONALLY LEFT BLANK

RECORD OF TEMPORARY REVISIONS



Component Maintenance Manual with Illustrated Parts List

DC Starter-Generator, 23046 Series I

SERVICE BULLETIN LIST

[illegible]

Retain this record in the front of the CMM. On receipt of service bulletins, record the service bulletin number, status (valid or invalid), revision number (if applicable) and the date issued.



Component Maintenance Manual with Illustrated Parts List
DC Starter-Generator, 23046 Series I

THIS PAGE INTENTIONALLY LEFT BLANK



Component Maintenance Manual with Illustrated Parts List

DC Starter-Generator, 23046 Series I

SERVICE INFORMATION LETTER LIST

[illegible]

Retain this record in the front of the CMM. On receipt of Service Information Letters, record the Service Information Letter number, status, revision number (if applicable) and the date issued.

SERVICE INFORMATION LETTER LIST



Component Maintenance Manual with Illustrated Parts List
DC Starter-Generator, 23046 Series I

THIS PAGE INTENTIONALLY LEFT BLANK



Component Maintenance Manual with Illustrated Parts List
DC Starter-Generator, 23046 Series I

LIST OF EFFECTIVE PAGES

<u>Title</u>	<u>Page</u>	<u>Date</u>
EXCLUSIVE DISTRIBUTOR	ii	Mar 23/10
RECORD OF REVISIONS	iii	Mar 23/10
	iv	Blank
RECORD OF TEMPORARY REVISIONS	v	Mar 23/10
	vi	Blank
SERVICE BULLETIN LIST	vii	Mar 23/10
	viii	Blank
SERVICE INFORMATION LETTER LIST	ix	Mar 23/10
	x	Blank
LIST OF EFFECTIVE PAGES	xi	Mar 23/10
	xii	Mar 23/10
	xiii	Mar 23/10
	xiv	Mar 23/10
	xv	Mar 23/10
	xvi	Mar 23/10
TABLE OF CONTENTS	xvii	Mar 23/10
	xviii	Blank
LIST OF ILLUSTRATIONS	xix	Mar 23/10
	xx	Mar 23/10
	xxi	Mar 23/10
	xxii	
LIST OF TABLES	xxiii	Mar 23/10
	xxiv	Blank
INTRODUCTION	xxv	Mar 23/10
	xxvi	Mar 23/10
	xxvii	Mar 23/10
	xxviii	Mar 23/10
	xxix	Mar 23/10
	xxx	Blank
DESCRIPTION AND OPERATION	1	Mar 23/10
	2	Mar 23/10
	3	Mar 23/10
	4	Mar 23/10
	5	Mar 23/10
	6	Mar 23/10
	7	Mar 23/10
	8	Mar 23/10
	9	Mar 23/10
	10	Blank

LIST OF EFFECTIVE PAGES



Component Maintenance Manual with Illustrated Parts List
DC Starter-Generator, 23046 Series I

LIST OF EFFECTIVE PAGES (Continued)

<u>Title</u>	<u>Page</u>	<u>Date</u>
TESTING AND FAULT ISOLATION.	1001	Mar 23/10
	1002	Mar 23/10
	1003	Mar 23/10
	1004	Mar 23/10
	1005	Mar 23/10
	1006	Mar 23/10
	1007	Mar 23/10
	1008	Mar 23/10
	1009	Mar 23/10
	1010	Mar 23/10
	1011	Mar 23/10
	1012	Mar 23/10
	1013	Mar 23/10
	1014	Mar 23/10
	1015	Mar 23/10
	1016	Mar 23/10
	1017	Mar 23/10
	1018	Mar 23/10
	1019	Mar 23/10
	1020	Mar 23/10
	1021	Mar 23/10
	1022	Mar 23/10
	1023	Mar 23/10
	1024	Mar 23/10
	1025	Mar 23/10
	1026	Mar 23/10
	1027	Mar 23/10
	1028	Blank
SCHEMATICS AND WIRING DIAGRAMS.	2001	Mar 23/10
	2002	Blank
DISASSEMBLY	3001	Mar 23/10
	3002	Mar 23/10
	3003	Mar 23/10
	3004	Mar 23/10
	3005	Mar 23/10
	3006	Mar 23/10
	3007	Mar 23/10
	3008	Mar 23/10

LIST OF EFFECTIVE PAGES



Component Maintenance Manual with Illustrated Parts List
DC Starter-Generator, 23046 Series I

LIST OF EFFECTIVE PAGES (Continued)

<u>Title</u>	<u>Page</u>	<u>Date</u>
DISASSEMBLY (Cont)	3009	Mar 23/10
	3010	Mar 23/10
	3011	Mar 23/10
	3012	Mar 23/10
	3013	Mar 23/10
	3014	Mar 23/10
	3015	Mar 23/10
	3016	Blank
CLEANING	4001	Mar 23/10
	4002	Mar 23/10
	4003	Mar 23/10
	4004	Mar 23/10
	4005	Mar 23/10
	4006	Mar 23/10
CHECK	5001	Mar 23/10
	5002	Mar 23/10
	5003	Mar 23/10
	5004	Mar 23/10
	5005	Mar 23/10
	5006	Mar 23/10
	5007	Mar 23/10
	5008	Mar 23/10
	5009	Mar 23/10
	5010	Mar 23/10
	5011	Mar 23/10
	5012	Mar 23/10
	5013	Mar 23/10
	5014	Mar 23/10
	5015	Mar 23/10
	5016	Mar 23/10
	5017	Mar 23/10
	5018	Mar 23/10
	5019	Mar 23/10
	5020	Mar 23/10
	5021	Mar 23/10
	5022	Mar 23/10
	5023	Mar 23/10
	5024	Mar 23/10

LIST OF EFFECTIVE PAGES



Component Maintenance Manual with Illustrated Parts List
DC Starter-Generator, 23046 Series I

LIST OF EFFECTIVE PAGES (Continued)

<u>Title</u>	<u>Page</u>	<u>Date</u>
CHECK (Contd)	5025	Mar 23/10
	5026	Mar 23/10
	5027	Mar 23/10
	5028	Mar 23/10
	5029	Mar 23/10
	5030	Mar 23/10
	5031	Mar 23/10
	5032	Blank
REPAIR	6001	Mar 23/10
	6002	Mar 23/10
	6003	Mar 23/10
	6004	Mar 23/10
.....	6005	Mar 23/10
	6006	Mar 23/10
	6007	Mar 23/10
	6008	Mar 23/10
	6009	Mar 23/10
	6010	Mar 23/10
	6011	Mar 23/10
	6012	Mar 23/10
	6013	Mar 23/10
	6014	Mar 23/10
	6015	Mar 23/10
	6016	Mar 23/10
	6017	Mar 23/10
	6018	Mar 23/10
ASSEMBLY	7001	Mar 23/10
	7002	Mar 23/10
	7003	Mar 23/10
	7004	Mar 23/10
	7005	Mar 23/10
	7006	Mar 23/10
	7007	Mar 23/10
	7008	Mar 23/10
	7009	Mar 23/10
	7010	Mar 23/10
	7011	Mar 23/10
	7012	Mar 23/10

LIST OF EFFECTIVE PAGES



Component Maintenance Manual with Illustrated Parts List
DC Starter-Generator, 23046 Series I

LIST OF EFFECTIVE PAGES (Continued)

<u>Title</u>	<u>Page</u>	<u>Date</u>
ASSEMBLY (Cont)	7013	Mar 23/10
	7014	Mar 23/10
	7015	Mar 23/10
	7016	Mar 23/10
	7017	Mar 23/10
	7018	Mar 23/10
	7019	Mar 23/10
	7020	Mar 23/10
	7021	Mar 23/10
	7022	Mar 23/10
	7023	Mar 23/10
	7024	Mar 23/10
FITS AND CLEARANCES	8001	Mar 23/10
	8002	Mar 23/10
	8003	Mar 23/10
	8004	Mar 23/10
	8005	Mar 23/10
	8006	Blank
SPECIAL TOOLS, FIXTURES, AND EQUIPMENT	9001	Mar 23/10
	9002	Mar 23/10
	9003	Mar 23/10
	9004	Mar 23/10
	9005	Mar 23/10
	9006	Mar 23/10
	9007	Mar 23/10
	9008	Mar 23/10
	9009	Mar 23/10
	9010	Mar 23/10
	9011	Mar 23/10
	9012	Mar 23/10
	9013	Mar 23/10
	9014	Mar 23/10
	9015	Mar 23/10
	9016	Mar 23/10
SPECIAL PROCEDURES	11001	Mar 23/10
	11002	Blank
REMOVAL	12001	Mar 23/10
	12002	Blank

LIST OF EFFECTIVE PAGES



Component Maintenance Manual with Illustrated Parts List
DC Starter-Generator, 23046 Series I

LIST OF EFFECTIVE PAGES (Continued)

<u>Title</u>	<u>Page</u>	<u>Date</u>
INSTALLATION	13001	Mar 23/10
	13002	Blank
SERVICING	14001	Mar 23/10
	14002	Blank
STORAGE	15001	Mar 23/10
	15002	Mar 23/10
	15003	Mar 23/10
	15004	Blank
REWORK	16001	Mar 23/10
	16002	Blank
ILLUSTRATED PARTS LIST	10001	Mar 23/10
	10002	Mar 23/10
	10003	Mar 23/10
	10004	Mar 23/10
	10005	Mar 23/10
	10006	Mar 23/10
	10007	Mar 23/10
	10008	Mar 23/10
	10009	Mar 23/10
	10010	Mar 23/10
	10011	Mar 23/10
	10012	Mar 23/10
	10013	Mar 23/10
	10014	Mar 23/10
	10015	Mar 23/10
	10016	Mar 23/10
	10017	Mar 23/10
	10018	Mar 23/10
	10019	Mar 23/10
	10020	Mar 23/10
	10021	Mar 23/10
	10022	Mar 23/10

LIST OF EFFECTIVE PAGES



Component Maintenance Manual with Illustrated Parts List
DC Starter-Generator, 23046 Series I

TABLE OF CONTENTS

<u>Title</u>	<u>Page</u>
EXCLUSIVE DISTRIBUTOR	.ii
RECORD OF REVISIONS	iii
RECORD OF TEMPORARY REVISIONS	v
SERVICE BULLETIN LIST	vii
SERVICE INFORMATION LETTER LIST	ix
LIST OF EFFECTIVE PAGES	xi
TABLE OF CONTENTS	xvii
LIST OF ILLUSTRATIONS	xix
LIST OF TABLES	xxiii
INTRODUCTION	xxv
DESCRIPTION AND OPERATION	1
TESTING AND FAULT ISOLATION	1001
SCHEMATICS AND WIRING DIAGRAMS	2001
DISASSEMBLY	3001
CLEANING	4001
CHECK	5001
REPAIR	6001
ASSEMBLY	7001
FITS AND CLEARANCES	8001
SPECIAL TOOLS, FIXTURES, AND EQUIPMENT	9001
SPECIAL PROCEDURES	11001
REMOVAL	12001
INSTALLATION	13001
SERVICING	14001
STORAGE	15001
REWORK	16001
ILLUSTRATED PARTS LIST	10001



Component Maintenance Manual with Illustrated Parts List
DC Starter-Generator, 23046 Series I

THIS PAGE INTENTIONALLY LEFT BLANK



Component Maintenance Manual with Illustrated Parts List
DC Starter-Generator, 23046 Series I

LIST OF ILLUSTRATIONS

<u>Figure</u>	<u>Title</u>	<u>Page</u>
1	Parts of the 23046 Series I DC Starter-Generator	1
2	Outline Drawing of Model 23046-001	5
3	Outline Drawing of Models 23046-007 and -007M	5
4	Outline Drawing of Model 23046-009	6
5	Outline Drawing of Model 23046-017	7
6	Outline Drawing of Models 23046-019 and -020	8
7	Outline Drawing of Model 23046-028	9
1001	Electrical Connections	1002
1002	Test Set-Up for Starter Test	1006
1003	Test Set-Up for Generator Test	1006
1004	Equalizing Voltage Acceptance Limits, 23046-001, -007, -007M, -009 and -017	1008
1005	Equalizing Voltage Acceptance Limits, 23046-019 and -020	1009
1006	Equalizing Voltage Acceptance Limits, 23046-028	1010
1007	Correct Viewing Angle	1012
3001	Removing Brushes	3005
3002	Removing Cover Assembly or Fan Cover and Air Inlet Assembly	3006
3003	Removing Drive Shaft	3007
3004	Dampener Hub Removal	3008
3005	Dampener Backplate Removal	3009
3006	Removing Bearing and Brush Support Assembly with Attached Armature	3010
3007	Removing Armature from Bearing and Brush Support Assembly	3012
3008	Removing Bearing from Drive End of Armature Shaft	3013
3009	Removing Bearing from Bearing and Brush Support Assembly	3014
3010	Removing Brush Holders	3015
5001	Inspecting End Bell, Drive End Pad	5006
5002	Typical Replacement Identification Plate	5007
5003	Brush Access Cover	5008
5004	Two Part Fan	5010
5005	Anti-Drive End Bearing Shield	5011
5006	Inspecting Drive Shaft and Dampener Hub	5012
5007	Dampener Plate	5013
5008	Friction Ring	5014
5009	Inspecting Dampener Backplate	5015
5010	Drive End Bearing Support Assembly	5016
5011	Inspecting Baffle Disc	5017
5012	Bearing Shield	5017
5013	Armature Check	5020

LIST OF ILLUSTRATIONS



Component Maintenance Manual with Illustrated Parts List
DC Starter-Generator, 23046 Series I

LIST OF ILLUSTRATIONS (Continued)

<u>Figure</u>	<u>Title</u>	<u>Page</u>
5014	Bearing and Brush Support Assembly	5024
5015	Determine Spring Force	5025
5016	Inspecting Bearing Retainer	5025
5017	Stator and Housing Assembly	5027
5018	Terminal Block	5029
6001	Helical Coil Insert Replacement	6006
6002	Drive End End Bell Machining Specifications	6007
6003	Anti-Drive End End Bell Machining Specifications	6008
6004	Armature Repair	6011
6005	Polarization Connection Diagram	6014
6006	Replacement of the Brush Access Cover Rivet Nut and Insulating Tape	6016
6007	Repair of the Mounting Adapter Guide Pin	6018
7001	Brush Spring Assembly	7004
7002	Bearing and Brush Support Assembly Detail (with Bolts)	7007
7003	Ball Bearing Installation into Bearing and Brush Support Assembly	7008
7004	Attaching Bearing Retainer	7009
7005	Baffle Disc or Bearing Shield and Ball Bearing Installation on Drive End of Armature Shaft	7010
7006	Installing Outer Baffle Disc and Retaining Ring	7011
7007	Attaching Bearing and Brush Support Assembly and Armature	7012
7008	Assembling Bearing and Brush Support Assembly with Stator and Housing Assembly	7014
7009	Dampener Plate and Driver	7016
7010	Installing Drive Shaft into Armature Shaft	7017
7011	Installing Fan	7018
7012	Installing Brushes	7019
8001	Measuring Bearing Liner Diameters	8004
8002	Measuring Armature Bearing Journal Diameters	8005
9001	Armature Shaft Adapter, Drive End	9002
9002	Armature Shaft Adapter, Anti-Drive End	9002
9003	Commutation Viewing Adapters	9003
9004	Inner Race Bearing Driver	9004
9005	Outer Race Bearing Driver	9005
9006	Commutator Turning Fixture	9006
9007	Armature Support	9007
9008	Bearing and Brush Support Assembly Support	9008
9009	Drive End Hub Support	9009
9010	Horizontal Stator Support	9010

LIST OF ILLUSTRATIONS



Component Maintenance Manual with Illustrated Parts List
DC Starter-Generator, 23046 Series I

LIST OF ILLUSTRATIONS (Continued)

<u>Figure</u>	<u>Title</u>	<u>Page</u>
9011	Vertical Stator Support	9011
9012	Dampener Hub Driver	9012
9013	Dampener Plate Driver	9013
9014	Spline Wrench	9014
9015	Anti-Drive End Bearing Hub Support	9015
9016	PlusNut® Fastener Header	9016
10001	DC Starter-Generator Model 23046 Series I	10010



Component Maintenance Manual with Illustrated Parts List
DC Starter-Generator, 23046 Series I

THIS PAGE INTENTIONALLY LEFT BLANK



Component Maintenance Manual with Illustrated Parts List
DC Starter-Generator, 23046 Series I

LIST OF TABLES

<u>Table</u>	<u>Title</u>	<u>Page</u>
i-1	Hazardous Materials	xxvii
1	Leading Particulars.	2
2	Model Particulars	3
3	Overhung Moment and Shaft Shear Torque	3
4	MOD Status	4
1001	Performance Test Conditions.	1001
1002	Test Equipment Requirements and Specifications.	1002
1003	Setpoint Tolerances	1004
1004	Equalizing Voltage Graphs.	1008
1005	Fault Isolation Table	1016
3001	Disassembly Tools	3001
3002	Disassembly Materials	3002
4001	Cleaning Materials and Equipment	4001
4002	Cleaning Procedures	4003
5001	Inspection Tools	5001
5002	Terminology	5030
6001	Repair Tools	6002
6002	Repair Materials	6003
6003	Refinishing Procedures	6013
7001	Assembly Tools	7001
7002	Assembly Materials	7002
8001	Torque Values	8001
8002	Acceptance Limits	8002
9001	Special Tools, Fixtures, and Equipment.	9001
15001	Packaging Material.	15001



Component Maintenance Manual with Illustrated Parts List
DC Starter-Generator, 23046 Series I

THIS PAGE INTENTIONALLY LEFT BLANK



Component Maintenance Manual with Illustrated Parts List
DC Starter-Generator, 23046 Series I

INTRODUCTION

1. Purpose

This Component Maintenance Manual (CMM) provides detailed instructions for overhaul and service repair of the Goodrich Corporation 23046 Series I DC Starter-Generators. Line maintenance is supported by Maintenance Manual (MM) 23700. A DC Starter-Generator overhaul includes:

- replacement of non-reusable parts such as bearings, brushes and miscellaneous hardware
- cleaning of subassemblies
- detailed inspection of parts, electrical check of all electromagnetic parts (including; insulation integrity check), Non-Destructive Test (NDT) inspections, and complete acceptance testing

NOTE: Magnetic particle inspection of the parts as specified in the CHECK section is only required when an overhaul of the starter-generator is being done.

- reconditioning of surface finishes as required
- repair/rework of parts as required
- re-certification including final assembly, records, and release tags.

Only an overhaul and an acceptance test authorize assignment of zero operating hours time since overhaul (TSO) to a DC Starter-Generator. Repairs performed that are partial of an overhaul do not affect TSO and equipment is released on a continue time basis.

An illustrated parts list is included at the back of this CMM for component identification and ordering of approved replacement parts. The structure and content of this manual is in general accordance with the requirements established by the Air Transport Association of America (ATA) for CMMs. This manual is formatted to conform to ATA Specification No. 100.

Give careful attention to applicable warnings and cautions. Before starting any overhaul procedure, become thoroughly familiar with the capabilities and limitations of the equipment. Make sure all necessary safety equipment, test equipment, repair materials, special tools, and fixtures are on hand.

All weights and measures used in this manual are in English units, followed by the Système Internationale (SI) equivalent in parentheses.

If an error, omission, or other technical discrepancy is found in this manual, fill out a copy of the Technical Publication Comment Form found at the back of this manual and send to Supervisor of Technical Publications, Goodrich Corporation or provide information to Technical Publications e-mail at twinsburg.techpubs@goodrich.com.



Component Maintenance Manual with Illustrated Parts List
DC Starter-Generator, 23046 Series I

2. User Qualification and Certification Requirements

The information contained in this manual is intended for persons authorized to perform maintenance, preventative maintenance, rebuilding, and alterations in accordance with Federal Aviation Regulation (FAR), Part-43.3.

3. Quality Assurance Requirements

The repair stations quality assurance activity is responsible for the correct performance of all tests and inspections specified in this manual. The quality assurance activity shall also be responsible for maintaining all necessary test, inspection, and maintenance records for each unit received for service.

All instrumentation and inspection equipment shall be calibrated and controlled in accordance with International Standards Organization (ISO) standard 10012, with all standards traceable to the National Bureau of Standards or an equivalent standards regulatory authority.

The quality assurance activity shall be responsible for certifying that personnel, skills, and materials meet the requirements of the work to be performed. Components of the DC Starter-Generator undergoing overhaul that are recovered as products of disassembly shall be examined 100% to determine serviceability.

The quality assurance activity shall maintain documented evidence that specifications applicable to special processes such as soldering, nondestructive testing (NDT), plating, etc. have been complied with during repair and/or overhaul of the DC Starter-Generator.

4. Safety Advisory

This manual describes physical and chemical processes that require the use of chemicals or other commercially available materials that require precautionary attention.

The user of this manual should obtain Material Safety Data Sheets and Occupational Safety and Health Act (OSHA) Form 20 or equivalent from the manufacturers or suppliers of materials to be used. The user must become thoroughly familiar with and follow all manufacturer/supplier procedures, recommendations, warnings, and cautions for the safe use, handling, storage, and disposal of materials that require precautionary attention. Users of this manual are also advised to refer to the applicable safety information contained in the "NIOSH Occupational Guideline for Chemical Hazards" published by the United States Department of Labor.

WARNING: ALERTS OPERATING AND MAINTENANCE PERSONNEL TO POTENTIAL HAZARDS THAT COULD RESULT IN PERSONAL INJURY; WARNINGS DO NOT REPLACE THE MANUFACTURER'S RECOMMENDATIONS.

CAUTION: ALERTS OPERATING AND MAINTENANCE PERSONNEL TO CONDITIONS THAT COULD RESULT IN EQUIPMENT DAMAGE.



Component Maintenance Manual with Illustrated Parts List
DC Starter-Generator, 23046 Series I

5. Materials List

The materials listed in this section are necessary for processes throughout this manual. A warning and/or caution will precede the use of materials listed in the following table:

WARNING: BEFORE USING ANY OF THE FOLLOWING MATERIALS, BE AWARE OF ALL HANDLING, STORAGE, AND DISPOSAL PRECAUTIONS RECOMMENDED BY THE MANUFACTURER OR SUPPLIER. FAILURE TO COMPLY WITH MANUFACTURER OR SUPPLIER RECOMMENDATIONS MAY RESULT IN SERIOUS INJURY, PHYSICAL DISORDER, OR DEATH.

Material	Used in
Acrylic Coating	ASSEMBLY
Anti-seize Compound	ASSEMBLY
Chemical Film Solution	REPAIR
Corrosion Preventative	CLEANING
Detergent	CLEANING
Dichromate Solution	REPAIR
Enamel Paint	REPAIR
Epoxy Primer	REPAIR
Insulating Enamel	ASSEMBLY
Isopropyl Alcohol	DISASSEMBLY, CLEANING, CHECK, REPAIR, ASSEMBLY
Loctite Grade D and Grade N	ASSEMBLY
Machine Oil	REPAIR
Zinc Chromate Primer	REPAIR
Zinc Phosphate Coating	REPAIR

Table Intro -1 - Hazardous Materials

6. Non-Goodrich Corporation Authorized Components and Processes Policy

Goodrich Corporation authorizes the use of Goodrich Corporation spare parts which meet stringent engineering design specifications and quality standards, and have traceability to having been procured and certified to design specifications by Goodrich Corporation Quality Assurance incoming and in process inspection systems. Other than the type certification holder (airframe manufacturer), Goodrich Corporation Equipment Service Centers are the only authorized distributors of Goodrich Corporation replacement parts and complete units.

It is the obligation of all repair and service facilities to provide the FAA, or any other in-country air authority, with proper traceability documentation indicating approval of all spare parts, materials, and processes to ensure configuration compliance and continued air worthiness.



Component Maintenance Manual with Illustrated Parts List
DC Starter-Generator, 23046 Series I

The use of any non-Goodrich Corporation authorized part, or any parts not having been submitted to Goodrich Corporation Quality Assurance inspection system will invalidate all factory warranties. All Goodrich Corporation warranties are automatically voided on any Goodrich Corporation designed unit that has been modified by the installation of any unauthorized parts, materials, or unapproved processes supplied by other outside services. The repair station's quality assurance activity shall assume product liability for all units that have been modified in this fashion.

Damage resulting from the use of non-Goodrich Corporation replacement parts, materials, or processes is not covered by the Goodrich Corporation warranty or service policy for any product or application.

7. Abbreviations

The following abbreviations are used in this manual:

A	- Ampere
CCW	- Counter Clockwise
CMM	- Component Maintenance Manual
DC	- Direct Current
GCU	- Generator Control Unit
IAW	- in accordance with
lbf.in.	- pounds force inches
IPL	- Illustrated Parts List
kPa	- kilopascals
MHz	- Megahertz
N·m	- Newton-meter(s)
NDT	- Non-Destructive Test
psig	- pounds per square inch gage
QAD	- Quick attach/detach
RMS	- RMS = 0.707 peak-to-peak for sine wave of AC waveform.
rpm	- Revolutions Per Minute
SI	- Système Internationale (SI)
SPD	- Standard Practice Document



Component Maintenance Manual with Illustrated Parts List
DC Starter-Generator, 23046 Series I

TIR	- Total Indicator Reading
TSO	- time since overhaul
V	- Volt
VAC	- Voltage Alternating Current
VDC	- Voltage Direct Current
Ω	- Ohm



Component Maintenance Manual with Illustrated Parts List
DC Starter-Generator, 23046 Series I

THIS PAGE INTENTIONALLY LEFT BLANK



Component Maintenance Manual with Illustrated Parts List
DC Starter-Generator, 23046 Series I

DESCRIPTION AND OPERATION

1. Purpose

The 23046 Series I DC Starter-Generators are designed to operate as a motor providing torque for engine starting and (after engine start-up) generating DC electrical power.

Some models feature a quick attach/detach (QAD) kit easing starter-generator removal/installation on the aircraft. A QAD kit consists of a mounting adapter attaching to the engine gearbox accessory drive pad, and a V-retainer coupling securing the starter-generator to the mounting adapter.

The 23046 Series I Starter-Generators consist of eight models, -001, -007, -007M, -009, -017, -019, -020 and -028. Examples of inter-relations between models: Model 23046-019 is a Model 23046-020 with a 23046-516 QAD kit installed and Model 23046-007M is a Model 23046-007 with new brushes.

The starter-generators consist of an armature enclosed by, and rotating within, the stator and housing assembly. See Figure 1. The armature is supported on two bearings, one at the drive end bearing support assembly, and a second at the bearing and brush support assembly. The drive shaft is installed into the hollow armature shaft. At the drive end, a dampener assembly absorbs torsional vibration generated by changes in engine gearbox speed and electrical load conditions. The dampener assembly consists of a dampener plate, a friction ring and a dampener backplate.

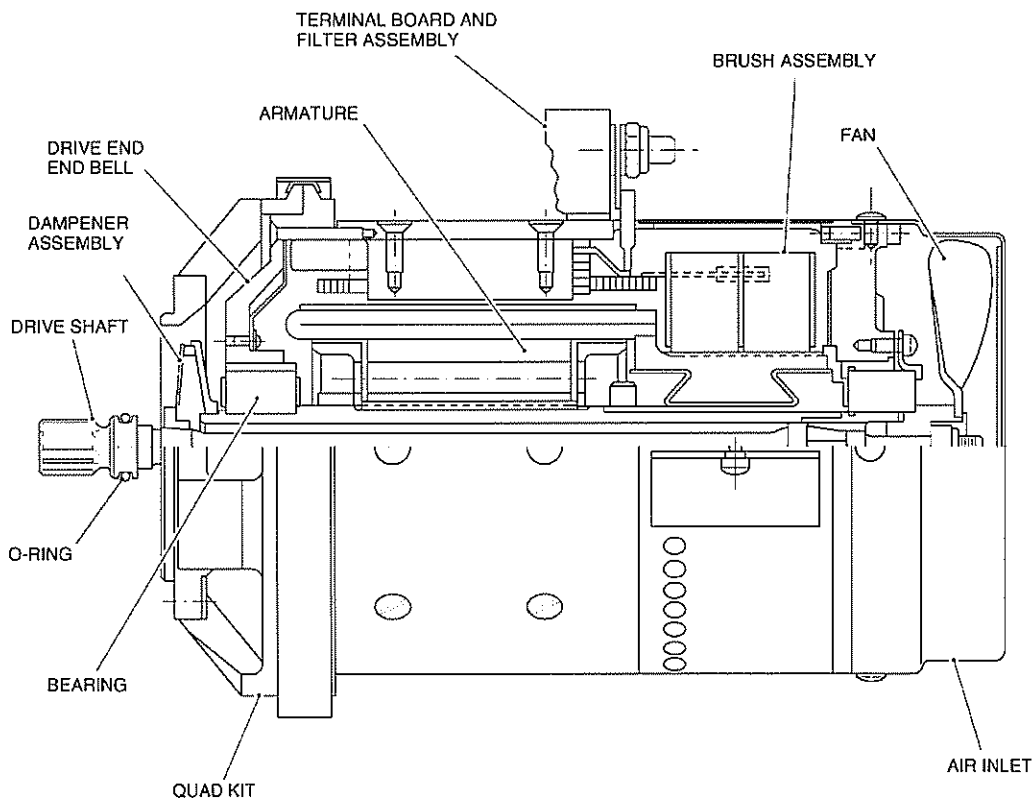


Figure 1 - Parts of the 23046 Series I DC Starter-Generator



Component Maintenance Manual with Illustrated Parts List DC Starter-Generator, 23046 Series I

This Starter-Generator is a self-excited machine incorporating four main poles and four interpoles. During on-ground operation, a fan attached to the anti-drive end of the drive shaft supplies forced-air cooling. In flight, the starter-generator, in addition to the fan, is blast-cooled through the air inlet and aircraft's air duct system. The fan is protected by a screened cover. Electrical and mechanical characteristics are given in Table 1.

Outline drawings for all models are shown in Figure 2 through Figure 7.

2. Operation

As an engine starter, the machine can be energized either by batteries or by ground-power units. The Starter-Generator utilizes a series starting field and should be started with shunt field deenergized.

CAUTION: THE START SWITCH MUST NOT BE HELD CLOSED FOR MORE THAN THE FEW SECONDS NEEDED FOR A NORMAL ENGINE START; OTHERWISE, THE MACHINE MAY BE DAMAGED BY EXCESSIVE TEMPERATURES CAUSED BY THE LARGE STARTING CURRENTS.

As a generator, the machine provides rated DC output when driven at speeds within the rated speed range. The power supply for starter operation should first be disconnected, and the generator should then be connected to the load bus. The generator will operate satisfactorily with either a carbon-pile or static type regulator conforming to BuWep Drawing E-1597-1 and MIL-R-6809. A reverse-current relay is normally used for automatically disconnecting the generator from the load bus, if generator voltage drops below a pre-determined minimum.

Characteristic	Value
Rating	6 kW
Continuous load within speed range	200 A
Voltage (gen. output)	30 V
Speed range (rated load)	6,200 to 12,000 rpm
Minimum speed for regulation	7,800 rpm
Direction of rotation (viewing drive end)	CCW
Operating altitude	35,000 feet (10,668 m) max.
Cooling	Self-Cooled
Terminal Designations	
Positive	B+
Negative	E-
Positive field	A+
Equalizer	D
Starting	C+
External start power	28 V max.
See Table 2 for starting current	

Table 1 - Leading Particulars



Component Maintenance Manual with Illustrated Parts List
DC Starter-Generator, 23046 Series I

Model No.	Weight lbs. (kg)	Spline and Mounting Flange Specification	Spline No. of Teeth	Spline Pitch Dia. inch (mm)	Cooling Opening	Starting Current (A)
23046-001	25 (No QAD) (11,34)	AND 10261, Type XI-B	12	0.600 (15,24)	Screened Cover	500
23046-007	27.8* (12,61)	AND 10262, Type XII-A	16	0.800 (20,32)	Screened Cover	500
23046-007M	27.8* (12,61)	AND 10262, Type XII-A	16	0.800 (20,32)	Screened Cover	500
23046-009	28.6* (12,97)	AND 10261, Type XI-B	12	0.600 (15,24)	Screened Cover	500
23046-017	28.4* (12,88)	AS972A spline, MS3326-2	12	0.600 (15,24)	Screened Cover	500
23046-019	30* (13,61)	AS972 spline, MS3326-2	12	0.600 (15,24)	Equivalent to 3 inch (76,2mm) Cover	800
23046-020	30* (13,61)	AS972 spline, MS3326-2	12	0.600 (15,24)	Equivalent to 3 inch (76,2 mm) Cover	800
23046-028	28.3* (12,84)	AND 10262, Type XII-A	16	0.800 (20,32)	Screened Cover	500
* with QAD kit						

Table 2 - Model Particulars

Model No	Overhung Moment lbf.in (N•m)	Shaft Shear Section Torque lbf.in (N•m) Max.
23046-001	105 (11,86)	1500 (169,5)
23046-007	120 (13,56)	1500 (169,5)
23046-007M	120 (13,56)	1500 (169,5)
23046-009	116 (13,11)	1500 (169,5)
23046-017	125 (14,12)	1100 (124,3)
23046-019	135 (15,25)	1600 (180,8)
23046-020	135 (15,25)	1600 (180,8)
23046-028	122 (13,78)	1500 (169,5)

Table 3 - Overhung Moment and Shaft Shear Torque

DESCRIPTION AND OPERATION



Component Maintenance Manual with Illustrated Parts List
DC Starter-Generator, 23046 Series I

3. **MOD Status**

See Table 4 below for the latest MOD level of each Starter-Generator model.

23046-								Description
001	007	007M	009	017	019	020	028	
A	A	A	A	-	A	A	A	03-6010-15 Bearings
B	B	B	B	A	B	B	B	03-6010-18 Bearings (SB 23046-0XX-24-03 or SB 23046-0XX-24-04)
-	-	-	-	-	C	C	-	23088-1324 Brushes (SB 23046-0XX-24-05)
-	-	-	-	-	D	D	-	23080-3111/-3112 Shields and 23048-1486 Fan (SB 23046-0XX-24-06)
-	-	-	-	-	E	E	-	23093-1300 Brushes (SB 23046-0XX-24-07)

Table 4 - MOD Status



Component Maintenance Manual with Illustrated Parts List DC Starter-Generator, 23046 Series I

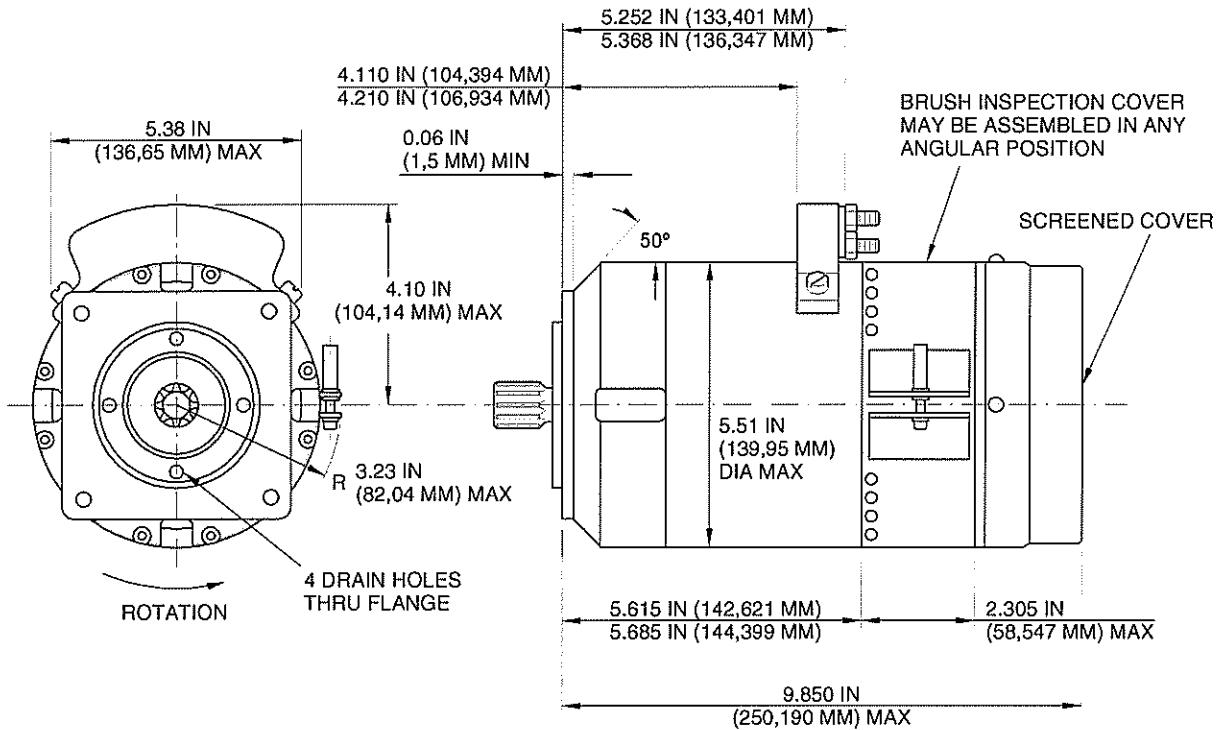


Figure 2 - Outline Drawing of Model 23046-001

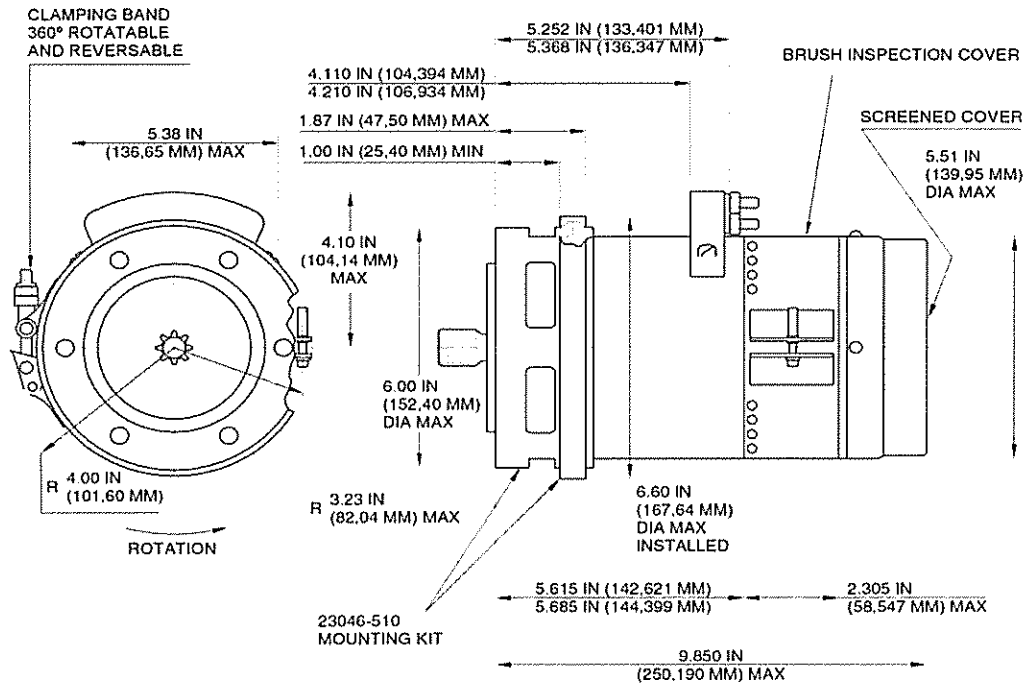


Figure 3 - Outline Drawing of Models 23046-007 and -007M

DESCRIPTION AND OPERATION



Component Maintenance Manual with Illustrated Parts List DC Starter-Generator, 23046 Series I

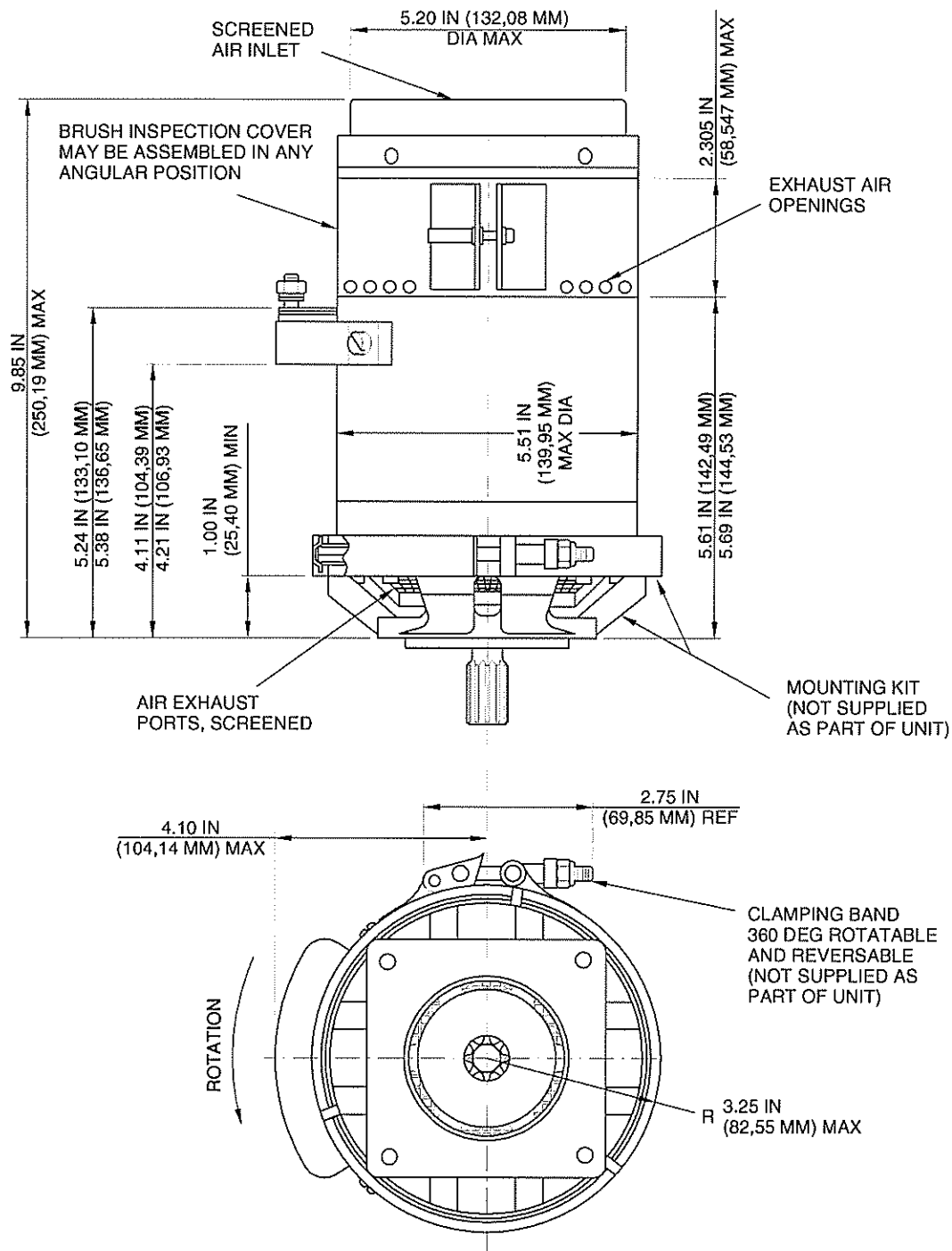


Figure 4 - Outline Drawing of Model 23046-009

DESCRIPTION AND OPERATION



Component Maintenance Manual with Illustrated Parts List
DC Starter-Generator, 23046 Series I

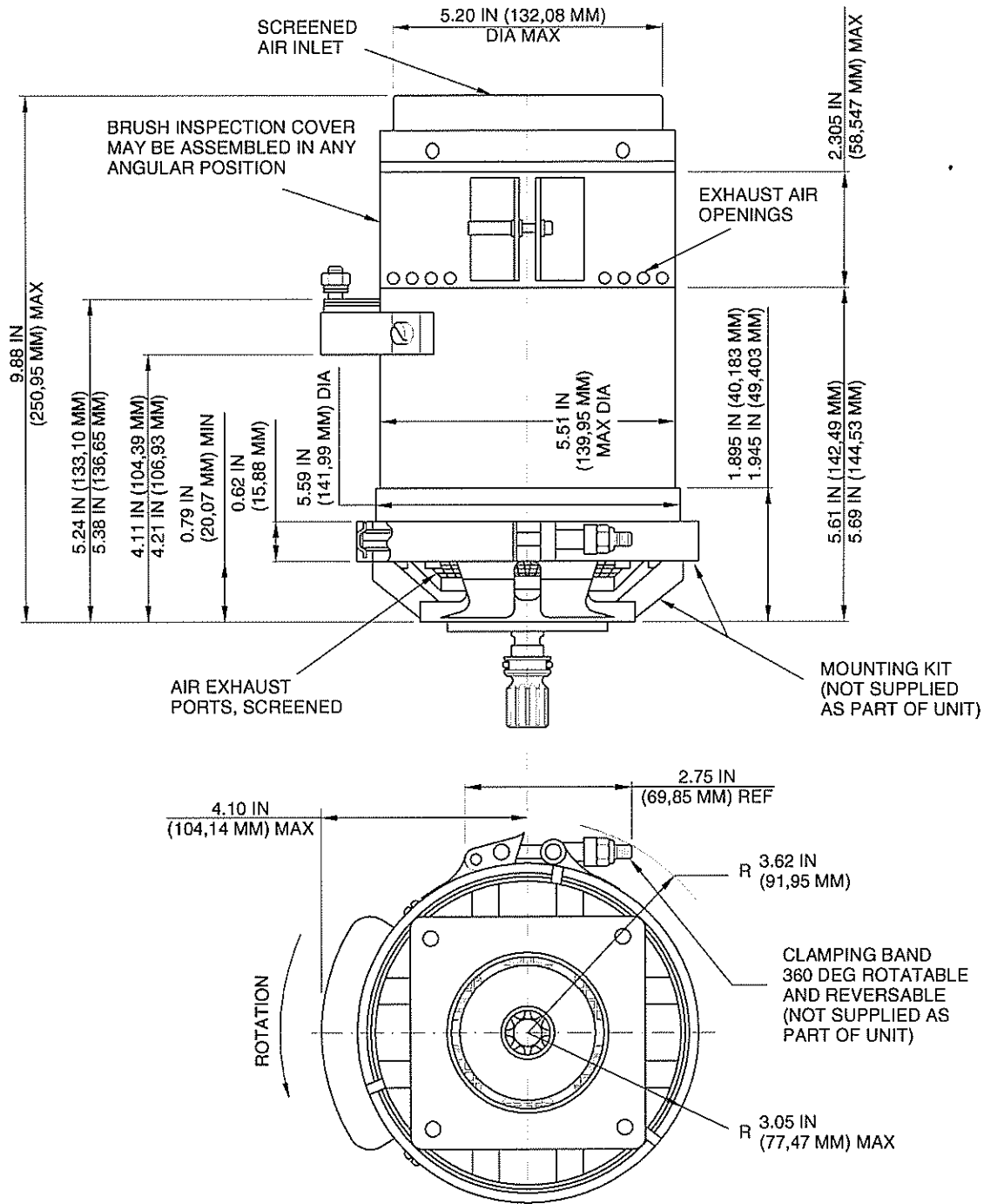


Figure 5 - Outline Drawing of Model 23046-017

DESCRIPTION AND OPERATION



Component Maintenance Manual with Illustrated Parts List
DC Starter-Generator, 23046 Series I

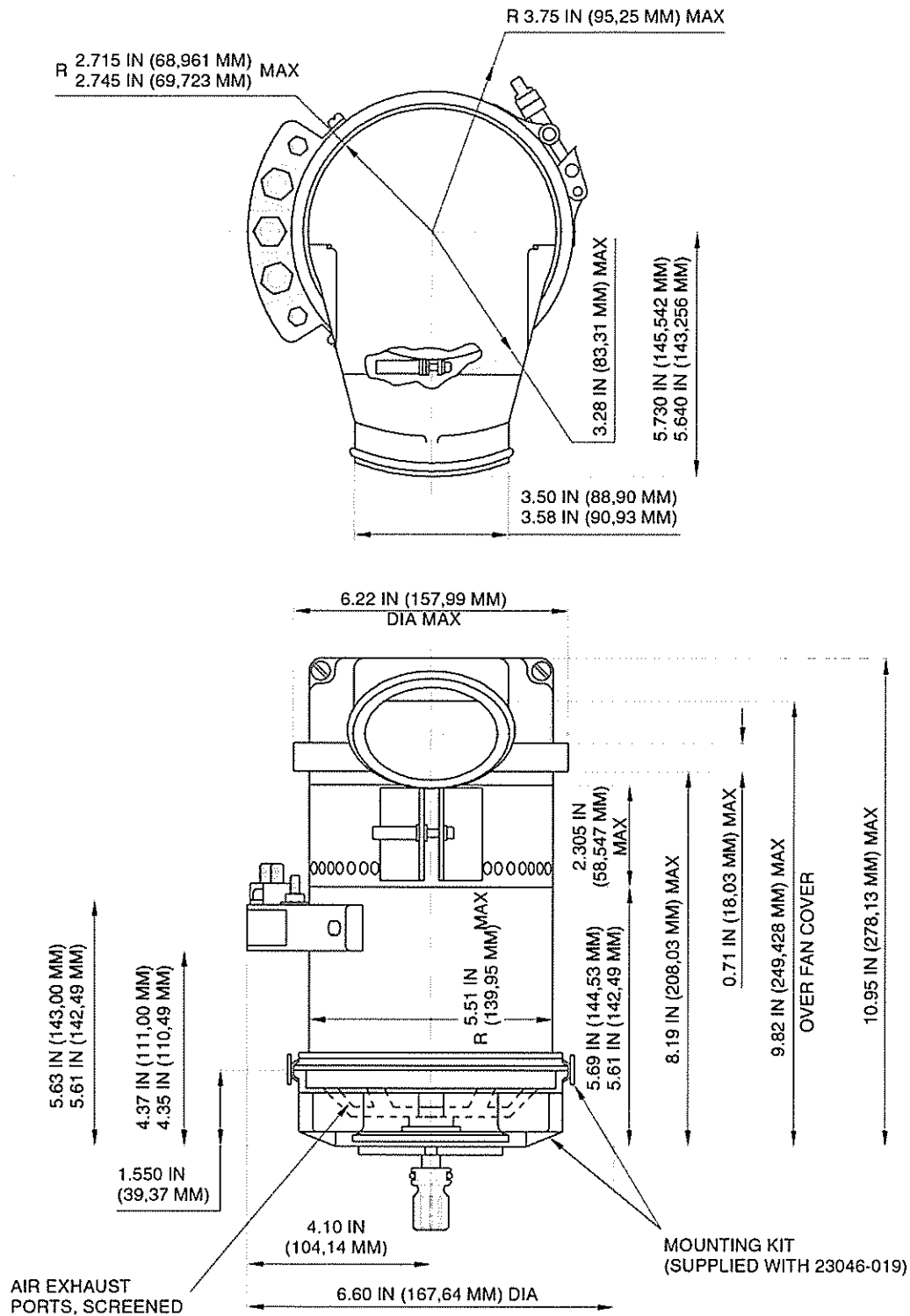


Figure 6 - Outline Drawing of Models 23046-019 and -020

DESCRIPTION AND OPERATION



Component Maintenance Manual with Illustrated Parts List

DC Starter-Generator, 23046 Series I

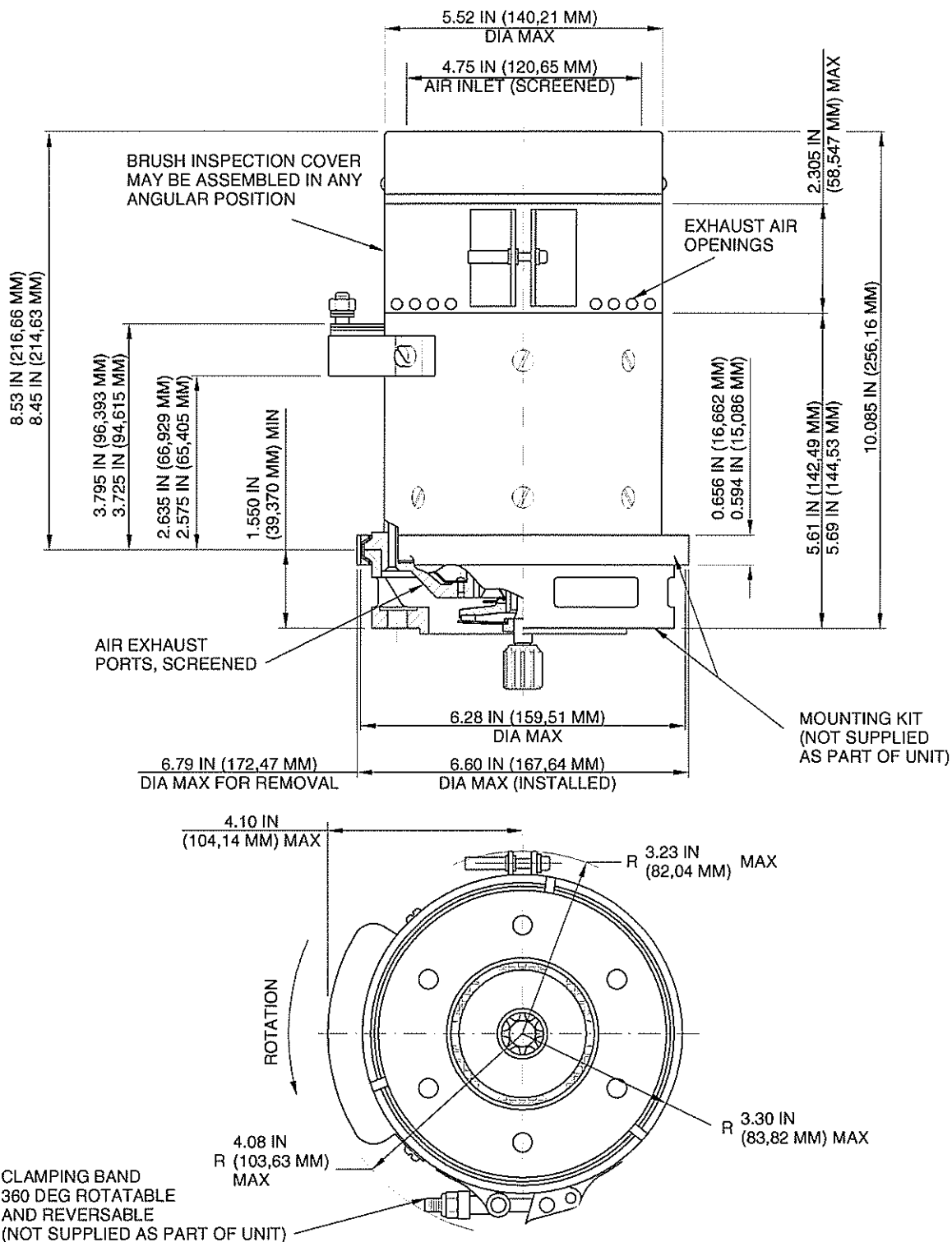


Figure 7 - Outline Drawing of Model 23046-028

DESCRIPTION AND OPERATION



Component Maintenance Manual with Illustrated Parts List
DC Starter-Generator, 23046 Series I

THIS PAGE INTENTIONALLY LEFT BLANK



Component Maintenance Manual with Illustrated Parts List
DC Starter-Generator, 23046 Series I

TESTING AND FAULT ISOLATION

1. Introduction

WARNING: THE DC STARTER-GENERATOR CAN PRODUCE HIGH OUTPUT CURRENT CAPABLE OF CAUSING SEVERE SHOCK OR DEATH. MAKE SURE ALL POWER IS SHUT OFF TO GENERATOR BEFORE REMOVING OR REPLACING TEST EQUIPMENT, INSTRUMENTS, OR ASSEMBLIES. TAKE EXTREME CARE WHEN PERFORMING "LIVE CIRCUIT" TESTS AND FAULT ISOLATION PROCEDURES.

The procedures provided in this section are performance tests and are classified as either verification tests or acceptance tests. A verification test is conducted to assist in fault isolation or to confirm the cause for removal before the repair or overhaul of the DC Starter-Generator. An acceptance test is conducted after the repair or overhaul of the unit. Record all test results on a copy of the data sheet(s) provided at the end of this section.

Verification Testing: A starter-generator requiring confirmation of the cause for removal or qualification for continued service must be inspected as detailed in Paragraph 3. of the CHECK section before testing can begin. A unit that passes initial inspections can be tested in accordance with this section to determine performance or to isolate a particular fault. When a fault is identified during testing, refer to the fault isolation tables in this section to determine the probable cause.

Acceptance Testing: A unit that has been overhauled or repaired and is ready to be returned to service must be tested in accordance with this section to verify minimum performance standards. Only when an acceptance test is fully completed, and in the order presented in this section, can zero operating hours time since overhaul (TSO) be assigned to an overhauled unit.

Refer to Figure 1001 for proper electrical connections, and Figure 1002 and Figure 1003 for typical test set-ups.

2. Test Conditions

Parameter	Operating Condition
Ambient Temperature	50° to 104° F (10° to 40° C)
Barometric Pressure	27 to 33 inch (914 to 1118 millibar) of mercury
Mounting	Drive shaft (rotational axis) horizontal
Cooling	Self-cooling

Table 1001 - Performance Test Conditions

Component Maintenance Manual with Illustrated Parts List
DC Starter-Generator, 23046 Series I

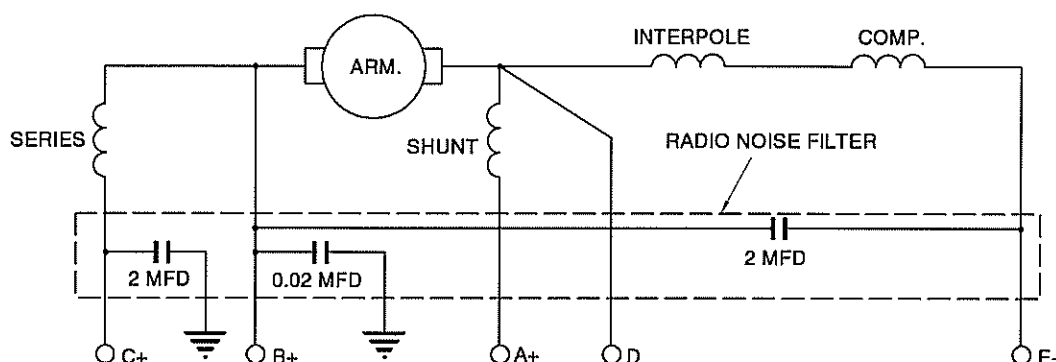


Figure 1001 - Electrical Connections

WARNING: DURING GENERATOR OPERATION, THE STARTER-GENERATOR CAN PRODUCE A HIGH OUTPUT CURRENT THAT CAN CAUSE SEVERE SHOCK OR DEATH. MAKE SURE ALL POWER IS SHUT OFF TO THE STARTER-GENERATOR BEFORE REMOVING OR REPLACING TEST EQUIPMENT, INSTRUMENTS, OR ASSEMBLIES. TAKE EXTREME CARE WHEN PERFORMING “LIVE CIRCUIT” TESTS AND FAULT ISOLATION PROCEDURES.

3. Required Test Equipment

Use instruments with accuracy and calibration in compliance with ISO-10012-1.

Equipment Type (Quantity)	Range and Accuracy or Equipment Rating	Generator Test Setup Reference
Commutation Viewing Adapter	-	Figure 9003
Drive Test Stand	Capable of driving starter-generator 6500 to 14,000 rpm	Not Illustrated
High Potential Tester	250 VAC rms, 50 to 60 Hz	Not Illustrated
Ohmmeter	-	Not Illustrated
Oscilloscope, Dual Tone	Digital: Bandwidth of 100 MHz, Sampling Rate of 100 ms/second Analog: Bandwidth of 60 to 100 MHz	Not Illustrated

Table 1002 - Test Equipment Requirements and Specifications

TESTING AND FAULT ISOLATION



Component Maintenance Manual with Illustrated Parts List
DC Starter-Generator, 23046 Series I

Equipment Type (Quantity)	Range and Accuracy or Equipment Rating	Generator Test Setup Reference
Power Supply, DC, Variable	0 to 30 VDC	Not Illustrated
QAD Mounting Kit (See IPL)	-	Not Illustrated
Starter Test Stand	Capable of measuring a locked rotor torque of 20 lbf.ft (27,1 N•m) and having a power supply capacity of 800 ADC at 22 V.	Not Illustrated
Thermometers or equivalent temperature measuring devices	Range: 50 to 300° F (10 to 150° C) Accuracy: ± 1° F (± 0.5° C)	Not Illustrated
Ammeter, DC	0 to 800 A	A ₁
Ammeter, DC	0 to 15 A	A ₂
Load Bank, Variable	30 V, 0 to 200 A	Variable Load Bank
Regulator, Voltage	Range: 0 to 36 VDC	Voltage Regulator
Resilient Cradle (with foam rubber cushion and restraining strap)	-	Not illustrated
Shunt, Precision	800 A, 500 mV DC	SH ₁
Strobe Light	-	Not Illustrated
Vibration Meter	Metrix vibration meter with probe magnet or equivalent	Not Illustrated
Voltmeter, DC	Range: 0 to 30 VDC Accuracy: ± 1% of reading	V ₁
Voltmeter, DC	Range: 0 to 3 VDC Accuracy: ± 1% of reading	V ₂

Table 1002 - Test Equipment Requirements and Specifications

4. Generator Thermal Stabilization

Stabilization is reached when either of the following conditions is met:

- Temperature of shunt field winding (A-E), as determined by its resistance, rises no more than 2° F (1,1° C) in five minutes.
- Frame temperature measured on side opposite terminal block, does not rise more than 2° F (1,1° C) in five minutes.



Component Maintenance Manual with Illustrated Parts List
DC Starter-Generator, 23046 Series I

5. Test Condition Setpoint Tolerances

Test condition setpoint must be set within tolerances indicated in Table 1003, exclusive of measurement accuracy.

Parameter	Test Condition Setpoint Tolerance
Speed (rpm)	± 20 rpm
DC Voltage	± 0.1 VDC
DC Current	± 2.5 A
Cooling Air Pressure	± 0.2 inch (5,1 mm) - water
Torque	± 1.0 lb-ft

Table 1003 - Setpoint Tolerances

6. Test Preparation

NOTE: Unless otherwise specified, numbers in parentheses () refer to item numbers given in Figure 10001 of the ILLUSTRATED PARTS LIST.

A. Perform an initial inspection

Perform an initial inspection of the starter-generator in a brightly lit work area to determine its overall condition. Parameters for this initial inspection are given in the CHECK section.

B. Cleaning and Brush Seating

CAUTION: THE STARTER-GENERATOR MUST BE SUPPORTED AT ALL TIMES DURING INSTALLATION OR REMOVAL. DO NOT ALLOW UNIT TO HANG UNSUPPORTED.

- (1) Reference CLEANING section.
- (2) Perform brush seating procedures per SPD 1006. Brushes must be seated 100% in the direction of rotation and at least 90% in the axial direction, before test measurements are made.

C. Install commutation viewing adapter.

WARNING: DURING OPERATION, THE DC STARTER-GENERATOR CAN PRODUCE A HIGH OUTPUT CURRENT CAPABLE OF CAUSING SEVERE SHOCK OR DEATH. MAKE SURE ALL POWER IS OFF TO STARTER-GENERATOR BEFORE REMOVING OR REPLACING TEST EQUIPMENT, INSTRUMENTS, OR ASSEMBLIES. TAKE EXTREME CARE WHEN PERFORMING "LIVE CIRCUIT" TESTS AND FAULT ISOLATION PROCEDURES.

CAUTION: STARTER-GENERATOR MUST BE SUPPORTED AT ALL TIMES DURING INSTALLATION OR REMOVAL. DO NOT ALLOW UNIT TO HANG UNSUPPORTED. TOO MUCH BENDING LOAD ON DRIVE SHAFT CAN DAMAGE SHEAR SECTION.

- (1) Remove air inlet.

TESTING AND FAULT ISOLATION



Component Maintenance Manual with Illustrated Parts List
DC Starter-Generator, 23046 Series I

- (2) Remove brush access cover (130).
- (3) Install commutation viewing adapter.

D. Attach dummy terminal block to starter-generator or remove terminal block grounding lead.

CAUTION: THE STARTER-GENERATOR MUST BE SUPPORTED AT ALL TIMES DURING INSTALLATION OR REMOVAL. DO NOT ALLOW UNIT TO HANG UNSUPPORTED. EXCESSIVE BENDING LOADS ON THE DRIVE SHAFT CAN DAMAGE THE SHEAR SECTION.

CAUTION: MAKE SURE THAT YOU USE A DUMMY TERMINAL BLOCK OR REMOVE THE TERMINAL BLOCK GROUNDING LEAD FOR ACCEPTANCE TESTING. THE TEST VOLTAGE CAN DAMAGE THE FILTER CAPACITOR IN THE TERMINAL BLOCK.

- (1) Refer to DISASSEMBLY section and ASSEMBLY section for details.

NOTE: If the grounding lead is removed, make sure to cover the lead with electrical tape before doing a dielectric test.

E. Install starter-generator onto drive stand

- (1) Install appropriate QAD kit end bell drive end pad onto drive stand. Refer to ILLUSTRATED PARTS LIST.

CAUTION: THE STARTER-GENERATOR MUST BE SUPPORTED AT ALL TIMES DURING INSTALLATION OR REMOVAL. DO NOT ALLOW UNIT TO HANG UNSUPPORTED. TOO MUCH BENDING LOADS ON DRIVE SHAFT CAN DAMAGE SHEAR SECTION.

- (2) While supporting anti-drive end of starter-generator, align and install drive end to end bell drive end pad.
- (3) Make sure that drive stand and starter-generator mating splines are correctly engaged.
- (4) Install rim-clenching clamp and tighten self-locking hex nut to the value stamped on the clamp.
- (5) Turn OFF all power to drive stand.
- (6) Connect starter-generator to test circuit as shown in Figure 1002.

Component Maintenance Manual with Illustrated Parts List
DC Starter-Generator, 23046 Series I

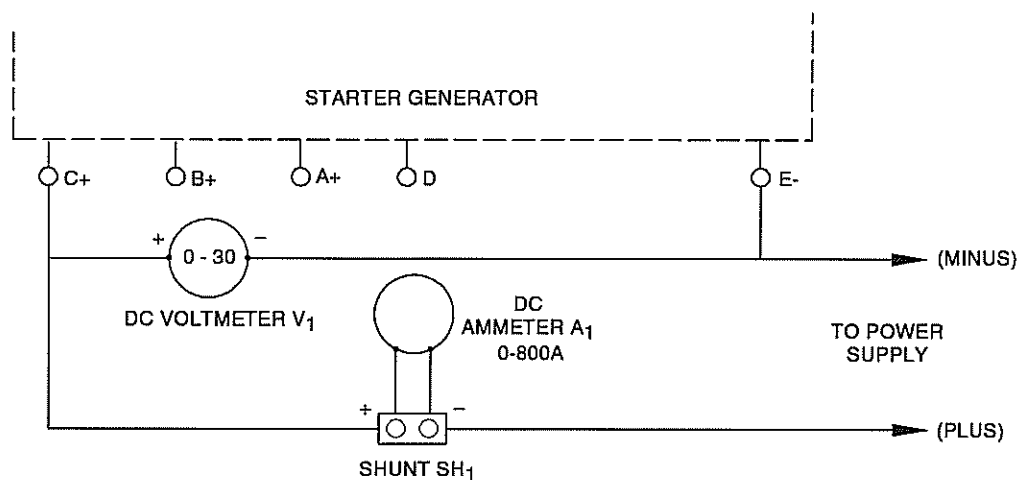


Figure 1002 - Test Set-Up for Starter Test

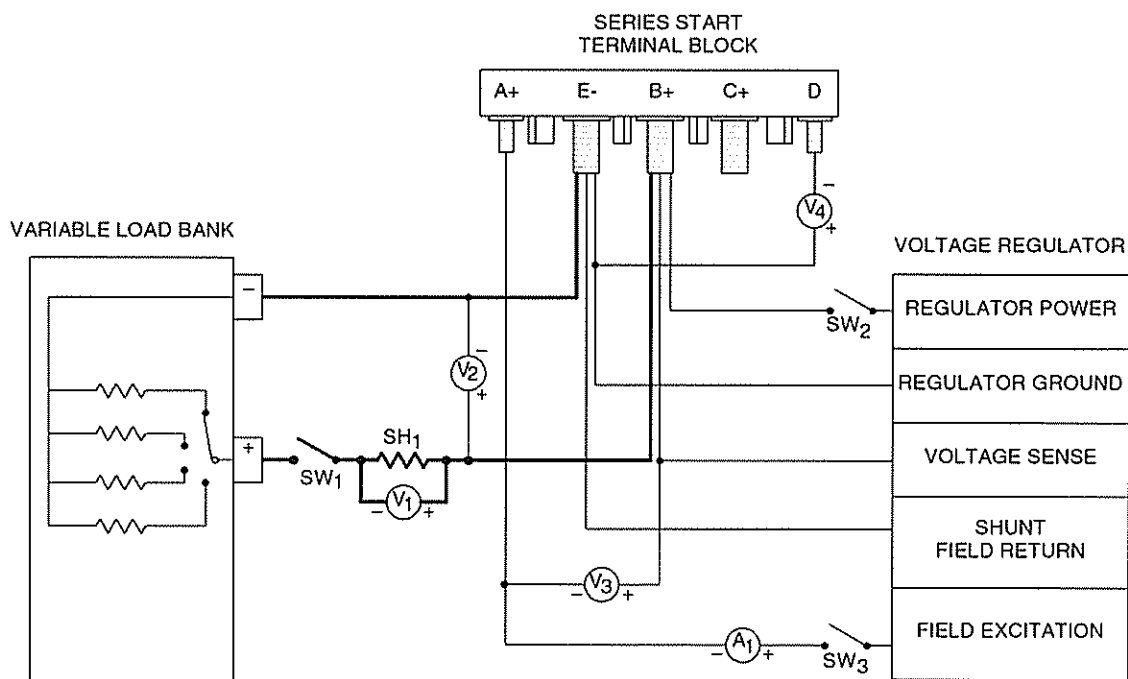


Figure 1003 - Test Set-Up for Generator Test



Component Maintenance Manual with Illustrated Parts List
DC Starter-Generator, 23046 Series I

7. Acceptance Tests

A. Maximum Speed for Regulation

(1) Procedure

Without operational warm-up and regulator connected to the Starter-Generator, operate Starter-Generator self-excited at 12,000 rpm for Models 23046-019 and -020, 13,000 rpm for all other models, 30 VDC, and no-load. Record field current, commutation, and frame temperature.

(2) Acceptance Limits

- (a) Shunt field current shall not be less than 0.81 A.
- (b) Commutation must be black.

B. Continuous Operating Speed and Equalizing Voltage

(1) Operate Starter-Generator at 12,000 rpm, 30 VDC and 200 A until stabilized.

NOTE: Stabilization is defined in Paragraph 4.

(2) Measure and record the voltage between terminals D and E, air inlet temperature and winding or frame temperature.

(3) Acceptance Limits.

- (a) The voltage between terminals D and E must be within the limits shown in Figure 1004, Figure 1005 or Figure 1006. Refer to Table 1004 for figure applicability.
- (b) For models 23046-001, -007, -007M, -009, -017 the winding or frame temperature must not be more than 175° F (97° C) higher than the air inlet temperature.



Component Maintenance Manual with Illustrated Parts List
DC Starter-Generator, 23046 Series I

Model	D to E Voltage Range
23046-001, 23046-007, 23046-007M, 23046-009, 23046-017	Figure 1004
23046-019, 23046-020	Figure 1005
23046-028	Figure 1006

Table 1004 - Equalizing Voltage Graphs

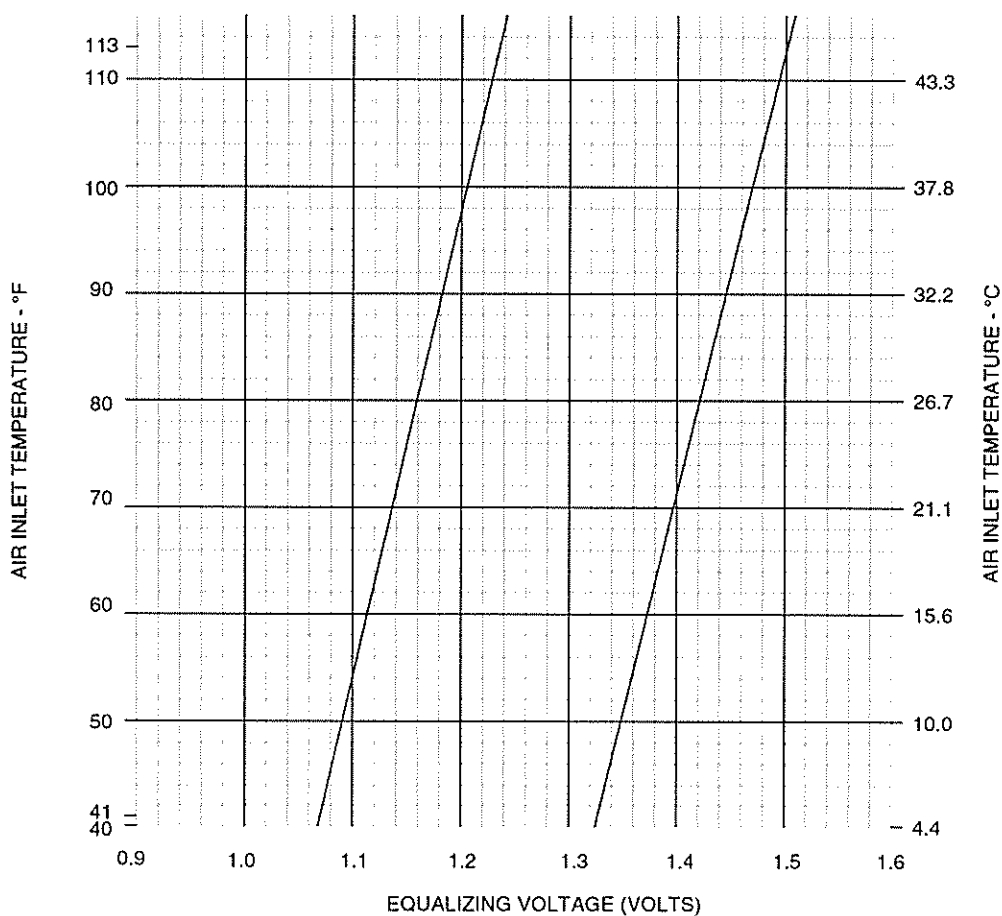


Figure 1004 - Equalizing Voltage Acceptance Limits, 23046-001, -007, -007M, -009 and -017



Component Maintenance Manual with Illustrated Parts List
DC Starter-Generator, 23046 Series I

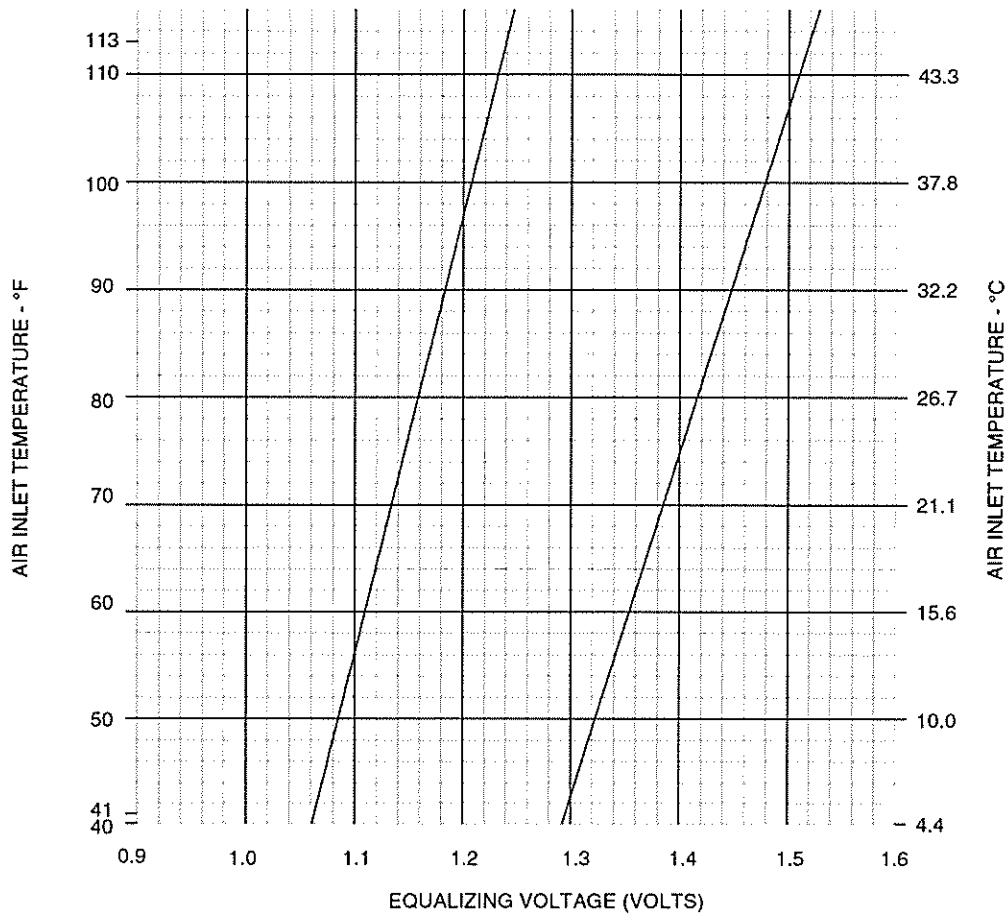


Figure 1005 - Equalizing Voltage Acceptance Limits, 23046-019 and -020



Component Maintenance Manual with Illustrated Parts List
DC Starter-Generator, 23046 Series I

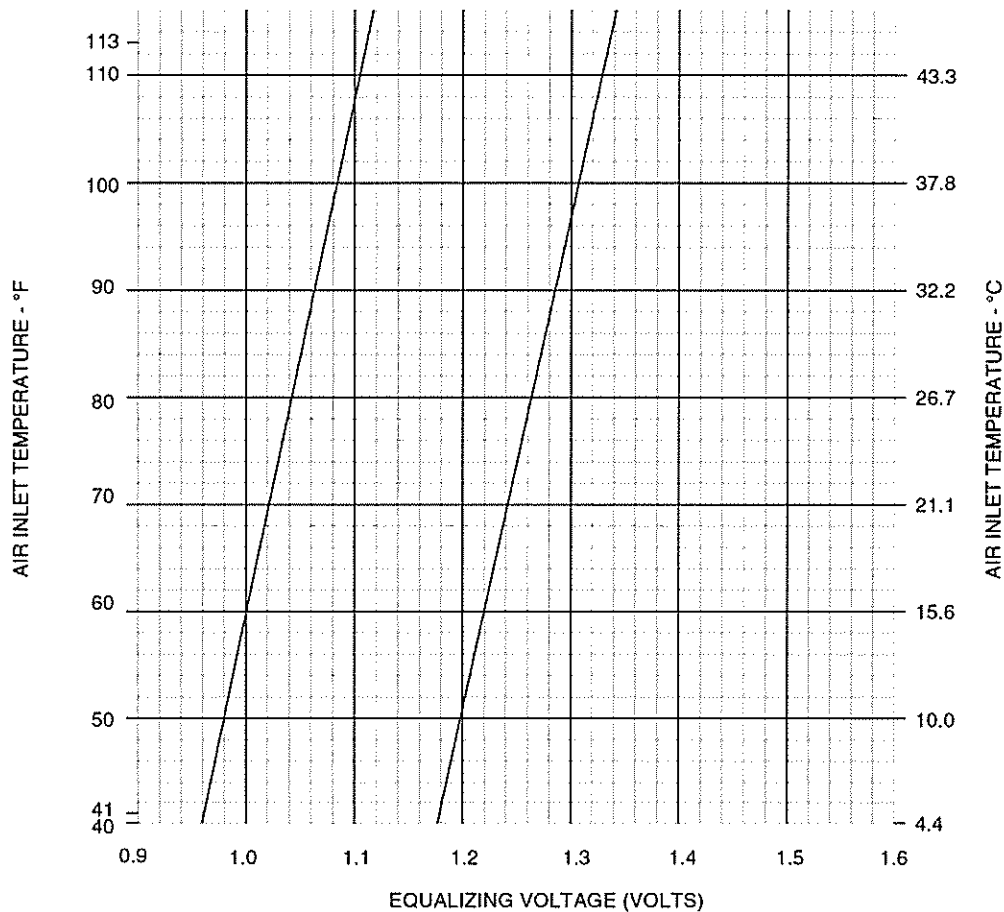


Figure 1006 - Equalizing Voltage Acceptance Limits, 23046-028

- (4) For models 23046-001, -007, -007M, -009, -017, reduce the load to 100 A and check the commutation.
- (5) Acceptance Limits
 - (a) Commutation must not exceed pinpoints along the edges of the brushes.



Component Maintenance Manual with Illustrated Parts List
DC Starter-Generator, 23046 Series I

C. Minimum Speed for Regulation

(1) Procedure

Operate as a generator at 7,800 rpm for Models 23046-019 and -020, 7,900 rpm for all other models, 30 VDC, with load of 200 A applied. Check external resistance and shunt field current.

(2) Acceptance Limits

(a) Shunt field current shall not exceed 8.0 A.

(b) External resistance (A to B volts divided by field current) shall not be less than 1.25 Ω .

(c) For models 23046-001, -007, -007M, -009, -017, commutation must not exceed pinpoints.

D. Minimum Speed

(1) Procedure

Operate as a generator at 7,000 rpm for Models 23046-019 and -020, 7,200 rpm for all other models, 26 VDC, with load of 200 A applied. Check external resistance and shunt field current.

(2) Acceptance Limits

(a) Shunt field current shall not exceed 8.0 A.

(b) External resistance shall not be less than 1.25 Ω .

E. Compounding

(1) Procedure

Operate at 12,000 rpm, 30 VDC. Check for compounding at 0, 50, 100, 150 and 200 A. Record shunt field current.

(2) Acceptance Limits

Shunt field current shall increase with load.

F. Overspeed

(1) Procedure

Operate starter-generator at no-load for five minutes at 14,000 rpm with field circuit open.

(2) Acceptance Limits

No mechanical failure shall occur as a result of this test.

Component Maintenance Manual with Illustrated Parts List
DC Starter-Generator, 23046 Series I

G. Commutation

(1) Procedure

- (a) Operate starter-generator at 12,000 rpm, 30 VDC, with load of 200 A.
- (b) Visually examine the condition of commutation.

NOTE: The proper angle for viewing commutation is approximately 30 to 45 degrees from the brush box as shown in Figure 1007

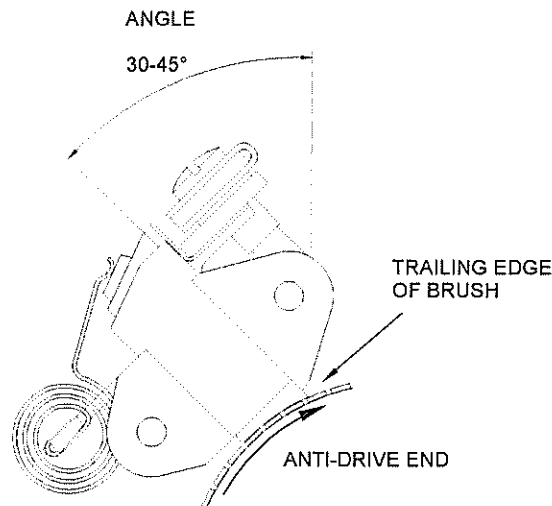


Figure 1007 - Correct Viewing Angle

(2) Acceptance Limits

- (a) Acceptable (pin point) commutation must be interpreted to permit sparking that extends approximately 0.12 inch. (3,0 mm) beyond the edge of the brush and occasional (no more than 10 per minute) single sparks that extend up to 0.25 inch (6,4 mm) beyond the edge of the brush.
- (b) Unacceptable commutation is considered to be continuous sparking or 'arcing' that extends 0.25 inch. (6,4 mm) beyond the edge of the brush.



Component Maintenance Manual with Illustrated Parts List
DC Starter-Generator, 23046 Series I

H. Starter-Generator Dielectric Test

WARNING: BEFORE USING HIGH VOLTAGE TESTER, MAKE CERTAIN THAT TESTER HAS BEEN TURNED "OFF" AND THAT NO ONE IS TOUCHING ANY PORTION OF CONNECTION POINTS OR PROBE LEADWIRES. FAILURE TO COMPLY WITH THIS WARNING CAN RESULT IN SEVERE ELECTRICAL SHOCK TO THE OPERATOR OR ASSISTANTS.

CAUTION: DO NOT PERFORM DIELECTRIC TESTING DURING MAINTENANCE INSPECTION OR TROUBLE SHOOTING OF THE UNIT AS IT MAY CAUSE CUMULATIVE DEGRADATION OF THE ELECTRICAL CIRCUIT INSULATION. THIS TEST IS RECOMMENDED ONLY FOR ACCEPTANCE TESTING OF AN OVERHAULED UNIT THAT HAS BEEN THOROUGHLY CLEANED.

CAUTION: MAKE SURE THAT YOU USE A DUMMY TERMINAL BLOCK OR REMOVE THE TERMINAL BLOCK GROUNDING LEAD FOR ACCEPTANCE TESTING. THE TEST VOLTAGE CAN DAMAGE THE FILTER CAPACITOR IN THE TERMINAL BLOCK.

(1) Operating Instructions

- (a) Attach a dummy terminal block, refer to DISASSEMBLY section and ASSEMBLY section for details.

NOTE: Alternatively, remove the screw that secures the terminal block to the housing that is adjacent to terminal D. Cover the terminal lug attached to the terminal block with electrical tape before doing a dielectric test.

- (b) Attach positive (red) lead of high pot tester to terminal B.
- (c) Attach negative (black) lead to high pot tester to an unfinished surface of the starter-generator frame.

WARNING: FAILURE TO USE NECESSARY SAFETY PRECAUTIONS WHEN HANDLING HIGH VOLTAGE ELECTRICAL LEADS DURING HIGH POTENTIAL TESTING CAN CAUSE SERIOUS INJURY OR DEATH.

CAUTION: INCREASE OR DECREASE THE TEST VOLTAGES SLOWLY (100 V PER SECOND, MAXIMUM). INCREASING OR DECREASING THE VOLTAGE TOO QUICKLY CAN CAUSE SERIOUS DAMAGE TO THE STARTER-GENERATOR.

- (d) Apply dielectric test voltage of 250 V RMS for one minute, or 300 V RMS for one second (commercial frequency) between the terminal and machine frame.
- (e) Slowly decrease voltage to zero.
- (f) Turn the power supply OFF.

TESTING AND FAULT ISOLATION



Component Maintenance Manual with Illustrated Parts List
DC Starter-Generator, 23046 Series I

(g) Disconnect the electrical test leads from the test unit.

(2) Acceptance Limits

Any arcing as evidenced by flashover (surface discharge), sparkover (air discharge), breakdown (puncture discharge) or leakage current exceeding 5mA shall be evidence of damp, dirty, weak or defective components

I. No Load Speed Test

(1) Procedure

(a) Operate as a starter with shunt field circuit open (terminals A and B) for all models except 23046-028, at no load with voltage applied between terminals C and E.

For Models other than 23046-019 and -020, connect 1.25 Ω between terminals A and B.

(b) Increase voltage until speed reaches 4,300 rpm for Models -019 and -020.

(c) Record voltage for Models -019 and -020.

(d) For all other models, apply 24 V from C to E.

(e) Record speed for all other models.

(2) Acceptance Limits

Voltage shall be less than 28 V for Models 23046-019 and -020. Speed shall be greater than 3,500 rpm for all other models.

J. Locked Rotor Test

CAUTION: DAMAGE TO THE EQUIPMENT MAY RESULT IF THE SHUNT FIELD WINDING IS NOT OPENED BEFORE ACCOMPLISHING THIS TEST. DO NOT ENERGIZE THE UNIT FOR MORE THAN 4 SECONDS.

(1) Procedure

(a) For all models except 23046-019 and -020, connect 1.25 Ω between terminals A and B.

(b) With shunt field open and the rotor locked, apply a variable DC voltage between terminals C(+) and E(-). Increase voltage until output torque measures 15 lbf.ft (20,3 N•m) for Models 23046-019 and -020, and 13 lbf.ft (17,6 N•m) for all other models. Record current and voltage.

(2) Acceptance Limits

(a) Current shall not exceed 800 A for Models 23046-019 and -020, and 500 A for all other models.



Component Maintenance Manual with Illustrated Parts List
DC Starter-Generator, 23046 Series I

- (b) Voltage shall not exceed 16 V for Models 23046-019 and -020, and 13 V for all other models.

K. Radial Vibration Test (for Models 23046-019 and -020 only)

CAUTION: MAKE SURE STARTER-GENERATOR IS RESTRAINED DURING INITIAL START UP.

- (1) Mount starter-generator in a resilient cradle (foam rubber cushioned) with the drive shaft supported (centred).
- (2) Mount the probe of the vibration measuring tool to the generator housing.
- (3) With no load, gradually increase starter voltage between terminals C and E until voltage reaches value in the table. The speed can be measured by pointing the strobe light at the fan.

Model	VDC	RPM
23046-019, 23046-020	30	12,000

- (4) Measure and record total frame vibration amplitude.
- (5) Acceptance Limits
 - (a) Total vibration amplitude shall not exceed 0.001 inch (0,025 mm).

L. Commutator Run-Out Check

- (1) Procedure

Rotate the armature on its own bearings. Record total indicated run-out and maximum bar-to-bar difference.
- (2) Acceptance Limits
 - (a) Total indicated run-out shall not exceed 0.0008 inch (0,020 mm)
 - (b) Bar-to-bar difference shall not exceed 0.0002 inch (0,005 mm)

8. Final Assembly After Acceptance Testing

After starter-generator has successfully completed and passed acceptance testing, refer to ASSEMBLY section for final assembly instructions.



Component Maintenance Manual with Illustrated Parts List
DC Starter-Generator, 23046 Series I

9. Fault Isolation Tables

The following tables list faults that may occur during acceptance testing, performance testing, or when in service. For each fault, probable causes are listed along with related corrective actions. When a fault is detected, perform the corrective actions necessary to return the starter-generator to a serviceable condition. A complete acceptance test is required after the corrective action is accomplished.

FAULT	PROBABLE CAUSE	CORRECTIVE ACTION
A. Maximum Speed for Regulation Test		
Shunt Field Current Low	Low voltage power source.	Check power source to make certain that full voltage is being applied to starter-generator terminals.
	Defective switch in power supply line.	Replace switch.
	Damaged armature.	Replace armature.
	Short circuited or open stator windings.	Replace stator and housing assembly.
	Improperly seated brushes.	Check brush seats and accomplish brush seating. See REPAIR section.
Excessive sparking at brushes	Short circuited or grounded field windings.	Replace stator and housing assembly.
	Excessive clearance in bearings or rough bearing races.	Replace bearings.
	Damaged or eccentric commutator.	Refinish commutator. See REPAIR section.
	Armature out of balance.	Balance or replace armature. See REPAIR section.
	Short circuit in armature windings.	Replace armature.
	Brushes incorrectly installed with top bevel reversed.	Reverse and reseal brushes. See SPD 1006.
	Severe overload or short circuit in electrical system.	Locate and correct fault.
	Brushes sticking in holders.	Free-up brushes. See SPD 1006.
	Weak brush springs.	Replace brush springs.

Table 1005 - Fault Isolation Table

TESTING AND FAULT ISOLATION



Component Maintenance Manual with Illustrated Parts List
DC Starter-Generator, 23046 Series I

FAULT	PROBABLE CAUSE	CORRECTIVE ACTION
B. Continuous Operating Speed and Equalizing Voltage		
D to E voltage exceeds limits or Winding or frame maximum temperature exceeded	Brushes are incorrectly seated or installed.	Refer to SPD 1006 for brush installation, seating, and run-in procedures.
	One or more brush leads are loose.	Remove brush access cover. Inspect all brush leads for open circuit. If open circuit exists, overhaul or repair as necessary
	Armature is shorted or grounded.	Disassemble starter-generator. Clean Armature. Dielectric test according to procedure in CHECK section. Replace armature if shorted or grounded.
	Stator windings shorted or grounded.	Disassemble starter-generator. Clean stator and housing assembly. Dielectric test stator and housing assembly according to procedure in CHECK section. Replace stator and housing assembly if shorted or grounded.

Table 1005 - Fault Isolation Table



Component Maintenance Manual with Illustrated Parts List
DC Starter-Generator, 23046 Series I

FAULT	PROBABLE CAUSE	CORRECTIVE ACTION
Excessive sparking at brushes	Short circuited or grounded field windings.	Replace stator and housing assembly.
	Excessive clearance in bearings or rough bearing races.	Replace bearings.
	Damaged or eccentric commutator.	Refinish commutator. See REPAIR section.
	Armature out of balance.	Balance or replace armature. See REPAIR section.
	Short circuit in armature windings.	Replace armature.
	Brushes incorrectly installed with top bevel reversed.	Reverse and reseal brushes. See SPD 1006.
	Severe overload or short circuit in electrical system.	Locate and correct fault.
	Brushes sticking in holders	Free-up brushes. See SPD 1006
	Weak brush springs.	Replace brush springs.

Table 1005 - Fault Isolation Table



Component Maintenance Manual with Illustrated Parts List
DC Starter-Generator, 23046 Series I

FAULT	PROBABLE CAUSE	CORRECTIVE ACTION
C. Minimum Speed for Regulation Test		
Field Current Limit is exceeded	An excessive load was applied during test.	Check and adjust applied load at load bank as necessary.
Low external field resistance	Brushes are not fully seated.	Refer to SPD 1006 for brush installation, seating and run-in procedures.
	Brushes hung up in brush holder assemblies.	DISASSEMBLE brush access cover from starter-generator. Inspect brush holder assemblies and brush leads for proper positioning. CLEAN and REPAIR starter-generator as necessary. Perform brush run-in procedure. Refer to SPD 1006 for brush installation, seating and run-in procedures.
	Surface of commutator incorrectly filmed or irregular.	DISASSEMBLE starter-generator. CHECK armature. REPAIR commutator if necessary. Perform brush run-in procedure. Refer to SPD 1006 for brush installation, seating and run-in procedures.
	Armature is either shorted or grounded.	DISASSEMBLE starter-generator. CLEAN armature. Dielectric test armature according to procedure in CHECK section. Replace armature if shorted or grounded.
	Stator windings shorted or grounded.	DISASSEMBLE starter-generator. CLEAN stator and housing assembly. Dielectric test stator and housing assembly according to procedure in CHECK section. Replace stator and housing assembly if shorted or grounded.

Table 1005 - Fault Isolation Table

TESTING AND FAULT ISOLATION



Component Maintenance Manual with Illustrated Parts List
DC Starter-Generator, 23046 Series I

FAULT	PROBABLE CAUSE	CORRECTIVE ACTION
Excessive sparking at brushes	Short circuited or grounded field windings.	Replace stator and housing assembly.
	Excessive clearance in bearings or rough bearing races.	Replace bearings.
	Damaged or eccentric commutator.	Refinish commutator. See REPAIR section.
	Armature out of balance.	Balance or replace armature. See REPAIR section.
	Short circuit in armature windings.	Replace armature.
	Brushes incorrectly installed with top bevel reversed.	Reverse and reseal brushes. See SPD 1006.
	Severe overload or short circuit in electrical system.	Locate and correct fault.
	Brushes sticking in holders.	Free-up brushes. See SPD 1006.
	Weak brush springs.	Replace brush springs.

Table 1005 - Fault Isolation Table



Component Maintenance Manual with Illustrated Parts List
DC Starter-Generator, 23046 Series I

FAULT	PROBABLE CAUSE	CORRECTIVE ACTION
D. Minimum Speed Test.		
Field Current Limit is exceeded or	An excessive load was applied during test.	Check and adjust applied load at load bank as necessary.
Low external field resistance	Brushes are not fully seated.	Refer to SPD 1006 for brush installation, seating and run-in procedures.
	Brushes hung up in brush holder assemblies.	DISASSEMBLE brush access cover from starter-generator. Inspect brush holder assemblies and brush leads for proper positioning. CLEAN and REPAIR starter-generator as necessary. Perform brush run-in procedure. Refer to SPD 1006 for brush installation, seating and run-in procedures.
	Surface of commutator incorrectly filmed or irregular.	DISASSEMBLE starter-generator. CHECK armature. REPAIR commutator if necessary. Perform brush run-in procedure. Refer to SPD 1006 for brush installation, seating and run-in procedures.
	Armature is either shorted or grounded.	DISASSEMBLE starter-generator. CLEAN armature. Dielectric test armature according to procedure in CHECK section. Replace armature if shorted or grounded.
	Stator windings shorted or grounded.	DISASSEMBLE starter-generator. CLEAN stator and housing assembly. Dielectric test stator and housing assembly according to procedure in CHECK section. Replace stator and housing assembly if shorted or grounded.

Table 1005 - Fault Isolation Table

TESTING AND FAULT ISOLATION



Component Maintenance Manual with Illustrated Parts List
DC Starter-Generator, 23046 Series I

FAULT	PROBABLE CAUSE	CORRECTIVE ACTION
E. Compounding Test		
Shunt field current does not increase with load.	Stator compensating windings bad.	Disassemble starter-generator. Replace stator and housing assembly.
F. Overspeed Test		
Noisy operation during overspeed test.	Cooling fan damaged.	<u>NOTE:</u> There are no authorized repair procedures to balance cooling fans. CHECK fan blades and air inlet/cover for damage. Replace cooling fan if damaged. REPAIR or replace air inlet/cover if damaged.
	Bearings are either defective and/or incorrectly installed.	Put starter-generator on vertical stator support. Remove brushes from brush holder assembly and inspect bearings. Disassemble starter-generator and replace bearings as necessary.
	Armature is striking against stator and housing assembly.	Disassemble starter-generator. CHECK armature, stator and housing for physical damage. Replace damaged parts as necessary.
Starter-generator vibrates.	Bearings are either defective and/or incorrectly installed.	Disassemble starter-generator. Replace bearings as necessary.
	Armature is out of balance.	Disassemble starter-generator. CHECK armature balance. Repair armature as necessary. Replace armature if damage is not repairable.

Table 1005 - Fault Isolation Table



Component Maintenance Manual with Illustrated Parts List
DC Starter-Generator, 23046 Series I

FAULT	PROBABLE CAUSE	CORRECTIVE ACTION
G. Commutation Test.		
Excessive sparking	Brushes are incorrectly seated or installed.	If commutation exceeds pinpoints, re-seat brushes. Refer to SPD 1006.
	One or more of brush holders are loose.	DISASSEMBLE bearing and brush support assembly from starter-generator. CHECK hardware that attaches brush holder assemblies to bearing and brush support assembly. If damage is found, REPAIR and ASSEMBLE bearing and brush support assembly. Refer to SPD 1004. Dielectric test bearing and brush support assembly according to procedure in CHECK section.
	Brush spring pressure is below limit.	DISASSEMBLE starter-generator. CHECK brush spring pressure. Refer to SPD 1006. Replace brush spring(s) that do not meet limits in the FITS AND CLEARANCES section.
	An armature winding is partially or completely open	DISASSEMBLE starter-generator. Replace armature.
	Surface of commutator is incorrectly filmed or irregular	DISASSEMBLE starter-generator. REPAIR commutator surface if possible. Replace armature if damage is not repairable.

Table 1005 - Fault Isolation Table



Component Maintenance Manual with Illustrated Parts List
DC Starter-Generator, 23046 Series I

FAULT	PROBABLE CAUSE	CORRECTIVE ACTION
	Stator and housing assembly is shorted or grounded.	DISASSEMBLE starter-generator. CLEAN stator and housing assembly Dielectric test stator and housing assembly according to procedure in CHECK section. Replace stator and housing assembly if stator and housing fail dielectric test.
	Armature is out of balance	DISASSEMBLE starter-generator. CHECK armature. REPAIR armature. Replace armature if correct armature balance cannot be accomplished.
H. Dielectric Test.		
Starter-generator insulation breakdown.	Bearing and brush support assembly is grounded.	Disassemble starter-generator. CLEAN anti-drive end end bell assembly. Dielectric test bearing and brush support assembly according to procedure in CHECK section. If assembly fails dielectric test, replace insulating sleeves and washers.
	Armature grounded.	Disassemble starter-generator. Clean armature. Dielectric test armature according to procedure in CHECK section. Replace armature if grounded.

Table 1005 - Fault Isolation Table



Component Maintenance Manual with Illustrated Parts List
DC Starter-Generator, 23046 Series I

FAULT	PROBABLE CAUSE	CORRECTIVE ACTION
	Stator and housing assembly grounded.	Disassemble starter-generator. Clean stator and housing assembly. Dielectric test stator and housing assembly according to procedure in CHECK section. Replace stator and housing assembly if grounded.
I. No-Load Speed Test		
Voltage exceeds required maximum or speed not greater than 3,500 rpm.	Stator and housing assembly shorted or grounded.	Disassemble starter-generator. Clean stator and housing assembly. Dielectric test stator and housing assembly according to procedure in CHECK section. Replace stator and housing assembly if shorted or grounded.
	Armature shorted or grounded.	Disassemble starter-generator. Clean armature. Dielectric test armature according to procedure in CHECK section. Replace armature if shorted or grounded.
J. Locked Rotor Test		
Current level exceeded or Voltage level exceeded.	Short circuited or open stator winding.	Replace stator and housing assembly.
	Open circuit in armature.	Replace armature.

Table 1005 - Fault Isolation Table



Component Maintenance Manual with Illustrated Parts List
DC Starter-Generator, 23046 Series I

FAULT	PROBABLE CAUSE	CORRECTIVE ACTION
K. Radial Vibration Test.(for Models 23046-019 and -020 only)		
Vibration amplitude limit exceeded.	Bearings are either defective and/or incorrectly installed.	Disassemble starter-generator. Replace bearings as necessary.
	Armature is out of balance.	Disassemble starter-generator. CHECK armature balance. Repair armature as necessary. Replace armature if damage is not repairable.
L. Commutator Run-Out Check		
Bar-to-bar runout or total indicator runout exceeded.	Commutator bars have shifted.	Disassemble starter-generator. Check commutator. Repair commutator surface as required. Replace armature if commutator is not repairable.

Table 1005 - Fault Isolation Table



Component Maintenance Manual with Illustrated Parts List
DC Starter-Generator, 23046 Series I

VERIFICATION / FINAL ACCEPTANCE DATA SHEET

Model Number: 23046- Inspected by: _____
Serial Number: _____ Date: _____
Modification Level: _____

Inspection/Test	Requirements/Limits	Measurements	Accept	Reject
Visual Inspection	Check for physical damage, missing parts, finishes, etc.			
Brush Seating	Per SPD 1006			
Maximum Speed for Regulation Test	Frame temperature	° F (° C)		
	Field current (0.81 A min.)	A		
	Commutation must be black			
Continuous Operating Speed and Equalizing Voltage	Air inlet temperature	° F (° C)		
	Frame temperature (stabilized)	° F (° C)		
	"D" to "E" volts (within limits of Figure 1004, 1005 or 1006)	VDC		
	Commutation must not exceed pinpoints			
Minimum Speed for Regulation	Field current (not to exceed 8.0 A)	A		
	"B" to "A" external field resistance (1.25 Ω minimum)	Ω		
	Commutation must not exceed pinpoints			
Minimum Speed	External field circuit ("B" to "A") resistance (1.25 Ω min.)	Ω		
	Shunt field current (8 A max.)	A		
Compounding	Shunt field current (must increase with load)	A		
Overspeed Test	No indication of failure			
Commutation Test	Not to exceed pinpoints			
Dielectric Test	No sign of insulation breakdown			
	Leakage current must not exceed 5mA	mA		
Starter No Load Test	Voltage must be less than 28 VDC for 23046-019 and -020	VDC		
	Speed must be greater than 3,500 rpm for all other models	rpm		
Locked Rotor Test for Models -019 and -020	"C" to "E" voltage (not to exceed 16 VDC)	VDC		
	Line current (800 A max.)	A		
Locked Rotor Test for all other Models	"C" to "E" voltage (not to exceed 13 VDC)	VDC		
	Line current (500 A max.)	A		
Radial Vibration Models -019 and -020 only	Total radial vibration (0.001 inch (0,025 mm) max.)	inch (mm)		
Commutator Runout	Total indicator reading (0.0008 inch (0,020 mm) max.)	inch (mm)		
	Bar to bar runout (0.0002 inch (0,005 mm) max.)	inch (mm)		

TESTING AND FAULT ISOLATION



Component Maintenance Manual with Illustrated Parts List
DC Starter-Generator, 23046 Series I

THIS PAGE INTENTIONALLY LEFT BLANK



Component Maintenance Manual with Illustrated Parts List
DC Starter-Generator, 23046 Series I

SCHEMATICS AND WIRING DIAGRAMS

1. **Introduction**

This section is unassigned.



Component Maintenance Manual with Illustrated Parts List
DC Starter-Generator, 23046 Series I

THIS PAGE INTENTIONALLY LEFT BLANK



Component Maintenance Manual with Illustrated Parts List
DC Starter-Generator, 23046 Series I

DISASSEMBLY

1. Introduction

This section provides disassembly procedures for the 23046 Series I DC Starter-Generator. No further disassembly beyond that given in this section is necessary unless further disassembly is indicated in the CHECK or TESTING AND FAULT ISOLATION sections.

During overhaul, all bearings, brushes, insulating parts, retaining rings, self locking nuts and lock washers must be discarded regardless of their condition.

Do not open or desolder permanent connections unless replacement of worn, damaged, or defective parts is necessary.

NOTE: All soldering must be performed according to ANSI/IPC J-STD-001.

NOTE: When a starter-generator is removed for service, the QAD kit usually stays with the aircraft.

2. Disassembly Tools

Other than normal shop tools, those listed in Table 3001 are needed for starter-generator disassembly.

Tools	Reference
Anti-drive End Bearing Hub Support	Figure 9015
Armature Shaft Adapter, Drive End	Figure 9001
Armature Shaft Adapter, Anti-drive End	Figure 9002
Bearing and Brush Support Assembly Support	Figure 9008
Bearing Puller	Commercially available
Dampener Hub Driver	Figure 9012
Drive End Hub Support	Figure 9009
Foam Cushion	Commercially available
Horizontal Stator Support	Figure 9010
Inner Race Bearing Driver	Figure 9004
Plastic or Leather Mallet	Commercially available
Pliers, External Snap Ring	Commercially available
Spline Wrench	Figure 9014
Vertical Stator Support	Figure 9011
Wire Hook Tool	Commercially available

Table 3001 - Disassembly Tools



Component Maintenance Manual with Illustrated Parts List
DC Starter-Generator, 23046 Series I

3. Disassembly Materials

Materials necessary for starter-generator disassembly are listed in Table 3002.

NOTE: Equivalent substitutes may be used for the materials listed in Table 3002. Disassembly materials are not available from Goodrich Corporation. All items can be purchased commercially.

WARNING: BEFORE USING ANY OF THE FOLLOWING MATERIALS, BE AWARE OF ALL HANDLING, STORAGE, AND DISPOSAL PRECAUTIONS RECOMMENDED BY THE MANUFACTURER OR SUPPLIER. FAILURE TO COMPLY WITH THE RECOMMENDATIONS MAY RESULT IN SERIOUS INJURY, PHYSICAL DISORDER, OR DEATH.

Item	Description/Specification	Source (CAGE Code)
Isopropyl Alcohol See <u>WARNING</u> before using this material.	TT-I-735, Grade A Flash Point: 53° F (12° C), FLAMMABLE Refer to the Material Safety Data (MSD) Sheet for the material for the additional safety information.	Commercially Available
Masking Tape	N/A	Commercially Available

Table 3002 - Disassembly Materials

4. Disassembly of Starter-Generator

NOTE: Unless otherwise specified, numbers in parentheses () refer to item numbers in Figure 10001 of the ILLUSTRATED PARTS LIST.

A. Remove QAD mounting kit (10) from starter-generator (if present).

NOTE: When a starter-generator is removed for service, the QAD kit normally stays on the aircraft.

- (1) Loosen self-locking hex nut (-30) from T-bolt (-40) and remove rim-clenching clamp (20).
- (2) Remove end bell, drive end pad (50) from starter-generator.

B. Remove identification plate (70), information plate (90), decal (110) and FAA-PMA label (-120) from stator and housing assembly (680) as applicable.

NOTE: Do not remove identification plate (70), information plate (90), decal (110), or FAA-PMA label (-120) unless damaged or illegible.

NOTE: Minor scratches and superficial damage that do not affect the legibility are not reason to replace the identification plate.

- (1) Remove identification plate (70).



Component Maintenance Manual with Illustrated Parts List
DC Starter-Generator, 23046 Series I

- (a) Use a smooth, flat tool that will not scratch the stator housing.
- (b) Wedge the tool between the edge of the identification plate and the housing near each of the four drive screws.
- (c) Pry up the corners of the plate to remove the plate from the housing.

NOTE: This will break the light metal at each corner of the identification plate from the stator and housing assembly.

NOTE: Discard the old identification plate after the unit information has been transferred to the new replacement identification plate (70).

- (d) Remove the remaining metal from under the head of each drive screw.
- (e) Use a small pair of sidecutters with the flat face to the housing to wedge in under the head of each drive screw. Hold the shank of the screw and turn out counterclockwise.

NOTE: Use a dull pair of small sidecutters so that you do not cut the head of the drive screw off.

- (f) Discard drive screws (80) attaching identification plate (70).

(2) Remove information plate (90)

- (a) Repeat the procedure given in Paragraph 4.B.(1) to remove the information plate (90) and two drive screws (100).

(3) Remove instruction decal (110) and FAA-PMA label (-120) from stator and housing assembly.

- (a) Peel off and discard the instruction decal and/or the FAA-PMA label from the stator and housing assembly.

WARNING: ISOPROPYL ALCOHOL IS DANGEROUS TO PERSONS. USE ONLY IN AN AREA WITH A GOOD FLOW OF AIR. KEEP AWAY FROM SOURCES OF HEAT, SPARKS AND FLAME. DO NOT BREATHE THE FUMES. PREVENT SKIN CONTACT. PUT ON PROTECTIVE CLOTHING AND EYE PROTECTION.

- (b) Use isopropyl alcohol to remove adhesive residue.
- (c) Wipe dry with a soft dry cloth.

C. Remove brush access cover (130) from starter-generator.

- (1) Remove or sufficiently loosen screw (140) that secures brush access cover to stator and housing assembly.
- (2) Remove brush access cover (130).



Component Maintenance Manual with Illustrated Parts List
DC Starter-Generator, 23046 Series I

D. Remove brushes (160) from bearing and brush support assembly (-480).

NOTE: Identification of brush holder assemblies and brush sets is not necessary during overhaul since all used brushes are discarded.

NOTE: On older models a lead from the stator windings is connected to the negative brush holder. The screw that secures the brush lead and stator harness also secures this field coil lead.

(1) Remove screw (170) that secures brush lead to brush holder assembly.

CAUTION: RAISE AND LOWER BRUSH SPRINGS SLOWLY. DO NOT LIFT BRUSH SPRINGS MORE THAN NEEDED TO REMOVE BRUSH SETS FROM BRUSH HOLDER.

- (2) Use a wire hook tool to lift the two brush springs away from a brush set. Remove brush set from brush holder assembly. See Figure 3001.
- (3) Slowly return brush springs to a resting position on brush holder assembly.
- (4) Identify brush set with number on brush holder from which it was removed.
- (5) Repeat Paragraph 4.D.(1) thru Paragraph 4.D.(4) for each remaining brush holder assembly.



Component Maintenance Manual with Illustrated Parts List
DC Starter-Generator, 23046 Series I

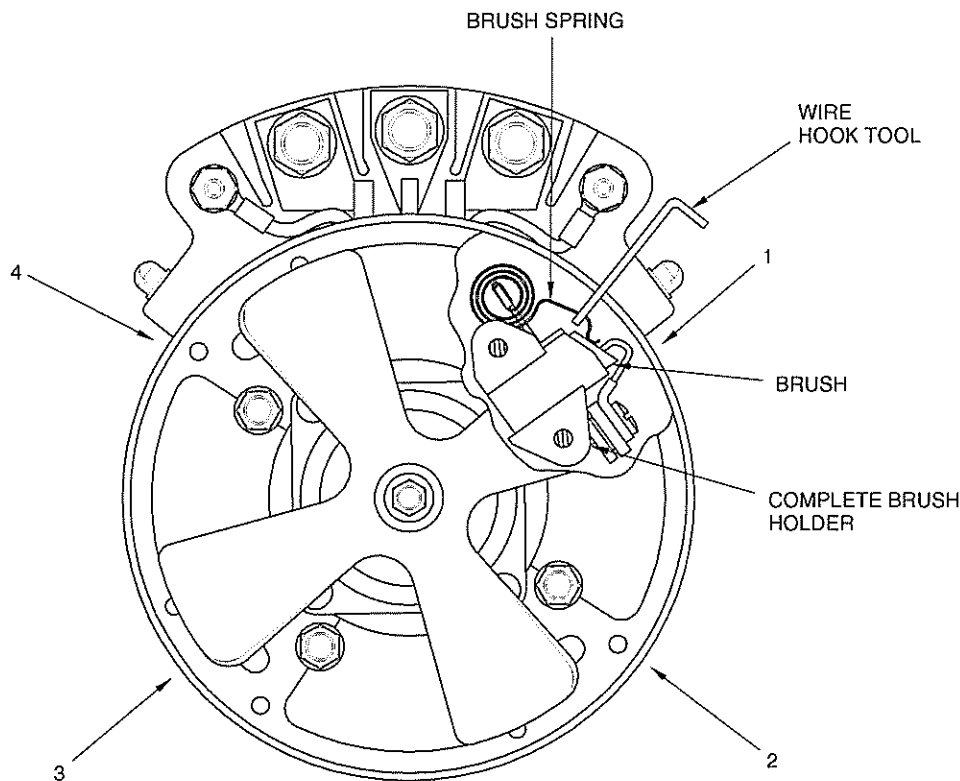


Figure 3001 - Removing Brushes

E. Remove air inlet assembly (200) and fan cover (240). (Models 23046-019 and 23046-020 only).

- (1) Loosen self-locking nut on T-bolt of rim-clenching clamp (210). Remove clamp. See Figure 3002.
- (2) Remove air inlet assembly (200) and fan cover (240) as a single unit from starter-generator.

NOTE: Do not perform the following step unless air inlet assembly or fan cover is worn or damaged and must be replaced.

- (3) Remove the two screws (220) and washers (230) that secure fan cover (240) to air inlet assembly (200).
- (4) Remove fan cover.



Component Maintenance Manual with Illustrated Parts List
DC Starter-Generator, 23046 Series I

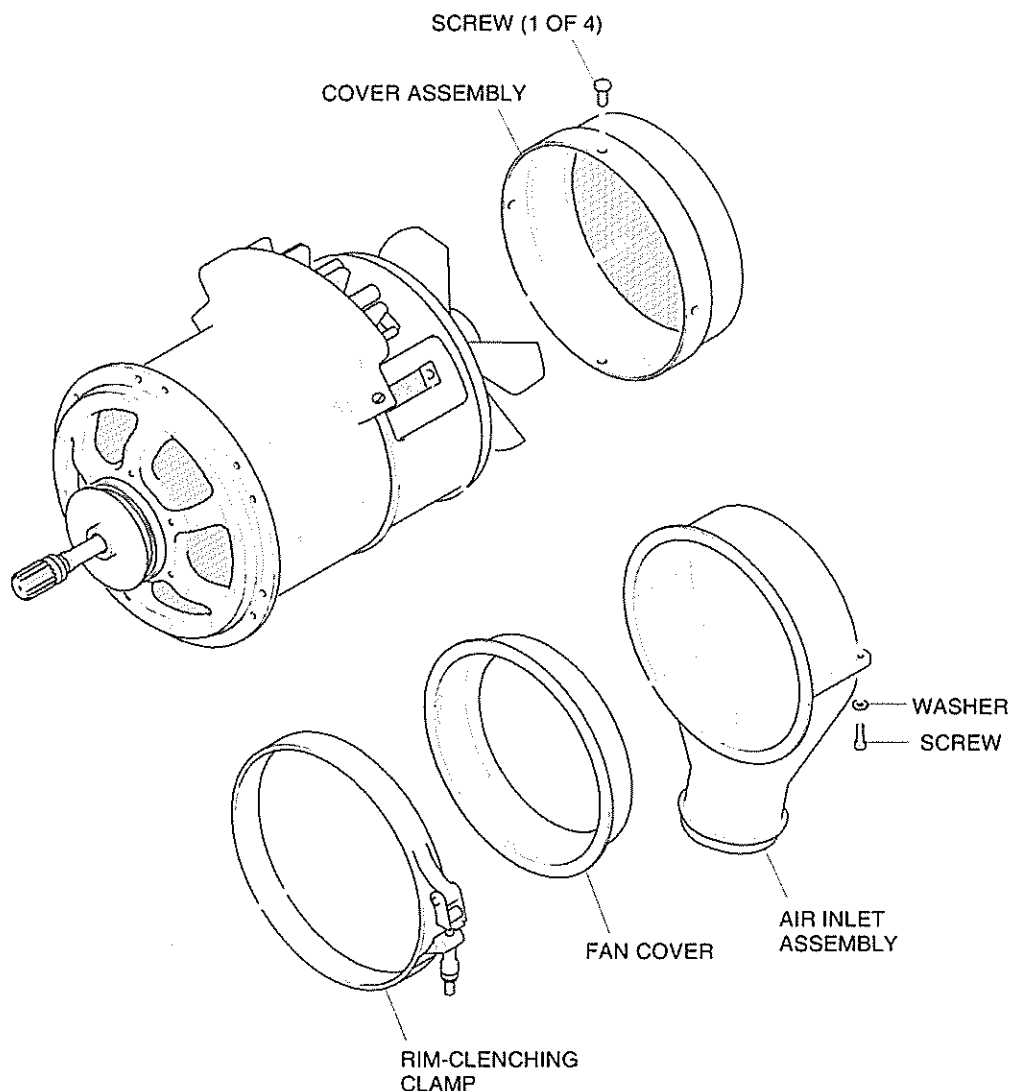


Figure 3002 - Removing Cover Assembly or Fan Cover and Air Inlet Assembly

F. Remove cover assembly (180). (All models except 23046-019 and 23046-020).

- (1) Remove four screws (190) that attach the cover assembly (180) to the starter-generator. See Figure 3002.
- (2) Remove cover assembly.

G. Remove fan (250) from drive shaft (290).

NOTE: Put a spline wrench on drive spline to prevent drive shaft from turning while removing self-locking nut.



Component Maintenance Manual with Illustrated Parts List
DC Starter-Generator, 23046 Series I

- (1) Remove self-locking nut (260).
- (2) Remove flat washer (270).
- (3) Slide fan off of drive shaft.
- (4) On models 23046-019 and 23046-020 at MOD D, remove anti-drive end bearing shield (280) from drive shaft.

H. Remove drive shaft (290) with attached dampener hub (310), dampener plate (320) and friction ring (330) from armature shaft.

CAUTION: EXCESSIVE FORCE WHEN TAPPING THE DRIVE SHAFT MAY DAMAGE THREADS.

- (1) Lightly tap anti-drive end of drive shaft with a plastic or leather mallet to disengage it from armature shaft. See Figure 3003.
- (2) Pull drive shaft out of drive end of starter-generator.
- (3) On models 23046-017, 23046-020 and 23046-028, remove O-ring (300) from drive shaft.

WARNING: FRICTION RING MAY CONTAIN ASBESTOS. DO NOT CUT, RIP, OR SAND ASBESTOS-CONTAINING MATERIALS. LEAVE UNDAMAGED MATERIALS ALONE AND, TO THE EXTENT POSSIBLE, PREVENT THEM FROM BEING DAMAGED, DISTURBED, OR TOUCHED. DISCARD MATERIAL BY FIRST CHECKING WITH LOCAL HEALTH, ENVIRONMENTAL, OR OTHER APPROPRIATE OFFICIALS TO FIND OUT ABOUT PROPER HANDLING AND DISPOSAL PROCEDURES.

- (4) Remove friction ring from drive shaft.
- (5) Remove dampener plate from dampener hub.

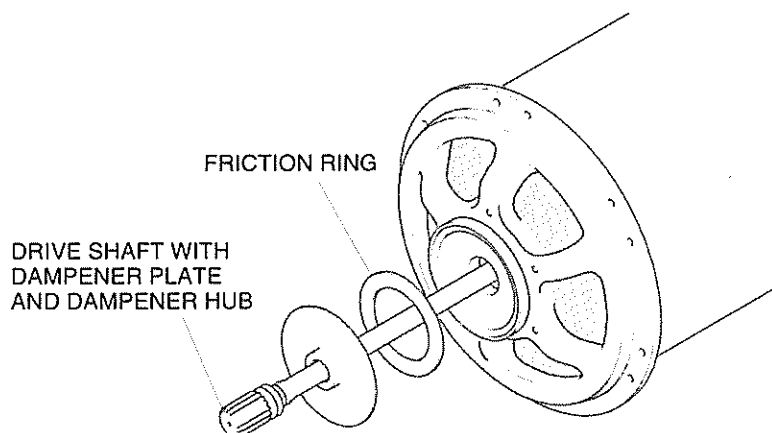


Figure 3003 - Removing Drive Shaft



Component Maintenance Manual with Illustrated Parts List
DC Starter-Generator, 23046 Series I

I. Remove dampener hub (310) from drive shaft (290). See Figure 3004.

NOTE: Do not remove dampener hub unless it is damaged.

CAUTION: FAILURE TO USE A FOAM CUSHION UNDER DRIVE SHAFT WHEN REMOVING DAMPENER HUB CAN DAMAGE DRIVE SHAFT.

- (1) Set a foam cushion and drive end hub support on an arbor press table.
- (2) Set a dampener hub driver on drive shaft on flat side of dampener hub.
- (3) Insert drive shaft through center of anti-drive end hub support until dampener hub driver is seated on anti-drive end hub support.
- (4) Press drive shaft down into anti-drive hub support until dampener hub disengages from drive shaft.

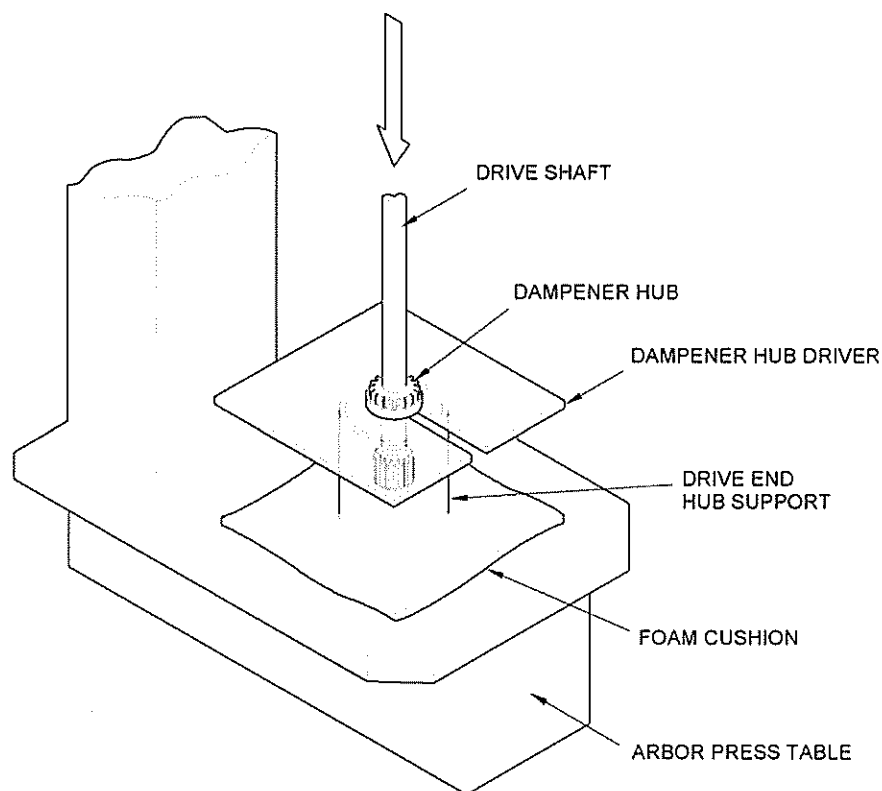


Figure 3004 - Dampener Hub Removal



Component Maintenance Manual with Illustrated Parts List
DC Starter-Generator, 23046 Series I

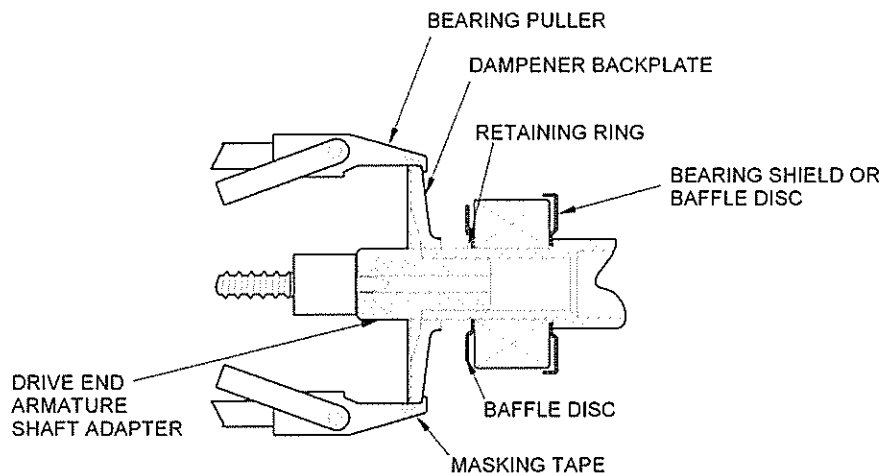
J. Remove dampener backplate (340) from armature (470) shaft. See Figure 3005.

CAUTION: FAILURE TO USE A DRIVE END ARMATURE SHAFT ADAPTER WHEN REMOVING DAMPENER BACKPLATE CAN PERMANENTLY DAMAGE ARMATURE SHAFT.

- (1) Put an armature shaft adapter in drive end of armature shaft.

CAUTION: TAPE JAWS OF BEARING PULLER TO PREVENT DAMAGE TO DAMPENER BACKPLATE DURING REMOVAL.

- (2) Remove dampener backplate from armature shaft using a suitable bearing puller.



* DRIVE END BEARING SUPPORT NOT SHOWN

Figure 3005 - Dampener Backplate Removal

K. Remove bearing and brush support assembly (-480) and attached armature (470) from stator and housing assembly (680).

- (1) Set starter-generator drive end down on a vertical stator support. See Figure 3006.
- (2) Remove eight screws (490), eight lock washers (500) and eight flat washers (510) that attach bearing and brush support assembly to stator and housing assembly.
- (3) Carefully lift bearing and brush support assembly, with attached armature, from stator and housing assembly.



Component Maintenance Manual with Illustrated Parts List
DC Starter-Generator, 23046 Series I

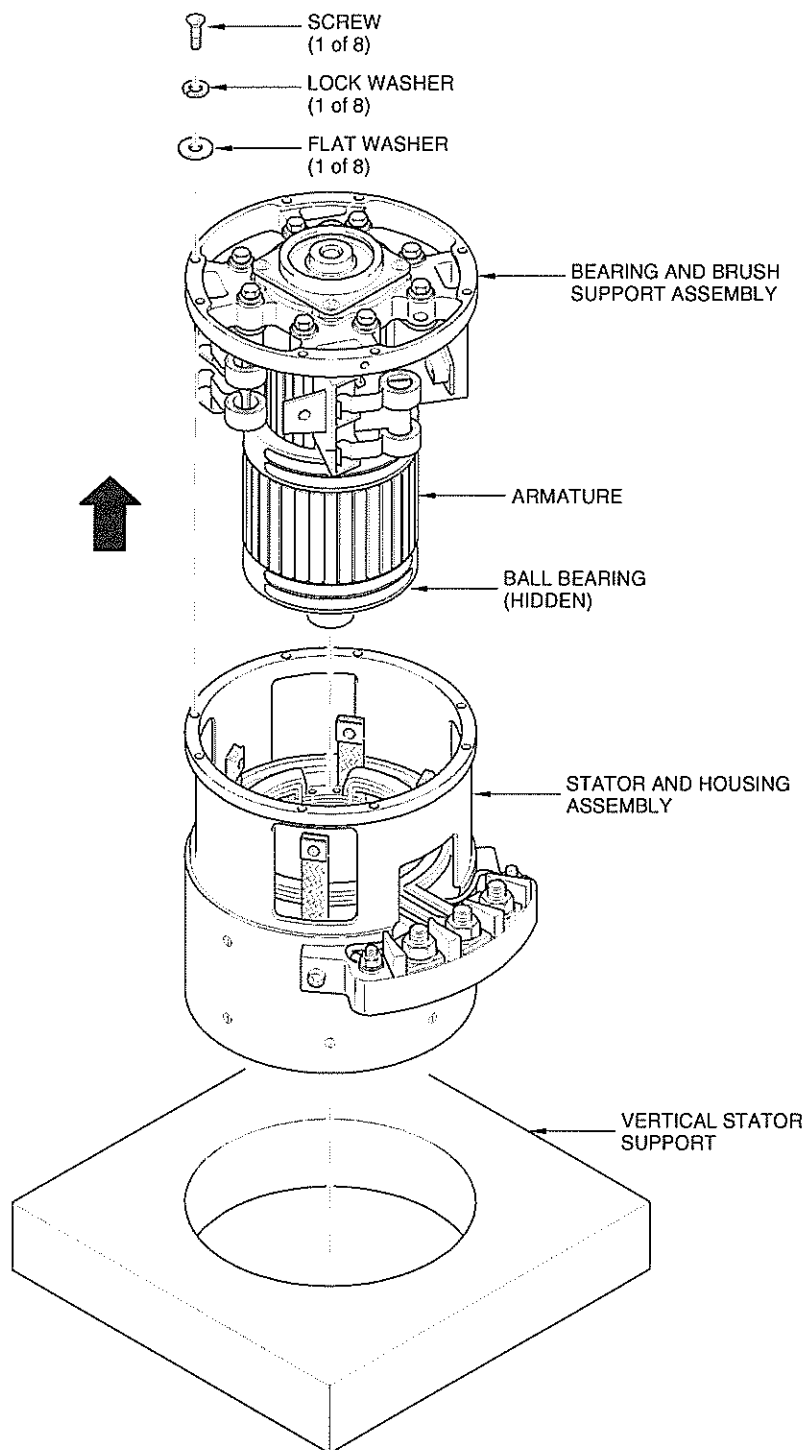


Figure 3006 - Removing Bearing and Brush Support Assembly with Attached Armature

DISASSEMBLY

24-30-63

Page 3010
Mar 23/10



Component Maintenance Manual with Illustrated Parts List
DC Starter-Generator, 23046 Series I

L. Remove drive end bearing support assembly (-350) from stator and housing assembly (680).

- (1) Set stator and housing assembly on a horizontal stator support.
- (2) For unit 23046-001 remove eight screws (360B), eight lock washers (370) and eight flat washers (380) that attach drive end end bell (390C) to stator and housing assembly (680).
- (3) For all other units, remove twelve screws (360) that attach drive end bearing support assembly (-350) to stator and housing assembly (680).

CAUTION: DO NOT USE EXCESSIVE FORCE WHEN TAPPING DRIVE END BEARING SUPPORT ASSEMBLY OR DAMAGE MAY RESULT TO THE MATING FLANGE OF THE STATOR AND HOUSING ASSEMBLY.

- (4) Insert a wooden stick, 8 inches (20 cm) or longer, through anti-drive end and hold it against hub of drive end end bell.
- (5) Lightly tap end of wooden stick using a plastic or leather mallet to loosen drive end bearing support assembly from stator and housing assembly.
- (6) Remove drive end bearing support assembly from stator and housing assembly.

M. Remove armature (470) from bearing and brush support assembly (-480).

- (1) Set a foam cushion and a bearing and brush support assembly support on an arbor press table. The foam cushion will help protect the armature shaft.

CAUTION: USE ONLY THE ANTI-DRIVE END ARMATURE SHAFT ADAPTER WHEN PRESSING ARMATURE FROM BEARING AND BRUSH SUPPORT ASSEMBLY TO PREVENT DAMAGE TO ARMATURE SHAFT.

- (2) Set bearing and brush support assembly with attached armature onto a bearing and brush support assembly support. See Figure 3007.
- (3) Insert an anti-drive end armature shaft adapter into end of armature shaft.

CAUTION: HOLD ARMATURE SECURELY WHEN IT IS PRESSED OUT OF THE BEARING AND BRUSH SUPPORT ASSEMBLY TO PREVENT POSSIBLE DAMAGE TO THE ARMATURE SHAFT.

- (4) While supporting armature assembly securely with one hand, carefully press armature shaft down and away from bearing and brush support assembly.
- (5) Carefully remove armature from inside of bearing and brush support assembly support.



Component Maintenance Manual with Illustrated Parts List
DC Starter-Generator, 23046 Series I

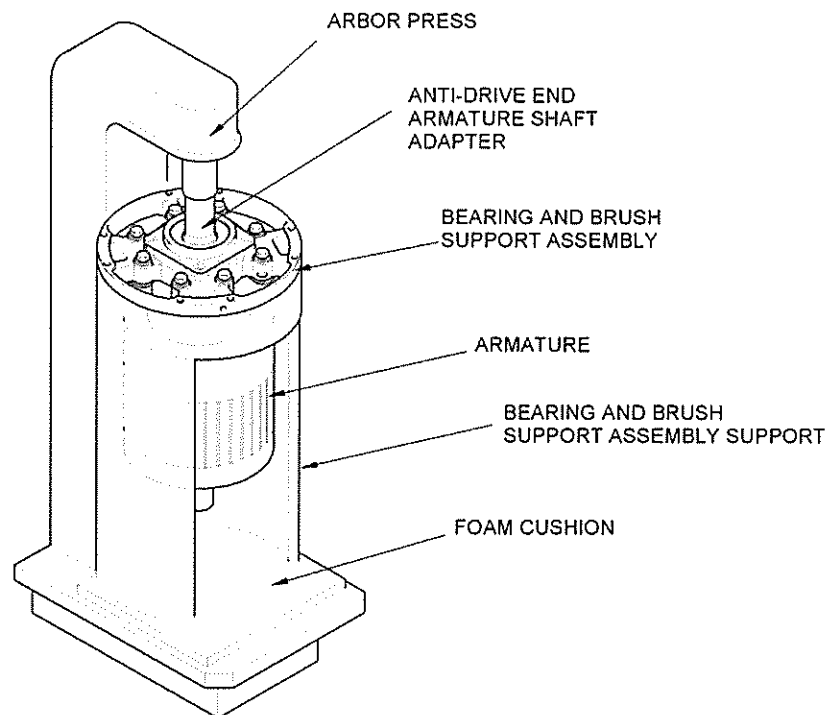


Figure 3007 - Removing Armature from Bearing and Brush Support Assembly

N. Remove ball bearing (450) and baffle disc (440) from drive end of armature (470) shaft. See Figure 3008.

- (1) Remove and discard retaining ring (430) using external snap ring pliers.
- (2) Remove baffle disc (440) from armature shaft.

CAUTION: USE DRIVE END ARMATURE SHAFT ADAPTER WHEN REMOVING BEARING TO PREVENT PERMANENT DAMAGE TO ARMATURE SHAFT.

- (3) Put a drive end armature shaft adapter into drive end of armature shaft.
- (4) Remove bearing from armature shaft using a suitable bearing puller.
- (5) Discard bearing.
- (6) On units 23046-019 and 23046-020 remove drive end bearing shield (460) from armature shaft.
- (7) On all other units remove baffle disc (440) from armature shaft.



Component Maintenance Manual with Illustrated Parts List
DC Starter-Generator, 23046 Series I

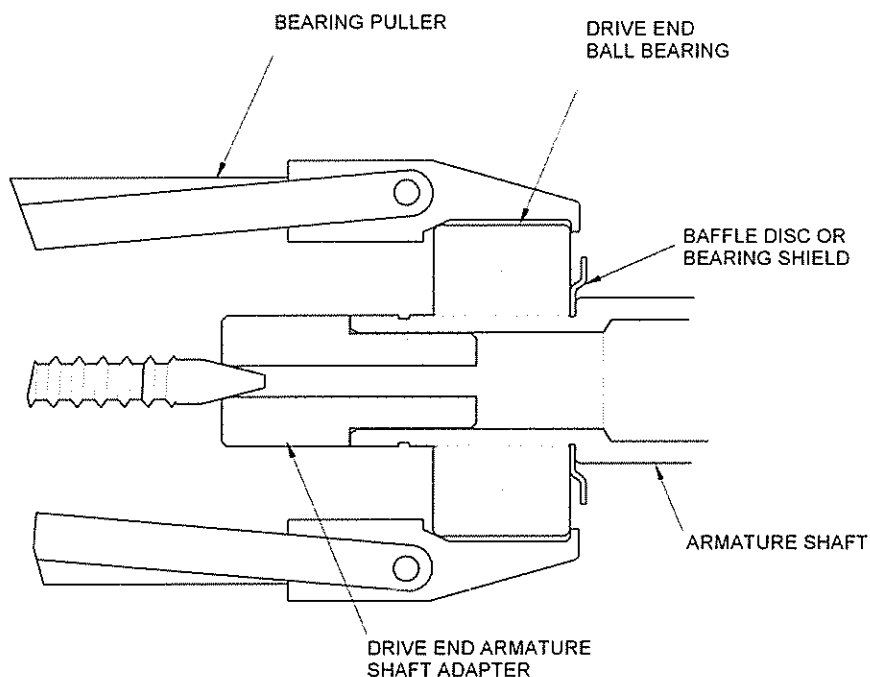


Figure 3008 - Removing Bearing from Drive End of Armature Shaft

O. Remove bearing retainer (660) from bearing and brush support assembly (-480).

- (1) Remove four screws (670) that attach bearing retainer (660) to bearing and brush support assembly (-480).
- (2) Remove bearing retainer.

P. Remove ball bearing (450) from bearing and brush support assembly (-480).

CAUTION: USE AN ANTI-DRIVE END BEARING HUB SUPPORT WHEN PRESSING THE BEARING FROM THE BEARING AND BRUSH SUPPORT ASSEMBLY TO PREVENT DAMAGE TO THE ASSEMBLY.

- (1) Set an anti-drive end bearing hub support on an arbor press table.
- (2) Set bearing and brush support assembly on anti-drive end bearing hub support with outboard side down.
- (3) Set an inner race bearing driver on inner race of ball bearing installed in bearing and brush support assembly. See Figure 3009.
- (4) Carefully press bearing from bearing and brush support assembly.
- (5) Remove bearing from inside anti-drive end bearing hub support. Discard bearing.



Component Maintenance Manual with Illustrated Parts List
DC Starter-Generator, 23046 Series I

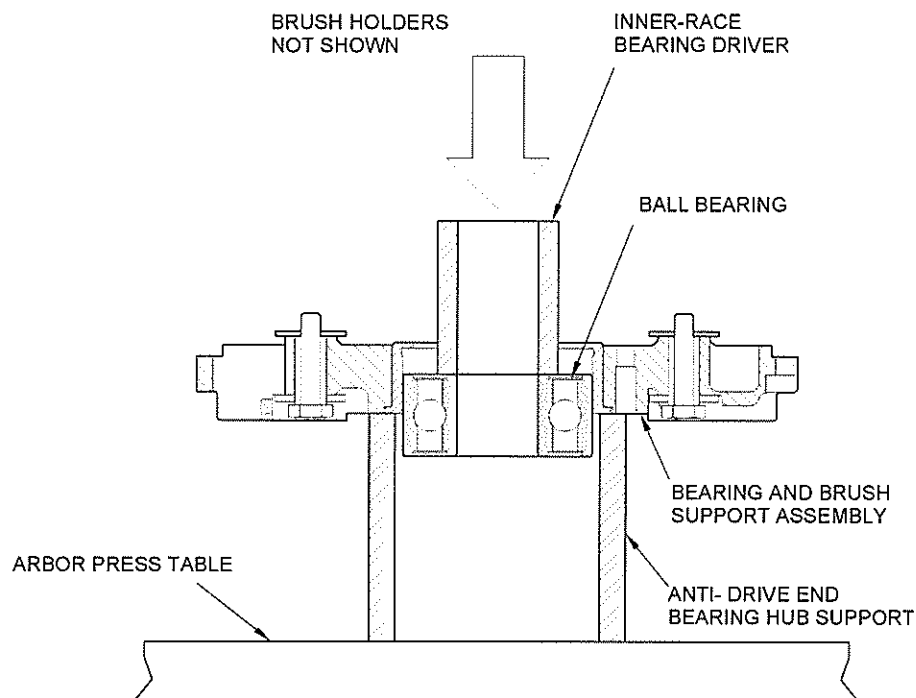


Figure 3009 - Removing Bearing from Bearing and Brush Support Assembly

Q. Remove brush holder assemblies (-550) from anti-drive end end bell (520).

NOTE: Do not remove brush holders and springs from anti-drive end end bell unless inspection reveals that replacement of part is required.

- (1) Remove eight bolts (560) or self-locking nuts (570) and flat washers (580) from anti-drive end end bell (520). See Figure 3010.
- (2) Remove eight non-metallic washers (590) and four insulation plates (650) and four brush holders (610) from anti-drive end end bell.
- (3) Remove insulation sleeves (640) from the through holes in anti-drive end end bell.
- (4) Discard non-metallic washers (590), insulation plates (650) and insulation sleeves (640).
- (5) Remove two brush springs (600) from each brush holder (610).



Component Maintenance Manual with Illustrated Parts List
DC Starter-Generator, 23046 Series I

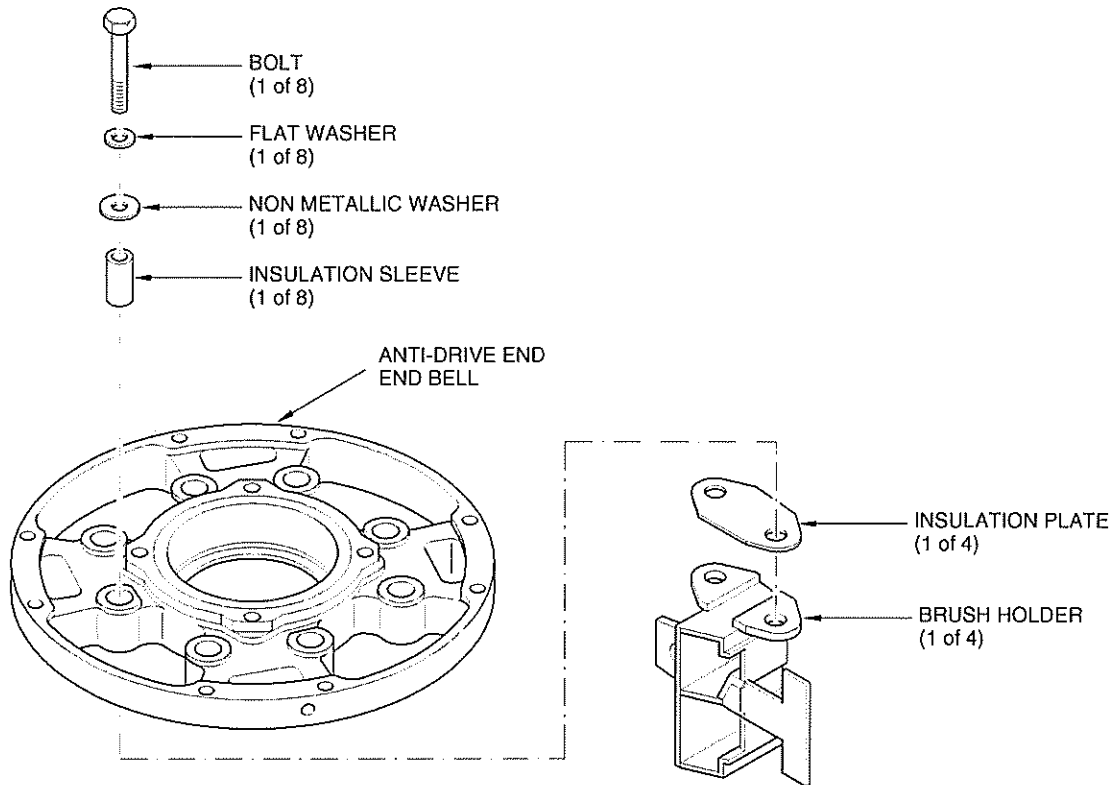


Figure 3010 - Removing Brush Holders

R. Disassembly of stator and housing assembly (680).

NOTE: Do not disassemble stator and housing assembly further than the removal of the terminal block to examine, repair, or replace parts determined to be unserviceable.

- (1) Remove three self-locking nuts (760) and flat washers (770) from terminal studs E, B and C. Discard the self-locking nuts and flat washers.
- (2) Remove two self-locking nuts (740) and flat washers (750) from terminal studs A and D. Discard the self-locking nuts and flat washers.
- (3) Remove two screws (700), lock washers (710) and flat washers (720) to release the terminal block (690) from the stator and housing assembly (680).
- (4) Carefully remove the terminal block from the stator leads.



Component Maintenance Manual with Illustrated Parts List
DC Starter-Generator, 23046 Series I

THIS PAGE INTENTIONALLY LEFT BLANK



Component Maintenance Manual with Illustrated Parts List
DC Starter-Generator, 23046 Series I

CLEANING

1. Introduction

This section contains the cleaning procedures for the 23046 Series I DC Starter-Generator.

2. Cleaning Materials

Table 4001 lists material and equipment required to perform the cleaning of assemblies, subassemblies and components.

NOTE: Equivalent substitutes may be used for materials listed in Table 4001.

NOTE: Refer to GSIL 2006-01 for the use of Brulin 815GD-NF

WARNING: YOU MUST OBEY THE MATERIAL/EQUIPMENT MANUFACTURERS WARNINGS AND CAUTIONS SHOWN ON PACKAGING, CONTAINERS AND/OR INSTRUCTION LEAFLETS. IF YOU DO NOT, YOU CAN CAUSE DEATH OR INJURY TO PERSONS AND DAMAGE TO EQUIPMENT OR PROPERTY.

WARNING: ISOPROPYL ALCOHOL IS DANGEROUS TO PERSONS. USE ONLY IN AN AREA WITH A GOOD FLOW OF AIR. KEEP AWAY FROM SOURCES OF HEAT, SPARKS AND FLAME. DO NOT BREATHE THE FUMES. PREVENT SKIN CONTACT. PUT ON PROTECTIVE CLOTHING AND EYE PROTECTION.

Material	Description/Specification	Source/CAGE Code
Bag, Polyethylene	Size that will easily enclose parts being washed.	Commercially available
Brush	Nonmetallic, soft bristle	Commercially available
Cleaning Pads/Wiping Cloths	Lint-free, soft fabric	Commercially available
Corrosion Preventive	Cold application, water displacing soft film MIL-C-16173, Grade 3.	Commercially available

Table 4001 - Cleaning Materials and Equipment



Component Maintenance Manual with Illustrated Parts List
DC Starter-Generator, 23046 Series I

Material	Description/Specification	Source/CAGE Code
De-ionized Water	For fresh water, Ion Exchange process to a resistivity not less than 750 kΩ/cm @ 77° F (25° C) min., when measured in accordance with ASTM D1125-95, Method A. For rinse water that is to be <u>reused</u>, check that the resistivity does not fall below 500 kΩ/cm @ 77° F (25° C) min., measured in accordance with ASTM D1125-95, Method A.	Commercially available
Desiccant	MIL-D-3464E Type II, Kraft bag, 4 unit bag size Englehard Corporation Desiccate 25 or equivalent	Commercially available
Detergent Refer to material technical data sheet for recommended dilution of water and detergent, and solution temperature. Equivalent detergents can be used to clean parts and assemblies. Detergents must not be destructive to the finish or material of part. Use only detergents specified in this table to clean IVD coated parts.	Formula 815 GD or 815 GD-NF Applications: Hot tanks - aluminum safe Steam Cleaning Pressure Sprayers Ultrasonic Cleaning Formula 1990 GD Application: Spray Wash Cleaning	The Brulin Corporation 2920 Dr. Andrew J Brown Ave. Indianapolis, Indiana 46205-4066 USA Phone: (1) 317/923-3211 FAX: (1) 317/925-4596 www.brulin.com (V94058)
Isopropyl Alcohol	TT-I-735, Grade A See <u>WARNING</u> before using this material. Flash Point: 53° F (12° C), FLAMMABLE Refer to the Material Safety Data (MSD) Sheet for the material for additional safety information	Commercially available

Table 4001 - Cleaning Materials and Equipment (Continued)

CLEANING

24-30-63

Page 4002
Mar 23/10



Component Maintenance Manual with Illustrated Parts List
DC Starter-Generator, 23046 Series I

3. Cleaning Procedures

Before you do procedures given in CHECK, REPAIR, ASSEMBLY, TESTING AND FAULT ISOLATION, and final inspection, the components must be cleaned, rinsed, and dried. All signs of moisture, oil, and other contamination must be removed from the internal structure of the generator to make sure all parts and circuits function correctly. Clean the starter-generator as follows:

A. Blow out dirt particles.

- (1) Put the parts into a ventilated air spray booth.

WARNING: COMPRESSED AIR USED FOR CLEANING MUST BE FREE OF OIL AND WATER. WHEN USING COMPRESSED AIR FOR CLEANING OR DRYING, CONTROL PRESSURE TO 29 PSIG (200 KPA) OR LESS. WEAR GOGGLES OR FACE SHIELD TO PROTECT EYES. TAKE PRECAUTIONS TO AVOID INJURY TO OTHER PERSONNEL IN AREA.

CAUTION: MAKE SURE THAT COMPRESSED AIR USED TO CLEAN/DRY THE COMPONENTS IS FREE FROM OIL AND WATER. THIS WILL PREVENT CONTAMINATION OF THE COMPONENTS.

- (2) Blow away all loose dirt with clean, dry, compressed air.

B. Clean the parts.

WARNING: DETERGENT CAN BE DANGEROUS TO PERSONS. USE ONLY IN AN AREA WITH A GOOD FLOW OF AIR. DO NOT BREATHE THE FUMES. PREVENT SKIN CONTACT. PUT ON PROTECTIVE CLOTHING AND EYE PROTECTION.

Wash the parts with one or more of the approved cleaning procedures given in Table 4002

Procedure	Instructions
Spray	Put the parts into a high pressure spray wash booth using detergent Brulin 1990 GD.
	<u>CAUTION:</u> LIMIT THE PRESSURE TO CLEAN PARTS. HIGH PRESSURE CAN REMOVE OR DAMAGE SURFACE COATINGS, CONNECTIONS AND INSULATION.
	Spray the parts with a solution of detergent and de-ionized water. Refer to the manufacturers specification for the recommended dilution of water and detergent and solution temperature.

Table 4002 - Cleaning Procedures



Component Maintenance Manual with Illustrated Parts List
DC Starter-Generator, 23046 Series I

Procedure	Instructions
Ultrasonic	CAUTION: ULTRASONIC CLEANING CAN DAMAGE PLATING, BLACK OXIDE COATINGS, AND, IN ALUMINUM PARTS, DIMENSIONAL TOLERANCES.
	Put the parts into the ultrasonic cleaning tank using detergent Formula 815 GD or 815 GD-NF and de-ionized water. Refer to the manufacturers specification for the recommended dilution of water and detergent and solution temperature.
	Let the parts soak for 15 to 20 minutes maximum.
	Remove all evidence of dirt with a soft bristle brush or cleaning rag.
Agitation	Put all parts into an agitation tank and clean using detergent Brulin 1990 GD and de-ionized water. Refer to the manufacturers specification for the recommended dilution of water and detergent and solution temperature.
	Agitate the parts until all evidence of dirt can be removed with a soft bristle brush or cleaning rag.
Soak	Put all parts into a soaking tank and clean using detergent Formula 815 GD, 815 GD-NF or Brulin 1990 GD and de-ionized water. Refer to the manufacturers specification for the recommended dilution of water and detergent and solution temperature.
	Soak the parts until all evidence of dirt can be removed with a soft bristle brush or cleaning rag.

Table 4002 - Cleaning Procedures (Continued)

C. Rinse the parts.

- (1) Rinse the part(s) with de-ionized water.

D. Do the applicable cleaning procedure again until the parts are clean and free of dirt and carbon residue.

E. Dry the parts.

WARNING: ISOPROPYL ALCOHOL IS DANGEROUS TO PERSONS. USE ONLY IN AN AREA WITH A GOOD FLOW OF AIR. KEEP AWAY FROM SOURCES OF HEAT, SPARKS AND FLAME. DO NOT BREATHE THE FUMES. PREVENT SKIN CONTACT. PUT ON PROTECTIVE CLOTHING AND EYE PROTECTION.

- (1) Examine each part for carbon particles by wiping with isopropyl alcohol on a clean lint-free cloth.

WARNING: WEAR INSULATED GLOVES AND OBSERVE ALL SAFETY PRECAUTIONS WHEN HANDLING HEATED PARTS TO AVOID PERSONAL INJURY.

- (2) Put the cleaned parts in an oven at a temperature of 200° to 250° F (93,3° to 121° C) for 2 to 3 hours.

CLEANING

24-30-63

Page 4004
Mar 23/10



Component Maintenance Manual with Illustrated Parts List
DC Starter-Generator, 23046 Series I

- (3) Remove moisture that remains on the parts with a dry lint-free cloth.

4. Corrosion Preventive

NOTE: If the drive shaft is not expected to be re-installed into a starter-generator immediately after cleaning, apply a corrosion preventive compound to its black oxide coating (identified by a very dark finish).

WARNING: THE CORROSION PREVENTIVE COMPOUND IS FLAMMABLE AND TOXIC. EXTENDED EXPOSURE CAN CAUSE SKIN IRRITATION, DIZZINESS, AND HEADACHE. WHEN EXTENDED EXPOSURE IS NECESSARY, USE RESPIRATORY PROTECTION AND PROTECTIVE CLOTHING.

- (1) Apply a layer of MIL-C-16173, grade 3, corrosion preventive compound to the drive shaft.

NOTE: The drive shaft must be completely dry before you apply the corrosion preventive compound.

- (2) Let the coating air dry for four hours. The coating will remain soft after drying.

5. Cleaning After Liquid Penetrant Inspection

A. Description

This specifies the procedure for cleaning part(s) after liquid penetrant inspection. Cleaning is necessary to remove residual penetrant or developer that could cause interference in processing or service requirements. Rinse parts fully with water and fully dry to make sure there is no interference in processing or service requirements.

B. Procedure

- (1) Rinse the part(s) immediately after liquid penetrant inspection:
 - (a) The parts must be fully rinsed using water by manual or automated spray to remove the liquid penetrant.
 - 1 Fully rinse the part(s) and aggressively agitate the part while it is immersed.
 - 2 Pour the rinse water out from the part(s). Hold the part(s) vertical, open end down, for approximately one minute to drain.
 - (b) The part(s) surfaces must be visually examined under a black light after rinsing to make sure all the liquid penetrant has been removed.
 - 1 Part(s) which show remaining liquid penetrant after rinsing must be fully cleaned and examined again.



Component Maintenance Manual with Illustrated Parts List
DC Starter-Generator, 23046 Series I

- (2) Drying of part(s) after rinsing.

CAUTION: REMAINING WATER USED FROM THE RINSING PROCEDURE MUST BE REMOVED. FAILURE TO OBEY DRYING INSTRUCTIONS CAN RESULT IN CORROSION OF THE MATERIAL.

- (a) Remove the rinse water from the part(s) with a dry lint-free cloth.

WARNING: WHEN YOU USE COMPRESSED AIR, ADJUST THE PRESSURE TO 29 PSI (200 KPA) MAXIMUM. PUT ON EYE PROTECTION TO PREVENT INJURY.

CAUTION: MAKE SURE THAT COMPRESSED AIR FOR USE TO CLEAN/DRY THE COMPONENTS IS FREE FROM OIL AND WATER. THIS WILL PREVENT CONTAMINATION OF THE COMPONENTS.

- (b) Use compressed air to remove rinse water.

WARNING: FAILURE TO USE NECESSARY SAFETY PRECAUTIONS WHEN HANDLING HOT MATERIALS CAN CAUSE SEVERE BURNS TO SKIN. WEAR THERMAL PROTECTIVE CLOTHING WHEN HANDLING HEATED PARTS.

- (c) Immediately put the washed and rinsed part(s) in an oven at a temperature of 200 to 250° F (93,3 to 121° C) for 2 to 3 hours.

- 1 The part(s) is to be positioned horizontally on a rack or on a block that will hold it in position.

- (d) After the part(s) is fully dried and has cooled.

- 1 Check for indications of possible corrosion caused by the liquid penetrant.

a Reject the part(s) if corrosion is found.

- 2 If no corrosion is found, put the part(s) in a polyethylene bag with a bag of desiccant and seal it with tape, twist tie or rubber band.

- 3 The part(s) must be kept in a bag with desiccant until just before it is assembled on the starter-generator.



Component Maintenance Manual with Illustrated Parts List
DC Starter-Generator, 23046 Series I

CHECK

1. Introduction

This section provides initial and detailed component inspection procedures for starter-generators returning from service for scheduled and unscheduled maintenance. Complete an initial inspection of the starter-generator before beginning any testing or disassembly. The results of this inspection will determine if a starter-generator is in satisfactory condition for a verification test or if it will need further disassembly, inspection, and/or replacement of parts.

This section provides the information that is needed to perform a satisfactory inspection of the starter-generator components. During an overhaul of the starter-generator, examine all primary components to find out if they are serviceable.

Refer to the FITS AND CLEARANCES section for dimensional acceptance limits. Reject a part if wear or damage is outside acceptance limits or is not serviceable.

2. Necessary Tools and Materials

The following tools and materials are needed to inspect the starter-generator and its components.

Tool Description	Reference
Alcohol, Isopropyl	Commercially available
Ammeter	Commercially available
Brush, non-metallic, soft bristle	Commercially available
Cleaning pads/wiping cloths, lint-free, soft fabric	Commercially available
Dial Indicator	Commercially available
Dynamic Balancer	Goodrich Corporation SPD1001
Growler	Commercially available
High Potential (Dielectric) Tester	Table 1001
LCR Meter	Commercially available
Magnifier, 7X to 10X	Commercially available
Pull Scale	Commercially available
V-blocks	Commercially available

Table 5001 - Inspection Tools

CHECK

24-30-63

Page 5001
Mar 23/10



Component Maintenance Manual with Illustrated Parts List
DC Starter-Generator, 23046 Series I

A. Use of growler.

The use of a growler is necessary to check for shorted windings during inspection. When an alternating current is passed through a growler, it sets up a magnetic flux in the windings of the armature. As the flux passes through the windings, a potential is induced causing the current to flow if the winding is short-circuited. This current is detected with an iron feeler.

The most common way of using a growler is the "feeler method" in which the growler spans a slot containing a winding. A "feeler" of iron, such as a hack-saw blade, is held about 0.25 inch. (6 mm) above the slot containing the other side of the same winding. If the winding is shorted, the feeler will be pulled down to the slot and will stick and vibrate. The feeler can also be used on the same side of the winding that is spanned by the growler.

3. General Information

Check Acceptance Limit Measurements for individual components that are found in the FITS AND CLEARANCES section of this CMM. If part does not meet acceptance limits, discard. Use the following procedures for all inspections.

A. Examine Starter-Generator and its components visually for:

- signs of corrosion
- chipped-off enamel
- nicks, cracks, dents, scratches
- gouging, scoring or glazing mating surfaces
- fretting corrosion on mating surfaces
- loose or defective attaching parts
- distortion
- discoloration
- crossed or stripped threads
- torn or cracked seals
- galling or glazing of friction lining residue
- bent or distorted springs
- rounding, stripping or uneven wear on spline teeth

B. Examine electrical components visually for:

- loose or defective attaching parts
- damage caused by excessive heat

CHECK

24-30-63

Page 5002
Mar 23/10



Component Maintenance Manual with Illustrated Parts List
DC Starter-Generator, 23046 Series I

- electrical arcing paths
- corroded contacts or terminals
- loose or defective contacts or terminals
- loose, broken or shorted terminations

4. Initial Inspection

NOTE: If damage is found during inspection, stop inspection immediately and begin further disassembly of starter-generator using instructions in DISASSEMBLY section.

A. Visual Examinations

- (1) Visually examine starter-generator in a brightly lit work area in accordance with Paragraph 3.A. and Paragraph 3.B.
 - (a) If shipping or handling damage exists, stop inspection and notify supervisor.
 - (b) If operational damage exists, write down the components that need to be replaced or repaired and continue the inspection.
- (2) Examine seating between housing and both end bell assemblies. Make sure that all attaching hardware is in place.

B. Clean exterior of starter-generator.

WARNING: ISOPROPYL ALCOHOL IS DANGEROUS TO PERSONS. USE ONLY IN AN AREA WITH A GOOD FLOW OF AIR. KEEP AWAY FROM SOURCES OF HEAT, SPARKS AND FLAME. DO NOT BREATHE THE FUMES. PREVENT SKIN CONTACT. PUT ON PROTECTIVE CLOTHING AND EYE PROTECTION.

- (1) Clean surface of starter-generator using a lint-free cleaning cloth moistened with isopropyl alcohol.
- (2) If needed, loosen any dust particles or grease with a soft bristle brush.

5. Non-Destructive Test (NDT) Inspections.

A. Magnetic Particle Inspection

Unless otherwise specified, do magnetic particle inspections IAW ASTM-E1444. Unless otherwise specified, acceptance criteria must be IAW MIL-STD-1907 Table I, Grade A.

B. Liquid Penetrant Inspection

Unless otherwise specified, do liquid penetrant inspections IAW ASTM-E1417 type 1 Method A or B, sensitivity 2 or higher. Unless otherwise specified, acceptance criteria must be IAW MIL-STD-1907 Table II, Grade C.

CHECK

24-30-63

Page 5003
Mar 23/10



Component Maintenance Manual with Illustrated Parts List
DC Starter-Generator, 23046 Series I

6. Replacing Service Parts and Common Hardware

A. Overhaul

- (1) All bearings, brushes, insulating parts, retaining rings, O-rings, self-locking nuts, and lockwashers shall be discarded regardless of their condition, if removed during overhaul.

B. Service repair

- (1) During repair, inspection or servicing, replace all exposed hardware and flat washers that are damaged, deformed, corroded, or have other apparent defects. Always replace bearings, retaining rings, lockwashers, and self-locking nuts regardless of their condition if removed during disassembly.

7. Inspection of Starter-Generator Parts and Assemblies

A. Overhaul inspection

At starter-generator overhaul, examine all components of starter-generator.

B. Repair or service inspection

Examine only those components that have been disassembled for repair or service. Do not disassemble any component for inspection unless instructed to do so in this section or in the TESTING AND FAULT ISOLATION section.

NOTE: Brushes and commutator must be inspected according to inspection criteria in SPD1006 during repair inspection.

C. The following terms are referenced in this section:

• DISCARD -

If damage to the part is found or part is outside of acceptance limits in the FITS AND CLEARANCES section, part must be removed and replaced with a new one.

• REPAIR -

If damage to part is found or part is outside of acceptance limits in the FITS AND CLEARANCES section, part can be repaired by a procedure found in the REPAIR section.

• DISASSEMBLE -

If damage is found to a part in an assembly, or part of the assembly is outside of acceptance limits in the FITS AND CLEARANCES section, disassemble part from assembly and repair or replace damaged part(s).

CHECK

24-30-63



Component Maintenance Manual with Illustrated Parts List
DC Starter-Generator, 23046 Series I

• REPLACE -

If damage occurs that causes a part to be outside of acceptance limits in the FITS AND CLEARANCES section or there is no repair procedure for that part, replace part. Retain the following parts to submit for remanufacture: armature and stator and housing assembly.

D. Dimensions and points

- (1) Letters (i.e., A,B,C,D, etc.) used in an illustration in this section indicate dimensions or points.

E. Surfaces

- (1) Numbers (i.e., 1, 2, 3, 4, etc.) used in an illustration indicate surfaces.

8. Inspection of Parts and Assemblies

Unless otherwise specified, item numbers in parentheses () refer to items found in Figure 10001 in the ILLUSTRATED PARTS LIST.

NOTE: In cases where a repair is authorized but the repair does not work, discard and replace the item in question regardless of whether or not it is so spelled out in this manual.

A. Rim-clenching clamp (20)

Examine part IAW procedures found in Paragraph 3.A.

- (1) DISCARD clamp if cracks, dents, nicks, or scratches exist.
- (2) REPAIR T-bolt (-40) with two or less damaged thread turns. DISCARD if damage exceeds that.
- (3) DISCARD nut (-30) if removed.

B. End bell, drive end pad (50)

- (1) Examine part IAW procedures found in Paragraph 3.A.
- (2) DISCARD part if cracks are found.
- (3) REPAIR part if minor dents, nicks and scratches or mating surface gouging, scoring, or glazing exist.
- (4) Visually examine mating surfaces (See Figure 5001) for corroded surface coatings and pitting.

DISCARD drive end pad if pilot flange diameter is not within limits in the FITS AND CLEARANCES section.

CHECK

24-30-63

Page 5005
Mar 23/10



Component Maintenance Manual with Illustrated Parts List
DC Starter-Generator, 23046 Series I

- (5) If the visual inspection finds indications which can be cracks, continue the inspection under 10X magnification.
 - (a) If visual indications of cracks or other structural defects are observed under 10X magnification, DISCARD the part.
- (6) If the visual inspection under 10X magnification finds indications which can be cracks, perform a liquid penetrant inspection IAW Paragraph 5.B.

NOTE: After liquid penetrant inspection, the part(s) must have the liquid penetrant removed to prevent interference with processing or with service requirements. Refer to the CLEANING section.

 - (a) REPLACE if damage is found.
- (7) Examine pad for damaged or missing guide pins (-60).
 - (a) REPLACE guide pin(s) if damaged or missing.
 - (b) REPLACE guide pin(s) if guide pin height is not within limits in the FITS AND CLEARANCES section.

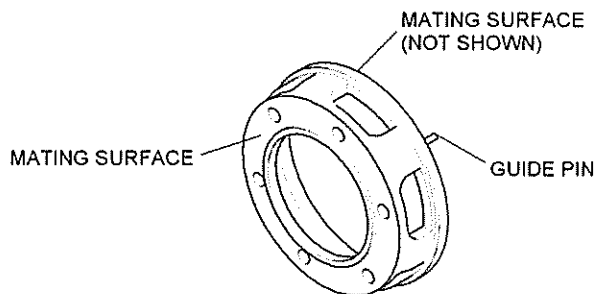


Figure 5001 - Inspecting End Bell, Drive End Pad

CHECK

24-30-63



Component Maintenance Manual with Illustrated Parts List
DC Starter-Generator, 23046 Series I

C. Identification plate (70). See Figure 5002.

NOTE: Identification plates with superficial nicks, dents and scratches that do not interfere with the legibility of the identification plate, can be reused providing they are sealed with acrylic coating.

- (1) Examine the part IAW the procedures found in Paragraph 3.A. If damage exists or the information is faded or unreadable:
 - (a) Retain the original identification plate.
 - (b) Transfer the information from the original identification plate to a replacement identification plate (70).
 - (c) DISCARD the original identification plate.
- (2) Make sure that drive screws (80) that attach identification plate are in place, and tight.
 - (a) DISCARD loose drive screws (80).

STARTER-GENERATOR, DC											
MODEL NO.				INSP				SPEC. NO.			
SER. NO.				MFD				CONTR.			
WT. LBS.				ROTATION				STOCK NO.			
EXT. START POWER				GENERATOR RATING							
VOLTS, MAX.				AMPS, MAX.				SPEED, RPM			
MOD				A B C D E F G H J K L M N P R T							
CAGE CODE 31435 MFR K4358				GOODRICH				THIS UNIT			
REPLACEMENT NAMEPLATE								OVERHAULED BY			

Figure 5002 - Typical Replacement Identification Plate

D. Brush access cover (130). See Figure 5003.

- (1) Examine the part IAW the procedures found in Paragraph 3.A.
 - (a) DISCARD the part if cracks or major damage exists.
 - (b) REPAIR the part if minor surface damage exists.
- (2) Examine the insulating tape for tears or loose edges.
 - (a) REPAIR the insulating tape if it is loose or worn.

CHECK

24-30-63

Page 5007
Mar 23/10



Component Maintenance Manual with Illustrated Parts List
DC Starter-Generator, 23046 Series I

- (3) Examine the screw (140) threads IAW the procedures found in Paragraph 3.A.
 - (a) REPAIR the part if two or less threads are damaged.
 - (b) DISCARD the part if more than two threads are damaged.
- (4) Make sure that the blind rivet nut (150) on the bracket is tightly attached.
 - (a) REPAIR the part if the nut is loose or missing.
- (5) Examine the blind rivet nut (150) threads IAW the procedures found in Paragraph 3.A.
 - (a) REPAIR the part if two or less threads are damaged.
 - (b) DISCARD the part if more than two threads are damaged.

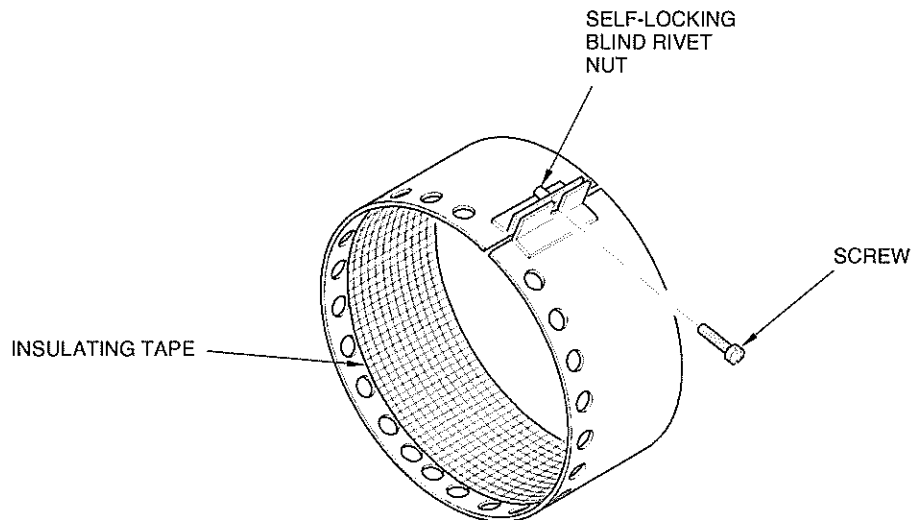


Figure 5003 - Brush Access Cover

E. Brushes (160)

NOTE: For repair only. Brushes must be replaced at each overhaul.

- (1) Examine part IAW procedures found in Paragraph 3.A. and Paragraph 3.B.
- (2) REPLACE if defective.
- (3) Replace brushes if remaining allowable wear will be exceeded before the next scheduled maintenance or overhaul.
- (4) For proper brush seating procedures, refer to SPD1006.

CHECK

24-30-63

Page 5008
Mar 23/10



Component Maintenance Manual with Illustrated Parts List
DC Starter-Generator, 23046 Series I

F. Cover assembly (180)

- (1) Examine part IAW procedures found in Paragraph 3.A.
- (2) DISCARD part if cracks are found.

G. Air inlet assembly (200) and fan cover (240)

- (1) Examine part IAW procedures found in Paragraph 3.A.
- (2) DISCARD part if cracks are found.

H. Rim-clenching clamp (210) for air inlet assembly

Examine part IAW procedures found in Paragraph 3.A.

- (1) DISCARD clamp if cracks, dents, nicks, or scratches exist.
- (2) REPAIR T-bolt with two or less damaged thread turns. DISCARD if damage exceeds that.
- (3) DISCARD self-locking nut if removed.

I. Fan (250)

- (1) One and two part fans.

- (a) Examine the part IAW the procedures found in Paragraph 3.A.

NOTE: Pay particular attention to the blade edges and surfaces and shaft mating surface.

- 1 DISCARD the part if cracks, scoring, gouging, glazing on mating surfaces, or major damage exists.
 - 2 REPAIR the fan if minor surface damage is found.

- (b) If the visual inspection finds indications which can be cracks, continue the inspection under 10X magnification.

- 1 If visual indications of cracks or other structural defects are observed under 10X magnification, DISCARD the part.

- (c) If the visual inspection under 10X magnification finds indications which can be cracks perform a liquid penetrant inspection IAW Paragraph 5.B.

NOTE: After liquid penetrant inspection, the part(s) must have the liquid penetrant removed to prevent interference with processing or with service requirements. Refer to the CLEANING section.

- 1 REPLACE if damage is found.

CHECK

24-30-63

Page 5009
Mar 23/10



Component Maintenance Manual with Illustrated Parts List
DC Starter-Generator, 23046 Series I

- (2) Two part fan, P/N 23046-1020
 - (a) Dimensionally inspect the fan assembly as follows. Refer to Figure 5004.
 - 1 Back edge of blade must not be less than 0.100 inch (2,54 mm) from the locating shoulder of the hub.
 - 2 Forward edge of blade must not be more than 0.830 inch (21,082 mm) from the locating shoulder of the hub.

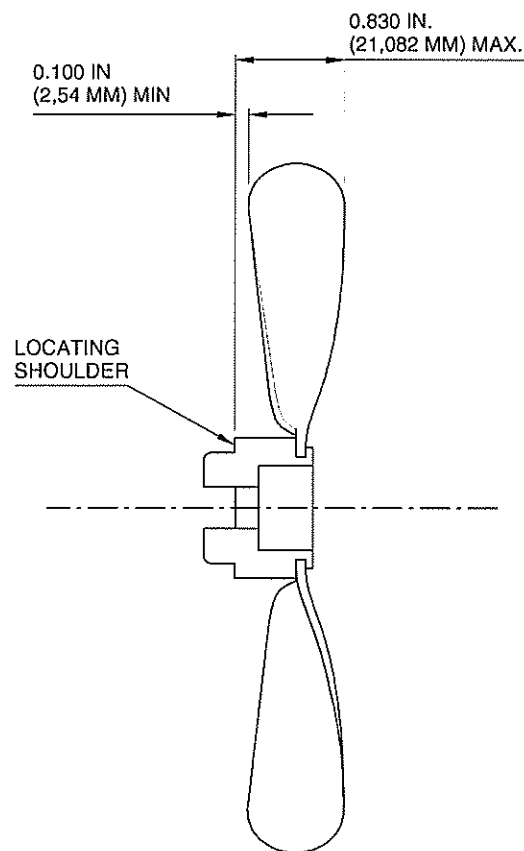


Figure 5004 - Two Part Fan

CHECK

24-30-63

Page 5010
Mar 23/10



Component Maintenance Manual with Illustrated Parts List
DC Starter-Generator, 23046 Series I

J. Anti-drive end bearing shield (280). See Figure 5005.

- (1) Examine the part IAW the procedures found in Paragraph 3.A.
 - (a) DISCARD the part if damaged.

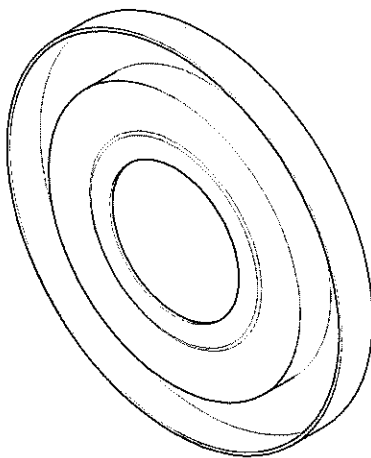


Figure 5005 - Anti-Drive End Bearing Shield

K. Drive shaft (290) and dampener hub (310). See Figure 5006.

- (1) Examine part IAW procedures found in Paragraph 3.A.
 - (a) DISASSEMBLE drive shaft (290) and dampener hub (310) if cracks or thread damage beyond two threads exist. DISCARD damaged part.
 - (b) REPAIR drive shaft and dampener hub if dents, nicks, and scratches exist.
- (2) Examine armature shaft mating spline, drive spline, and dampener hub for rounding, stripping, or uneven wear.

DISASSEMBLE drive shaft and dampener hub if damaged. DISCARD damaged part.
- (3) Measure drive spline diameter over two gage pins.
 - (a) DISASSEMBLE drive shaft and dampener hub if diameter is not within the limits of the FITS AND CLEARANCES section. DISCARD drive shaft.
- (4) Perform a magnetic particle inspection IAW procedures found in Paragraph 5.A.
 - (a) DISCARD drive shaft and dampener hub if damage is found.

CHECK

24-30-63

Page 5011
Mar 23/10



Component Maintenance Manual with Illustrated Parts List
DC Starter-Generator, 23046 Series I

- (5) Visually examine mating surfaces for gouging, scoring, or glazing.
 - (a) REPAIR damaged surface(s) if minor damage is found.
 - (b) DISCARD part(s) if part(s) fail visual inspection.

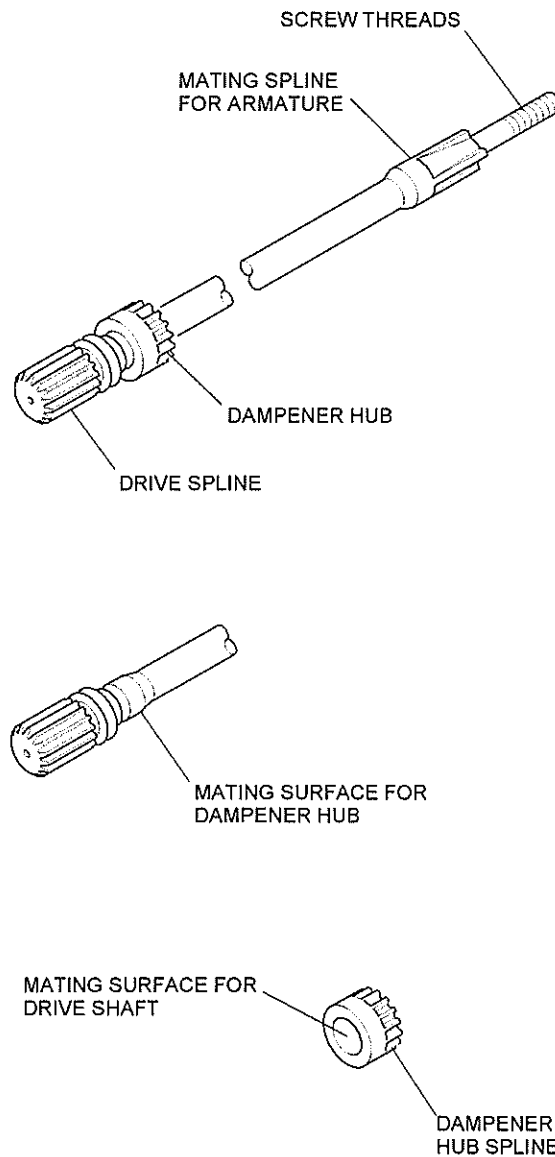


Figure 5006 - Inspecting Drive Shaft and Dampener Hub



Component Maintenance Manual with Illustrated Parts List
DC Starter-Generator, 23046 Series I

L. Dampener plate (320). See Figure 5007.

- (1) Examine part IAW procedures found in Paragraph 3.A.
 - (a) DISCARD part if cracks exist.
 - (b) REPAIR if minor nicks, scratches, gouging, scoring, or glazing exist.
- (2) Visually examine spline teeth for rounding, stripping, or uneven wear.
 - (a) DISCARD part if damaged.
- (3) Measure dampener plate thickness.
 - (a) DISCARD if thickness is below the acceptance limit in the FITS AND CLEARANCES section.
- (4) Measure the distance between pins of the spline teeth.
 - (a) DISCARD if distance is above the acceptance limit in the FITS AND CLEARANCES section.
- (5) Perform a magnetic particle inspection IAW the procedures found in Paragraph 5.A.
 - (a) DISCARD the part if damaged.

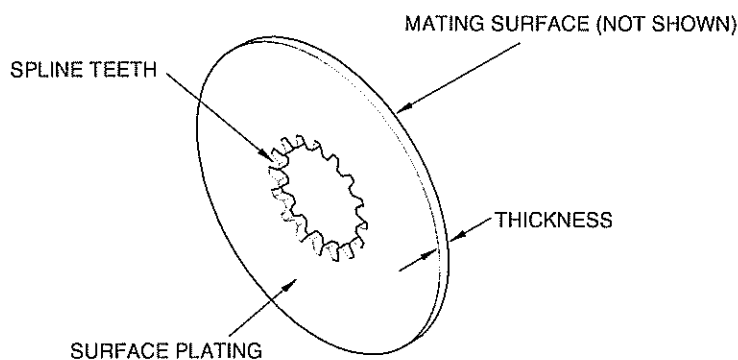


Figure 5007 - Dampener Plate

CHECK

24-30-63

Page 5013
Mar 23/10



Component Maintenance Manual with Illustrated Parts List
DC Starter-Generator, 23046 Series I

M. Friction ring (330). See Figure 5008.

WARNING: DO NOT CUT, RIP, OR SAND ASBESTOS-CONTAINING MATERIALS. LEAVE UNDAMAGED MATERIALS ALONE AND, TO THE EXTENT POSSIBLE, PREVENT THEM FROM BEING DAMAGED, DISTURBED, OR TOUCHED. DISCARD MATERIAL BY FIRST CHECKING WITH LOCAL HEALTH, ENVIRONMENTAL, OR OTHER APPROPRIATE OFFICIALS TO FIND OUT ABOUT PROPER HANDLING AND DISPOSAL PROCEDURES.

- (1) If friction ring is P/N 02-5600-05, it must be discarded because it contains asbestos. Install only new friction rings with no asbestos material that have a black color.
- (2) Examine the part IAW the procedures found in Paragraph 3.A.
 - (a) DISCARD the part if damage is found.
- (3) Measure the friction ring (330) thickness.
 - (a) DISCARD the part if thickness is not within limits in the FITS AND CLEARANCES section.

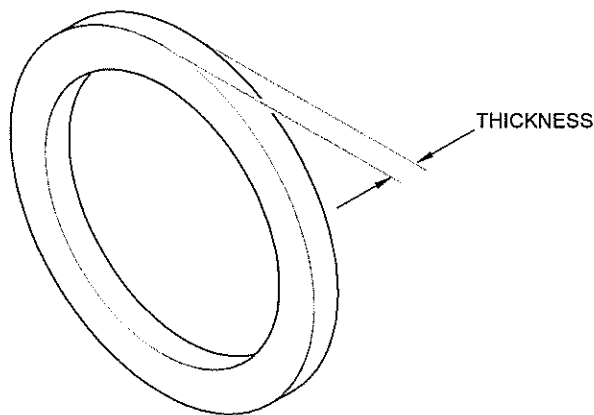


Figure 5008 - Friction Ring

N. Dampener backplate (340). See Figure 5009.

- (1) Examine part IAW procedures found in Paragraph 3.A.
 - (a) DISCARD part if cracks exist.
 - (b) REPAIR if minor nicks or scratches exist.
 - (c) DISCARD if gouging, scoring, or glazing exists on mating surfaces.

CHECK

24-30-63



Component Maintenance Manual with Illustrated Parts List
DC Starter-Generator, 23046 Series I

- (2) Perform a magnetic particle inspection IAW the procedures found in Paragraph 5.A.
 - (a) DISCARD part if cracks are found.

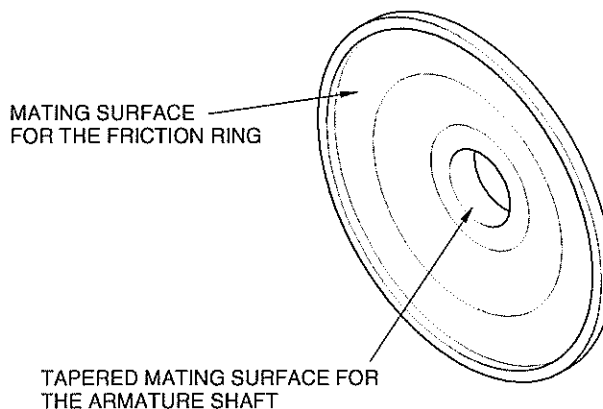


Figure 5009 - Inspecting Dampener Backplate

O. Drive end bearing support assembly (-350). See Figure 5010.

- (1) Examine part IAW procedures found in Paragraph 3.A.
 - (a) DISCARD part if cracks or fretting and/or corrosion on mating surface are found.
 - (b) REPAIR if minor dents, scratches and nicks or gouging, scoring, or glazing on the mating surfaces are found.
- (2) If the visual inspection finds indications which can be cracks, continue the inspection under 10X magnification.
 - (a) If visual indications of cracks or other structural defects are observed under 10X magnification, DISCARD the part.
- (3) If the visual inspection under 10X magnification finds indications which can be cracks, perform a liquid penetrant inspection IAW Paragraph 5.B.

NOTE: After liquid penetrant inspection, the part(s) must have the liquid penetrant removed to prevent interference with processing or with service requirements. Refer to the CLEANING section.

- (a) REPLACE if damage is found.

CHECK

24-30-63



Component Maintenance Manual with Illustrated Parts List
DC Starter-Generator, 23046 Series I

- (4) Measure bearing liner diameter using a gage and equipment with an accuracy of 0.00004 inch. (0,0010 mm).
 - (a) REPAIR part if diameter is not within the limits of the FITS AND CLEARANCES section.
- (5) Examine screen (400) for damage.
 - (a) DISCARD screen if damage is found.

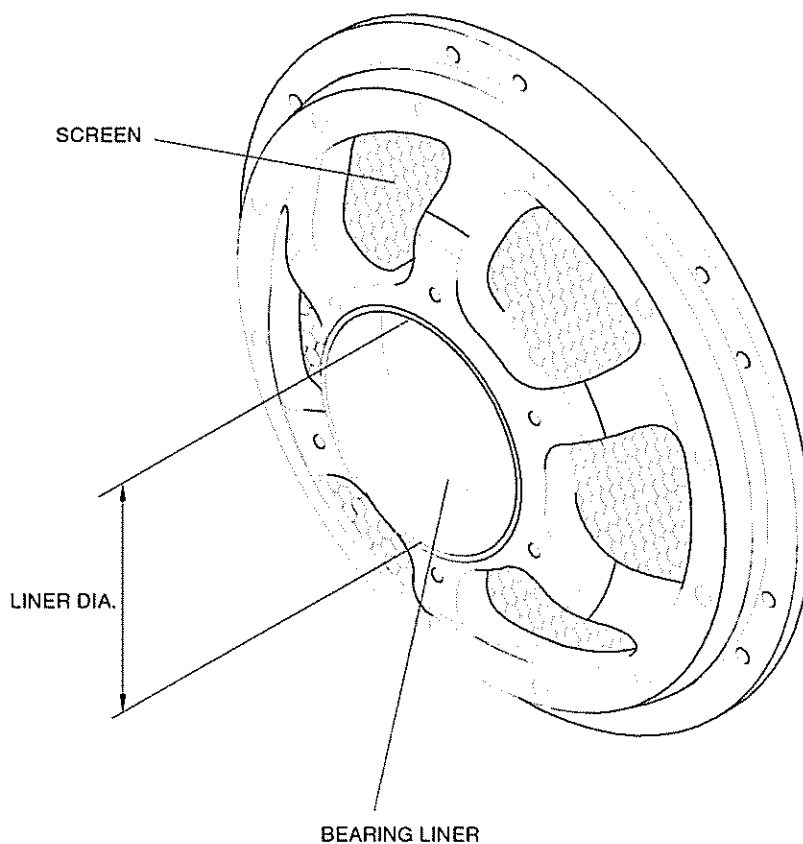


Figure 5010 - Drive End Bearing Support Assembly

P. Baffle disc (440). See Figure 5011.

- (1) Examine part IAW procedures found in Paragraph 3.A.
 - (a) DISCARD part if damaged.
- (2) Carefully examine mating surfaces 1, 2, and 3 for gouging, scoring, or glazing. See Figure 5011.
 - (a) DISCARD part if damaged.

CHECK

24-30-63



Component Maintenance Manual with Illustrated Parts List
DC Starter-Generator, 23046 Series I

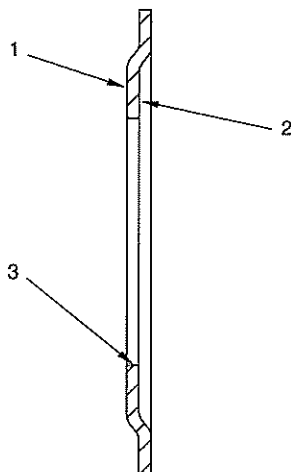


Figure 5011 - Inspecting Baffle Disc

Q. Drive end bearing shield (460). See Figure 5012.

- (1) Examine the part IAW the procedures found in Paragraph 3.A.
 - (a) DISCARD the part if damaged.

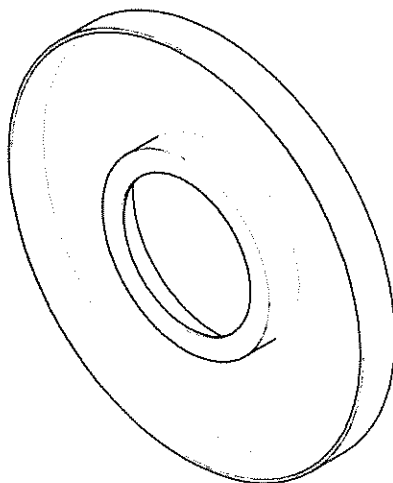


Figure 5012 - Bearing Shield

CHECK

24-30-63

Page 5017
Mar 23/10



Component Maintenance Manual with Illustrated Parts List
DC Starter-Generator, 23046 Series I

R. Armature (470). See Figure 5013.

NOTE: Use of Glyptol on either the armature (470) or stator and housing assembly (680) stator windings is strictly prohibited. Glyptol may be used on internal surfaces of the housing only, and only where other coatings are not required. Units having Glyptol coating on any other surfaces shall have these parts replaced prior to returning the units to service.

- (1) Before the commutator is refinished, measure commutator bar-to-bar run-out in a full circumference outside the brush paths (area where the brushes do not touch).
 - (a) REPLACE armature (470) if bar-to-bar run-out is more than the limits of the FITS AND CLEARANCES section before refinishing the armature. NO REPAIR IS PERMITTED.
- (2) Examine the part IAW procedures found in Paragraph 3.A.
 - (a) DISCARD the part if cracks or fretting and/or corrosion on mating surface are found.
 - (b) REPAIR if minor dents, scratches and nicks or gouging, scoring or glazing on mating surfaces are found.
- (3) Examine bearing journals for gouging, scoring or glazing.
 - (a) REPAIR bearing journals if damaged.
 - (b) REPLACE armature if damage is not repairable.
- (4) Measure bearing journal diameters, A and C, as shown in Figure 5013.
 - (a) REPAIR bearing journals if they do not meet limits in FITS AND CLEARANCES section. Refer to SPD 1000.
 - (b) REPLACE armature if damage is not repairable.
- (5) Measure commutator diameter B, as shown in Figure 5013.
 - (a) REPLACE armature if minimum commutator diameter "B" is below limit in FITS AND CLEARANCES section.
- (6) Examine internal spline for rounding, stripping, or uneven wear.
 - (a) Check the drive spline for too much wear by engaging a new drive shaft into the armature internal drive spline. Reject the armature shaft if too much play is found when engaging the drive shaft.

NOTE: An armature spline with too much wear will not engage tightly with a new drive shaft.
 - (b) REPLACE armature if damaged.

CHECK

24-30-63



Component Maintenance Manual with Illustrated Parts List
DC Starter-Generator, 23046 Series I

- (7) Examine commutator undercut for broken, damaged or missing mica.
 - (a) REPLACE armature if damaged.
- (8) Examine commutator bars for burning and discoloration.
 - (a) REPAIR commutator if damaged.
 - (b) REPLACE armature if damage is not repairable.
- (9) Measure commutator undercut.
 - (a) REPAIR commutator if mica undercut depth (before recut) is below limits in FITS AND CLEARANCES section.
 - (b) REPLACE armature if damage is not repairable.
- (10) After the commutator is refinished, measure commutator bar-to-bar and total indicator reading (TIR) run-out in a full circumference. Support the armature (470) on two "V" blocks.
 - (a) Measure commutator bar-to-bar run-out.
 - 1 Resurface armature if bar-to-bar run-out is more than the limits of the FITS AND CLEARANCES section.
 - (b) Measure commutator TIR run-out.
 - 1 Resurface armature if TIR is more than the limits of the FITS AND CLEARANCES section.
- (11) Check armature balance at two planes using a dynamic balancer. Refer to SPD 1001.
 - (a) REPAIR armature if balance does not meet limit in FITS AND CLEARANCES section.
- (12) Test armature windings for shorts using a growler and iron feeler. Refer to Paragraph 2.A. of this section for additional information about use of growler during inspection.
 - (a) REPLACE the armature (470) if a short exists.

CHECK

24-30-63

Page 5019
Mar 23/10



Component Maintenance Manual with Illustrated Parts List DC Starter-Generator, 23046 Series I

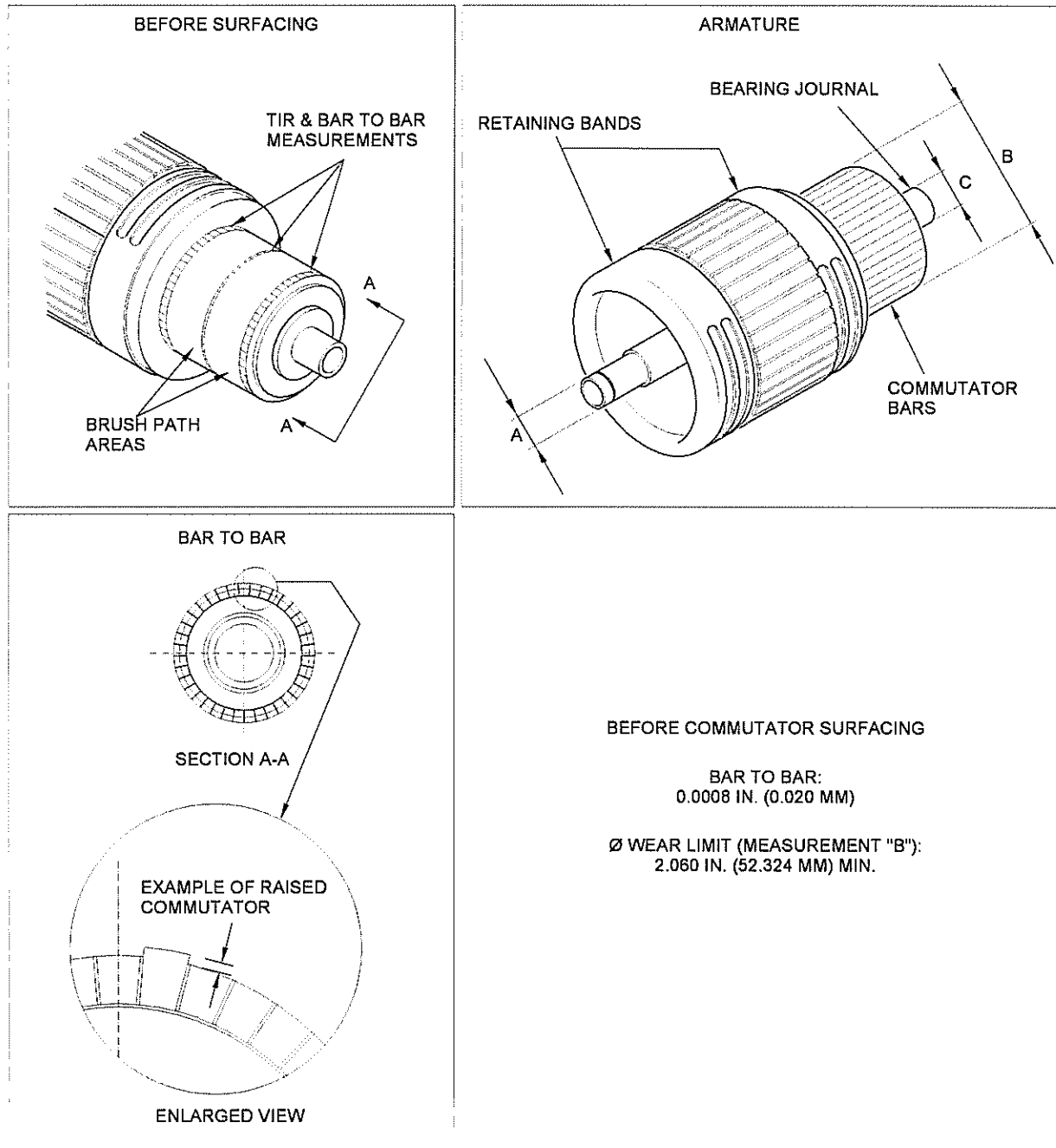


Figure 5013 - Armature Check

CHECK

24-30-63

Page 5020
Mar 23/10



Component Maintenance Manual with Illustrated Parts List
DC Starter-Generator, 23046 Series I

(13) Perform a dielectric test.

WARNING: BEFORE YOU USE THE HIGH VOLTAGE ELECTRICAL EQUIPMENT, MAKE SURE THAT THE POWER SWITCH IS IN THE 'OFF' POSITION. MAKE SURE THAT NO PERSON TOUCHES THE EQUIPMENT OR THE PROBES. THIS WILL PREVENT DEATH OR INJURY FROM ELECTRIC SHOCK.

CAUTION: FAILURE TO TURN OFF DIELECTRIC TESTER POWER BEFORE CONNECTING OR DISCONNECTING HIGH VOLTAGE ELECTRICAL LEADS CAN CAUSE SERIOUS DAMAGE TO THE ARMATURE.

CAUTION: ARMATURE (470) MUST BE THOROUGHLY CLEAN BEFORE PERFORMING A DIELECTRIC TEST.

- (a) With power OFF, connect positive test lead of dielectric tester to armature (470) shaft.
- (b) With power OFF, put negative test lead on commutator bar.
- (c) With dielectric tester output voltage at 0, turn power ON.
- (d) At a rate not to be more than 100 V/sec., slowly adjust output voltage to 250 V RMS, commercial frequency for one minute. Slowly decrease voltage back to 0.
- (e) Turn dielectric tester power OFF.
- (f) Remove negative test lead.
- (g) Connect negative lead to retaining band.
- (h) Repeat dielectric test between retaining bands and commutator bars.
- (i) Acceptance limits:
 - 1 Arcing as evidenced by flashover (surface discharge), spark over (air discharge), breakdown (puncture discharge), or leakage current that is more than 2mA is evidence of damp, dirty, weak or defective components and constitutes a failure.

If armature (470) fails dielectric test, clean armature and repeat dielectric test.

- 2 If armature (470) fails dielectric test after cleaning, replace armature.

CHECK

24-30-63

Page 5021
Mar 23/10



Component Maintenance Manual with Illustrated Parts List
DC Starter-Generator, 23046 Series I

S. Bearing and brush support assembly (-480). See Figure 5014 and Figure 5015.

- (1) Examine the part IAW the procedures found in Paragraph 3.A.
 - (a) DISCARD the part if cracks or fretting and/or corrosion on the mating surface are found.
 - (b) REPAIR if minor dents, scratches and nicks or gouging, scoring or glazing on the mating surfaces (marked 1, 2, 3, and 6 on Figure 5014) are found.
- (2) If the visual inspection finds indications which can be cracks, continue the inspection under 10X magnification.
 - (a) If visual indications of cracks or other structural defects are observed under 10X magnification, DISCARD the part.
- (3) If the visual inspection under 10X magnification finds indications which can be cracks perform a liquid penetrant inspection IAW Paragraph 5.B.

NOTE: After liquid penetrant inspection, the part(s) must have the liquid penetrant removed to prevent interference with processing or with service requirements. Refer to the CLEANING section.

 - (a) REPLACE if damage is found.
- (4) Measure bearing liner diameter 'A' using a gage with an accuracy of ± 0.00004 inch (0,0010 mm). See Figure 5014.
 - (a) DISASSEMBLE the bearing and brush support assembly (-480) if damage exists.
 - (b) REPAIR if bearing liner diameter does not meet the limits in the FITS AND CLEARANCES section.
- (5) Examine the helical coil inserts (540, 530) (marked 4 and 5 on Figure 5014) for damage.
 - (a) REPAIR helical coil inserts if damage is found.
 - (b) REPAIR damaged parts.
- (6) Hand tighten a machine screw into the thread in each brush holder (610) to make sure that the self-locking feature is functional. The machine screw should bind in the thread before it is fully engaged.
 - (a) REPAIR the brush holder (610) if the self-locking feature does not function properly, but the brush holder has a helical coil insert.
 - (b) REPLACE the brush holder (610) if the self-locking feature does not function properly, but the brush holder does not have a helical coil insert.

CHECK

24-30-63

Page 5022
Mar 23/10



Component Maintenance Manual with Illustrated Parts List
DC Starter-Generator, 23046 Series I

- (7) Examine the brush holders (610), including the brush spring supports and center supports for cracks, warping, and discoloration caused by electrical arcing.
 - (a) DISCARD the brush holder (610) if damage exists.
- (8) Measure brush spring (600) force using a pull scale with harness.
See Figure 5015.

NOTE: When measuring brush spring force, readings can vary widely depending on where measuring device touches spring. Measurements must be taken from curved tip of spring at point where spring touches brush.

- (a) Take six measurements.
 - (b) Calculate an average from the measurements.
 - (c) Discard brush spring if not within limits of FITS AND CLEARANCES.
- (9) Perform a dielectric test.

WARNING: BEFORE YOU USE THE HIGH VOLTAGE ELECTRICAL EQUIPMENT, MAKE SURE THAT THE POWER SWITCH IS IN THE 'OFF' POSITION. MAKE SURE THAT NO PERSON TOUCHES THE EQUIPMENT OR THE PROBES. THIS WILL PREVENT DEATH OR INJURY FROM ELECTRICAL SHOCK

CAUTION: FAILURE TO TURN OFF DIELECTRIC TESTER POWER BEFORE CONNECTING OR DISCONNECTING HIGH VOLTAGE ELECTRICAL LEADS CAN CAUSE SERIOUS DAMAGE TO BEARING AND BRUSH SUPPORT ASSEMBLY.

CAUTION: BEARING AND BRUSH SUPPORT ASSEMBLY (-480) MUST BE THOROUGHLY CLEAN BEFORE PERFORMING A DIELECTRIC CHECK.

- (a) With power OFF, connect positive test lead of dielectric tester to metal surface of brush holder.
 - (b) With power OFF, connect negative test lead to uncoated surface of bearing and brush support.
 - (c) With dielectric tester output voltage at 0, turn power ON.
 - (d) Slowly adjust output voltage (at a rate not more than 100 V/sec.) to 250 V RMS, commercial frequency for one minute. Adjust voltage back to 0.
 - (e) Turn dielectric tester power OFF.
 - (f) Disconnect test leads from bearing and brush support assembly.

CHECK

24-30-63

Page 5023
Mar 23/10



Component Maintenance Manual with Illustrated Parts List
DC Starter-Generator, 23046 Series I

(g) Acceptance limits:

- 1 Arcing as evidenced by flashover (surface discharge), spark over (air discharge), breakdown (puncture discharge), or leakage current that is more than 2mA is evidence of damp, dirty, weak or defective components and constitutes a failure.

If bearing and brush support assembly (-480) fails dielectric test, clean bearing and brush support assembly, and repeat dielectric test.

- 2 If bearing and brush support assembly fails dielectric test after cleaning, disassembly bearing and brush support assembly, replace all insulating materials (i.e., washers, sleeves, plates), and repeat dielectric test.

If bearing and brush support assembly (-480) fails dielectric test, **REPLACE** bearing and brush support assembly.

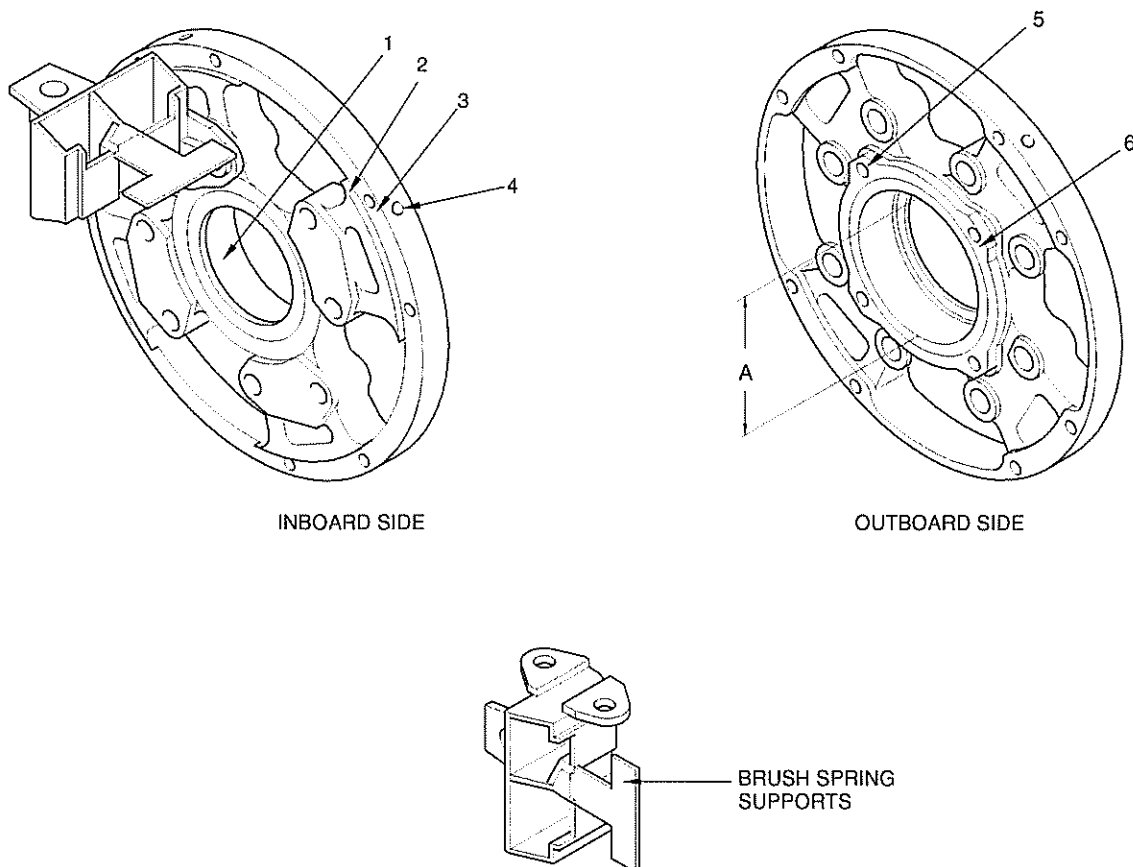


Figure 5014 - Bearing and Brush Support Assembly

CHECK

24-30-63

Page 5024
Mar 23/10



Component Maintenance Manual with Illustrated Parts List
DC Starter-Generator, 23046 Series I

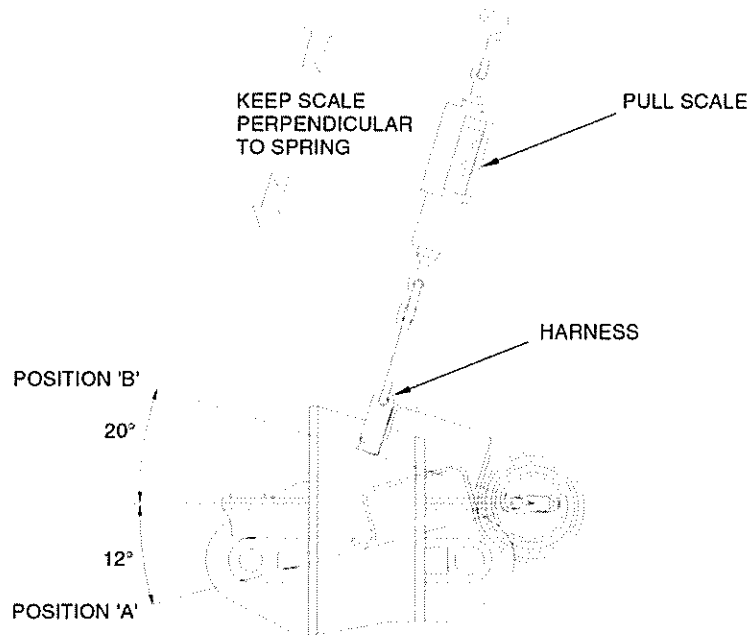


Figure 5015 - Determine Spring Force

T. Bearing retainer (660)

- (1) Examine part IAW procedures found in Paragraph 3.A.
 - (a) DISCARD part if damaged.
- (2) Carefully examine mating surfaces one and two for gouging, scoring, or glazing. See Figure 5016.
 - (a) DISCARD part if damaged.

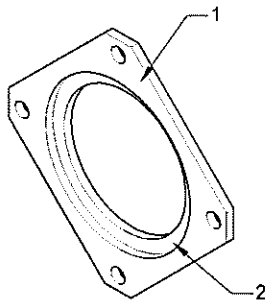


Figure 5016 - Inspecting Bearing Retainer

CHECK

24-30-63

Page 5025
Mar 23/10



Component Maintenance Manual with Illustrated Parts List
DC Starter-Generator, 23046 Series I

U. Stator and housing assembly (680). See Figure 5017.

NOTE: Use of Glyptol on either the armature (470) or stator and housing assembly (680) stator windings is strictly prohibited. Glyptol may be used on internal surfaces of the housing only, and only where other coatings are not required. Units having Glyptol coating on any other surfaces shall have these parts replaced prior to returning the units to service.

- (1) Examine the part IAW the procedures found in Paragraph 3.A. and Paragraph 3.B.
 - (a) DISCARD the part if cracks or other major damage exists.
 - (b) REPAIR thread damage of two turns or less.
 - (c) DISCARD the part if thread damage is more than two turns.
 - (d) REPAIR the part if other damage exists.
- (2) Examine the brush leads and stator leads for damage.
 - (a) If brush lead damage is more than 5 percent of the brush lead, REPLACE the stator and housing assembly (680).
 - (b) If stator lead damage is found, REPLACE the stator and housing assembly (680).
- (3) Perform a dielectric test.

WARNING: BEFORE YOU USE THE HIGH VOLTAGE ELECTRICAL EQUIPMENT, MAKE SURE THAT THE POWER SWITCH IS IN THE 'OFF' POSITION. MAKE SURE THAT NO PERSON TOUCHES THE EQUIPMENT OR THE PROBES. THIS WILL PREVENT DEATH OR INJURY FROM ELECTRIC SHOCK.

CAUTION: FAILURE TO TURN OFF THE DIELECTRIC TESTER POWER BEFORE CONNECTING OR DISCONNECTING HIGH VOLTAGE ELECTRICAL LEADS CAN CAUSE SERIOUS DAMAGE TO THE STATOR AND HOUSING ASSEMBLY.

CAUTION: STATOR AND HOUSING ASSEMBLY (680) MUST BE THOROUGHLY CLEAN BEFORE PERFORMING A DIELECTRIC CHECK.

CAUTION: USE A DUMMY TERMINAL BLOCK, OR DISCONNECT GROUNDING LEAD FROM ATTACHED TERMINAL BLOCK BEFORE TEST.

- (a) If grounding lead is disconnected from attached terminal block, cover the lead with electrical tape.
- (b) Jumper all stator leads together.

CHECK

24-30-63

Page 5026
Mar 23/10



Component Maintenance Manual with Illustrated Parts List
DC Starter-Generator, 23046 Series I

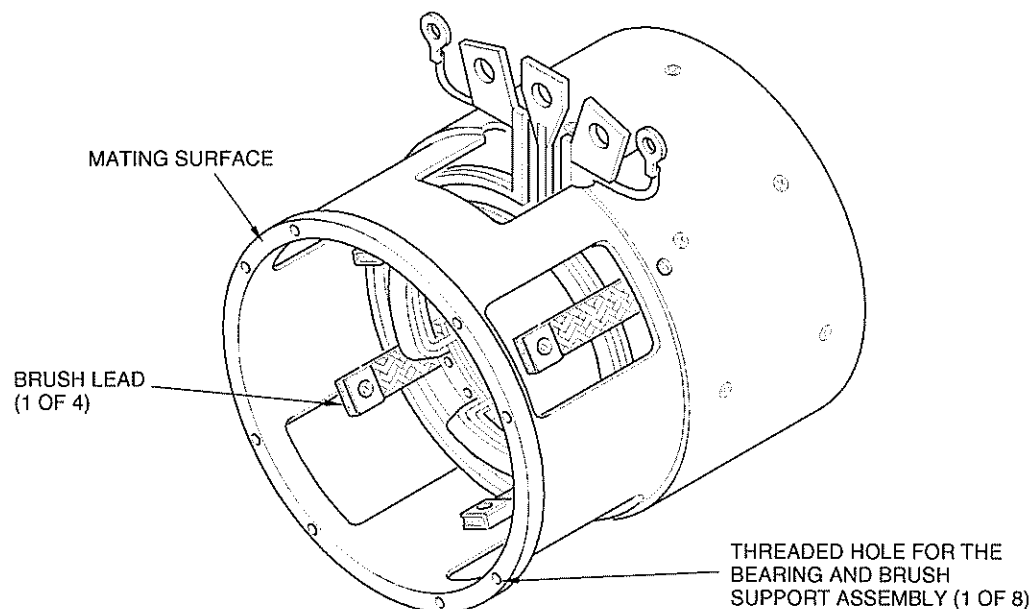
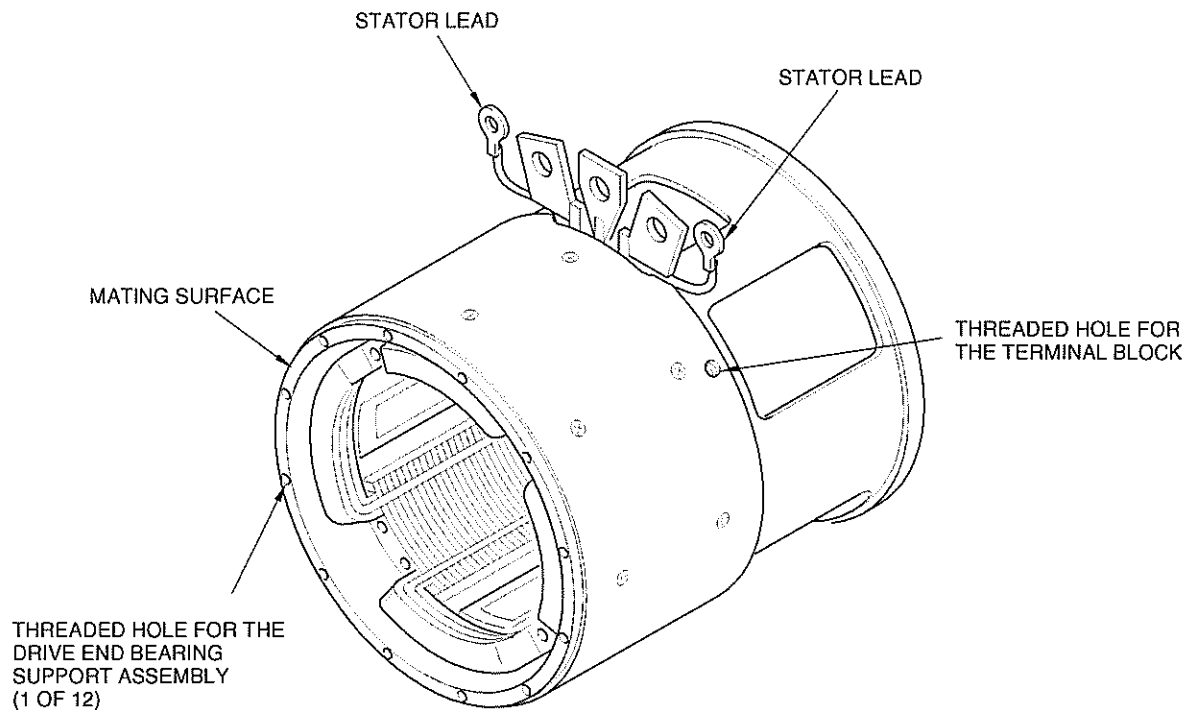


Figure 5017 - Stator and Housing Assembly

CHECK

24-30-63

Page 5027
Mar 23/10



Component Maintenance Manual with Illustrated Parts List
DC Starter-Generator, 23046 Series I

- (c) With power OFF, connect positive test lead of dielectric tester to jumpered stator leads.
- (d) With power OFF, connect negative test lead to uncoated surface of housing.
- (e) With dielectric tester output voltage at 0, turn power ON.
- (f) At a rate not to be more than 100 V/sec., slowly adjust output voltage to 250 V RMS, commercial frequency for one minute. Slowly decrease voltage back to 0.
- (g) Turn dielectric tester power OFF.
- (h) Disconnect test leads.
- (i) Disconnect jumper from stator leads.
- (j) Acceptance limits:
 - 1 Arcing as evidenced by flashover (surface discharge), spark over (air discharge), breakdown (puncture discharge), or leakage current that is more than 2mA is evidence of damp, dirty, weak or defective components and constitutes a failure.

If stator and housing assembly (680) fails dielectric test, clean stator and housing assembly and repeat dielectric test.
 - 2 If stator and housing assembly (680) fails dielectric test after cleaning, replace stator and housing assembly.

V. Terminal block (690). See Figure 5018.

CAUTION: THE USE OF RE-MANUFACTURED TERMINAL BLOCKS IS NOT AUTHORIZED BY GOODRICH CORPORATION. DAMAGED TERMINAL BLOCKS (OTHER THAN THOSE WITH REPAIRABLE THREAD DAMAGE) MUST BE DISCARDED.

- (1) Examine the part IAW the procedures found in Paragraph 3.A. and Paragraph 3.B.
 - (a) DISCARD the part if damage, other than thread damage, exists.
 - (b) REPAIR the part if thread damage of two turns or less exists.
 - (c) REPLACE the terminal block (690) if thread damage of more than two turns exists.

CHECK

24-30-63



Component Maintenance Manual with Illustrated Parts List
DC Starter-Generator, 23046 Series I

- (2) Using a LCR meter, measure the capacitance values between terminals B+ and E-, between C and ground and between B and ground at a frequency of 110 to 130 Hz, 77° F (25° C) $\pm$ 10%.
- (a) Measure terminals B+ to E-
- 1 DISCARD the part if capacitance values fall outside the required limits in the FITS AND CLEARANCES section.
- (b) To check whether the capacitor is holding a charge, connect an ammeter in series with the capacitor and voltage source.
- 1 Gradually apply voltage at the rated value of the capacitor listed in the FITS AND CLEARANCES section across terminals B+ to E- and measure current.
- a DISCARD the part if there is more than 1 μ A of leakage.
- (c) Discharge the capacitor after testing is completed using a current limiting resistor.
- (d) Repeat steps a thru c for C to ground.
- (e) Repeat steps a thru c for B to ground but measure the capacitance value at a frequency of 900 to 1100 Hz.

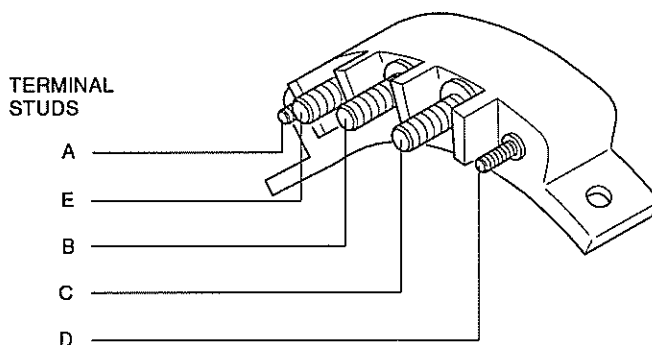


Figure 5018 - Terminal Block

CHECK

24-30-63

Page 5029
Mar 23/10



Component Maintenance Manual with Illustrated Parts List
DC Starter-Generator, 23046 Series I

9. Terminology

Table 5002 lists the definitions of terms used to describe damage conditions.

Term	Definition	Probable Cause
Corrosion	The chemical or electrochemical reaction between a material, usually a metal, and its environment that produces a deterioration of the material and its properties.	Environmental condition that causes deterioration.
Crack	A break in material.	Severe stress from overload or shock; possible extension of a scratch.
Dent	A small, smoothly rounded depression.	A sharp blow or too much pressure.
Fretting	Wear that occurs between tight-fitting surfaces subjected to cyclical relative motion of extremely small amplitude. Usually, fretting is accompanied by corrosion, especially of the very fine wear debris.	Vibration between mating surfaces.
Fretting corrosion	The accelerated deterioration at the interface between contacting surfaces as the result of corrosion and slight oscillatory movement between the two surfaces.	Vibration between mating surfaces.
Galling	Transfer of metal from one surface to another.	Result of localized lubrication break-down between sliding surfaces.
Glazing	Smoothing and creep of a surface.	Result of localized lubrication break-down between sliding surfaces.
Gouging	Removal of surface material, typified by rough and deep depressions.	Protruding objects, incorrectly aligned.
Nick	A sharp bottomed depression that can have rough outer edges.	Dropping, banging.
Rounding	Removal of surface metal at corners of dulling of edges.	Result of abrasion, vibration, or poor mating surfaces.

Table 5002 - Terminology

CHECK

24-30-63

Page 5030
Mar 23/10



Component Maintenance Manual with Illustrated Parts List
DC Starter-Generator, 23046 Series I

Term	Definition	Probable Cause
Scoring	A deep scratch following a path of part travel.	Result of localized lubrication break-down between sliding surfaces.
Scratch	A very shallow furrow or irregularity, usually longer than wide.	Movement of a sharp object across the surface.
Short	Evidenced by sparking, arcing, or high current. Electrical condition where two conductors are connected by a path of very low resistance.	The result when two carrying conductors are connected by a path of negligible resistance.
Stripped Thread	Thread of a nut, stud, bolt, or screw damaged by tearing away parts of the thread.	Incorrect installation of threaded mating parts.
Tear	Parting of material.	Too much tension, caused by external force.
Wear	Slow removal of material. Wear may not be visible to the naked eye.	Result of abrasive substances contacting rolling surfaces and acting as a lapping compound.

Table 5002 - Terminology (Continued)

CHECK

24-30-63

Page 5031
Mar 23/10



Component Maintenance Manual with Illustrated Parts List
DC Starter-Generator, 23046 Series I

THIS PAGE INTENTIONALLY LEFT BLANK



Component Maintenance Manual with Illustrated Parts List
DC Starter-Generator, 23046 Series I

REPAIR

1. Introduction

CAUTION: ANY USE OF PARTS, MATERIALS, OR PROCESSES NOT AUTHORIZED BY GOODRICH CORPORATION FOR MAINTENANCE OR OVERHAUL OF THE STARTER-GENERATOR CAN AFFECT CONTINUED FLIGHT WORTHINESS OR INVALIDATE CERTIFICATION.

Repairs are limited to:

- repairing damaged surfaces
- repairing damaged threads
- replacing helical coil inserts
- bearing liner and journal restoration
- commutator refinishing
- armature balancing
- restoring surface coatings of parts and assemblies
- correcting output voltage polarity
- replacing brush access cover insulating tape
- replacing brush access cover self-locking blind rivet nut
- replacing terminal lugs
- removal and replacement of mounting adapter guide pin

Repair operations not outlined or referenced in this manual are not authorized by Goodrich Corporation.

In addition to specific repairs authorized by Goodrich Corporation in this manual, standard repair procedures for starter-generators are described in the following Goodrich Corporation Standard Practice Documents (SPD).

- Selective (Brush Plating), Electrodeposition - Refer to SPD 1000.
- Armature Balancing for DC Starter-Generators - Refer to SPD 1001.
- Surface Repair and Coating - Refer to SPD 1002.
- Plastic Media Blasting - Refer to SPD 1003.
- Brush Holder Alignment Fixtures - Refer to SPD 1004.
- Brush Installation, Seating, and Run-In - Refer to SPD 1006.

REPAIR

24-30-63

Page 6001
Mar 23/10



Component Maintenance Manual with Illustrated Parts List
DC Starter-Generator, 23046 Series I

2. Repair Tools

In addition to normal shop tools, specific tools for repair of the starter-generator are listed in Table 6001. The tools listed are not available for purchase from Goodrich Corporation.

NOTE: Equivalent substitutes may be used for the tools listed in Table 6001.

WARNING: YOU MUST OBEY THE TOOL/EQUIPMENT MANUFACTURERS WARNINGS AND CAUTIONS SHOWN ON PACKAGING, CONTAINERS AND/OR INSTRUCTION LEAFLETS. IF YOU DO NOT, YOU CAN CAUSE DEATH OR INJURY TO PERSONS AND DAMAGE TO EQUIPMENT OR PROPERTY.

Tool Description	Source/Figure Reference
Arbor Press	Commercially Available
Bump Switch	Rating: 30 VDC, 10 A
Commutator Turning Fixture	Figure 9006
Crimp Tool	Commercially Available
Helical Coil Insert Removal and Insertion Tool	Commercially Available
India Stone	Commercially Available
PlusNut® Fastener Header P/N C1000-1032	Figure 9016
6 V Battery of Equivalent DC Power Source	Commercially Available
Thread Chasers	Commercially Available
V-blocks	Commercially Available

Table 6001 - Repair Tools



Component Maintenance Manual with Illustrated Parts List
DC Starter-Generator, 23046 Series I

3. Repair Materials

Materials necessary for starter-generator repair are listed in Table 6002.

NOTE: Equivalent substitutes can be used for the materials listed in Table 6002. Repair materials are not available from Goodrich Corporation. All items can be purchased commercially.

WARNING: BEFORE USING ANY OF THE MATERIALS LISTED IN Table 6002, BE AWARE OF ALL HANDLING, STORAGE, AND DISPOSAL PRECAUTIONS RECOMMENDED BY THE MANUFACTURER OR SUPPLIER. FAILURE TO COMPLY WITH MANUFACTURER'S OR SUPPLIER'S RECOMMENDATIONS MAY RESULT IN SERIOUS INJURY, PHYSICAL DISORDER OR DEATH.

Item	Description/Material Specification	Source (CAGE Code)
Abrasive paper	400/600 grit (non-aluminum oxide only)	Commercially Available
Black Enamel, Color 17038	TT-E-489G, Class B	Commercially Available
Chemical Film Solution Alodine 1200 or Pen, Touch-N-Prep	Chemical Film Solution MIL-C-5541, Class 1A or Class 3 or Alodine 1132 Touch-N-Prep Pen IAW MIL-DTL-81706B	Henkel Surface Technology Madison Heights, MI 48071-0000 www.hstna.com (V1N6B3)
Coating, Zinc Phosphate	TT-C-490, Type I	Commercially Available
Dichromate Touch Up Solution	ASM-M-3171, Type III	Commercially Available
Epoxy Primer, Yellow	MIL-P-23377F, Type I	Commercially Available
Isopropyl Alcohol	TT-I-735, Grade A See <u>WARNING</u> before using this material. Flash Point: 53° F (12° C), FLAMMABLE Refer to the Material Safety Data (MSD) Sheet for the material for additional safety information.	Commercially Available
Oil, Machine	N/A	Commercially Available
Pads, Cleaning	Lint-free cotton	Commercially Available

Table 6002 - Repair Materials

REPAIR

24-30-63

Page 6003
Mar 23/10



Component Maintenance Manual with Illustrated Parts List
DC Starter-Generator, 23046 Series I

Item	Description/Material Specification	Source (CAGE Code)
Primer, Zinc Chromate	TT-P-1757 Composition G, Color Yellow	Commercially Available
Tape, Insulating	P/N SG13-06 (R) 8 mil, PTFE coated fiberglass, acrylic adhesive	Saint Gobain New Haven, CT www.saint-gobain.com/us (V1DS07)
White Polyurethane Enamel, Color 17925	MIL-PRF-85285, Type I	Commercially Available

Table 6002 - Repair Materials (Continued)

4. Surface Repair

Repair formed metal components that have surface damage or corrosion and qualify for repair.

A. Repair procedure

- (1) Remove corrosion, sharp edges, burrs, nicks, or scratches from cast surfaces, machined mating surfaces, and sealing surfaces with india stone or abrasive paper. Remove only sharp edges or burrs that are above surface of part.
- (2) Remove minor raised edges, burrs, nicks, or scratches on any polished or load-bearing surface. Remove only material that extends above polished surface. Make sure that dimensional tolerances are within limits.

5. Thread Repair

A. Repair damaged threads as needed.

CAUTION: DO NOT USE THREAD CUTTING DIE, NON-REPAIRABLE DAMAGE CAN RESULT.

- (1) Use thread chaser to repair damaged threads.
- (2) Use india stone to remove remaining sharp edges or burrs.
- (3) To prevent corrosion apply a light coating of machine oil to repaired threads.

6. Helical Coil Insert Replacement

NOTE: Unless otherwise specified, item numbers in parentheses () refer to items found in Figure 10001 of the ILLUSTRATED PARTS LIST

A. Remove and replace damaged helical coil inserts (530, 540) in anti-drive end end bell (520) as needed.

- (1) Remove damaged helical coil insert with a helical coil insertion/removal tool.
- (2) Clean hole for helical coil insert. Refer to the CLEANING section.

REPAIR

24-30-63



Component Maintenance Manual with Illustrated Parts List
DC Starter-Generator, 23046 Series I

WARNING: KEEP ZINC CHROMATE PRIMER AWAY FROM SOURCES OF HEAT, SPARKS AND FLAME. ZINC CHROMATE PRIMER IS FLAMMABLE.

WARNING: DO NOT GET ZINC CHROMATE PRIMER ON YOUR SKIN AND DO NOT BREATHE THE FUMES. ZINC CHROMATE PRIMER IS A POISONOUS MATERIAL.

- (3) Apply thin coat of zinc chromate primer to outer surface of replacement helical coil insert.
- (4) Install helical coil insert to the depth shown in Figure 6001 below part surface while primer is still wet.
- (5) Break off helical coil insert installation tang.

WARNING: KEEP CHEMICAL FILM AWAY FROM SOURCES OF HEAT, SPARKS AND FLAME. CHEMICAL FILM IS FLAMMABLE.

WARNING: DO NOT GET CHEMICAL FILM ON YOUR SKIN AND DO NOT BREATHE THE FUMES. CHEMICAL FILM IS A POISONOUS MATERIAL.

- (6) Apply chemical film solution MIL-C-5541, Class 3 to bare metal.

B. Remove and replace damaged helical coil inserts (-620) in brush holder (-610B) as needed.

- (1) Remove damaged helical coil insert with a helical coil insertion/removal tool.
- (2) Clean hole for new helical coil insert. Refer to the CLEANING section.
- (3) Install helical coil insert in brush holder to the depth shown in Figure 6001 below part surface.
- (4) Break off helical coil insert installation tang.

WARNING: KEEP CHEMICAL FILM AWAY FROM SOURCES OF HEAT, SPARKS AND FLAME. CHEMICAL FILM IS FLAMMABLE.

WARNING: DO NOT GET CHEMICAL FILM ON YOUR SKIN AND DO NOT BREATHE THE FUMES. CHEMICAL FILM IS A POISONOUS MATERIAL.

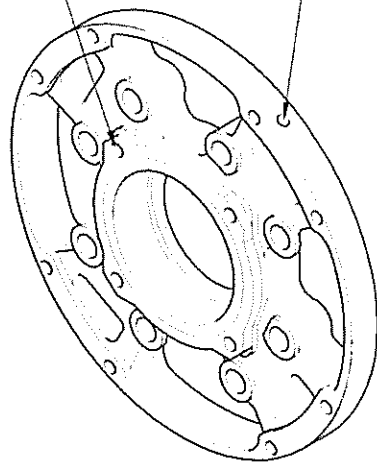
- (5) Apply chemical film solution MIL-C-5541, Class 3 to bare metal.



Component Maintenance Manual with Illustrated Parts List
DC Starter-Generator, 23046 Series I

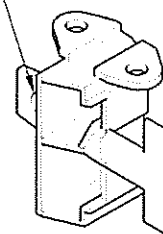
HELICAL COIL INSERT
IAW MS33537.
INSTALL 0.03 TO 0.05 INCH
(0,8 TO 1,3MM) BELOW SURFACE

HELICAL COIL INSERT IAW
MS33537 INSTALL 0.03 TO 0.05 INCH
(0,8 TO 1,3MM) BELOW SURFACE



ANTI-DRIVE END BELL

HELICAL COIL INSERT
INSTALL 0.02 TO 0.04 INCH
(0,5 TO 1,0 MM) BELOW
SURFACE



BRUSH HOLDER

Figure 6001 - Helical Coil Insert Replacement

7. Bearing Journal and Liner Restoration

A. Bearing journals

- (1) If the armature shaft bearing journals are worn beyond limits in the FITS AND CLEARANCES section, restoration may be possible by electrodeposition plating in accordance with Goodrich Corporation Standard Practice Document (SPD) 1000.

NOTE: If armature shaft is out of concentricity, machine shaft within concentricity before plating.

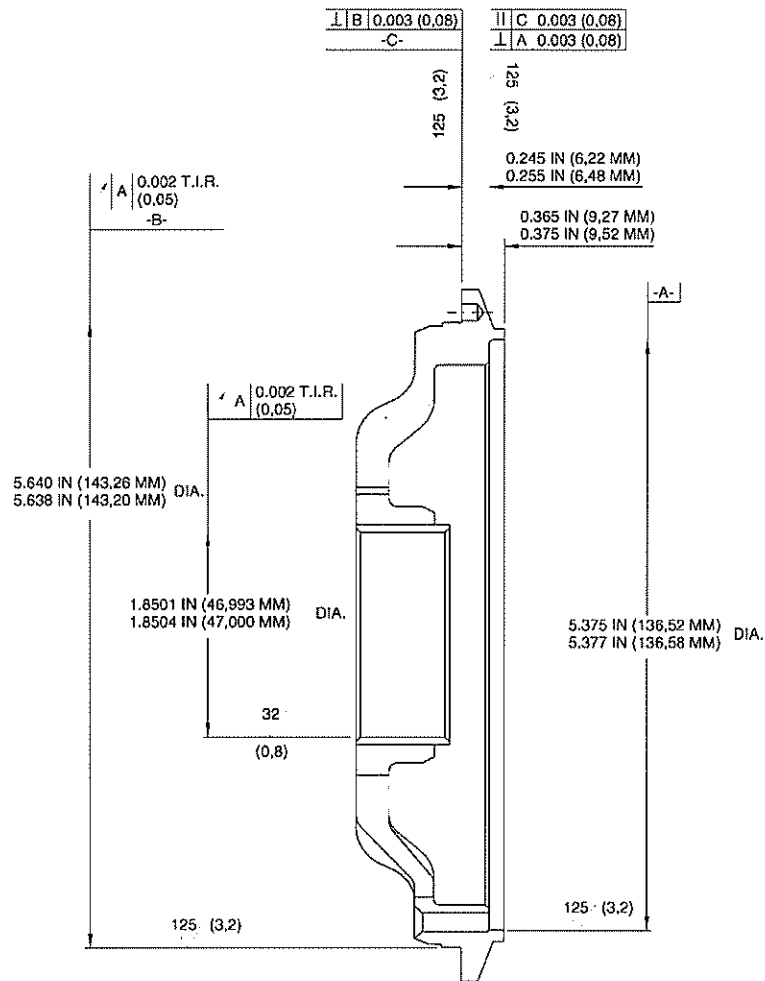
- (2) Maximum allowable thickness of plating shall be 0.01 inch (0,254 mm), or 0.02 inch (0,508 mm) measured across a diameter.
- (3) Armature shaft bearing journals may be plated to final dimensions, or overplated then ground to final dimensions listed in the FITS AND CLEARANCES section.



Component Maintenance Manual with Illustrated Parts List
DC Starter-Generator, 23046 Series I

B. Bearing liner restoration

- (1) If bearing liners are worn beyond limits in the FITS AND CLEARANCES section, restoration may be possible by electrodeposition plating in accordance with Goodrich Corporation Standard Practice Document (SPD) 1000.
- (2) Components not in compliance with the dimensions in Figure 6002 or Figure 6003 must be reworked or replaced.
- (3) The bearing liners in drive end end bells, part no. 23046-1191, 23046-1630 and 23046-2080 cannot be restored because of the bearing liner hardness.



UNLESS OTHERWISE SPECIFIED
ALL DIMENSIONS ARE IN INCHES
FOLLOWED BY MILLIMETERS
IN PARENTHESES

Figure 6002 - Drive End End Bell Machining Specifications

REPAIR

24-30-63



Component Maintenance Manual with Illustrated Parts List DC Starter-Generator, 23046 Series I

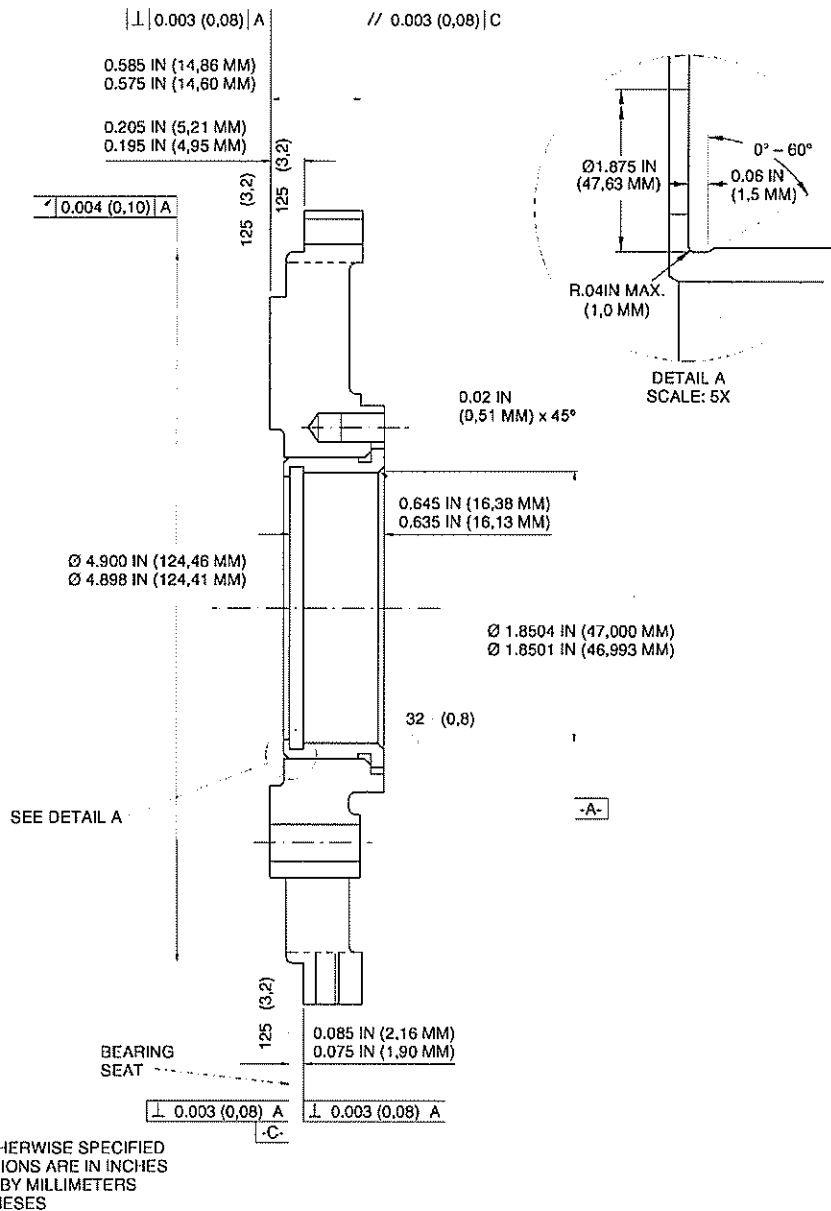


Figure 6003 - Anti-Drive End End Bell Machining Specifications

8. Commutator Refinishing

NOTE: Unless otherwise specified, numbers in parentheses () refer to item numbers in Figure 10001 of ILLUSTRATED PARTS LIST.

If inspection reveals that commutator is rough, pitted, scored or burned; refinish commutator on a lathe that is accurately set up and adjusted.

REPAIR

24-30-63

Page 6008
Mar 23/10



Component Maintenance Manual with Illustrated Parts List
DC Starter-Generator, 23046 Series I

A. Setup

WARNING: WHILE PREPARING MACHINE FOR COMMUTATOR RE-FINISHING, MAKE SURE THAT POWER TO LATHE IS SHUT OFF OR "LOCKED OUT". USE ALL SAFETY PRECAUTIONS WHILE WORKING WITH OR AROUND POWER MACHINE TOOLS. WEAR EYE PROTECTION.

- (1) Install bearing (450) into the commutator turning fixture.
- (2) Install commutator turning fixture into tail stock of lathe.

CAUTION: AVOID TOUCHING COMMUTATOR WITH BARE HANDS. SKIN ACIDS AND OILS CAN CONTAMINATE CONDUCTING SURFACES, CAUSING CORROSION AND POOR ELECTRICAL CONTACT.

- (3) Insert commutator end of armature into commutator turning fixture.
- (4) Position tail stock so that drive end of armature shaft slides into head stock chuck.
- (5) Secure tail stock.
- (6) Tighten head stock chuck until armature is secure.

B. Repair procedure. Refer to Figure 6004.

CAUTION: DO NOT REMOVE EXCESS MATERIAL FROM COMMUTATOR. REMOVAL OF TOO MUCH MATERIAL WILL RESULT IN SHORTENED COMMUTATOR LIFE AND REDUCTION OF COMMUTATOR DIAMETER.

CAUTION: DO NOT TOUCH THE COMMUTATOR WITH YOUR BARE HANDS. CONTAMINATION FROM YOUR SKIN CAN CAUSE CORROSION AND UNSATISFACTORY ELECTRICAL CONTACT.

- (1) Cut commutator to a surface finish of 64 to 100 micro inches (1,6 to 2,5 microns) rms. Finish surface must extend from end of full undercut to within 0.00 to 0.06 inches (0,0 to 1,5 mm) of the outboard end of the commutator.

NOTE: The suggested feed rate is 0.006 to 0.007 inches (0,15 to 0,18 mm) per revolution at a surface speed of 1000 to 1500 surface feet per minute (305 to 457 surface meters per minute).



Component Maintenance Manual with Illustrated Parts List
DC Starter-Generator, 23046 Series I

WARNING: WHEN USING COMPRESSED AIR FOR CLEANING OR DRYING, CONTROL PRESSURE TO 29 PSIG (200 KPA) OR LESS. WEAR GOGGLES OR FACE SHIELD TO PROTECT EYES. TAKE PRECAUTIONS TO AVOID INJURY TO OTHER PERSONNEL IN AREA.

CAUTION: MAKE SURE THAT COMPRESSED AIR USED TO CLEAN OR DRY COMPONENTS IS FREE FROM OIL AND WATER. THIS WILL PREVENT CONTAMINATION OF THE COMPONENTS.

- (2) Clean armature (470) surfaces with compressed air, 29 PSIG (200 kPa) maximum.
- (3) Measure the depth of the mica undercut between the commutator bars. Refer to the FITS AND CLEARANCES section for limits.
- (4) If the undercut is out of limits, use a 0.31 to 0.50 inch (7,9 to 12,7 mm) diameter cutter wheel to undercut the mica to a depth of 0.050 to 0.070 inch (1,27 to 1,78 mm) and a width of 0.035 to 0.045 inch (0,89 to 1,14 mm)

NOTE: The beginning of mica undercut must be 0.010 to 0.030 inch (0,25 to 0,76 mm) from the finished face of the risers. All cuts must be equally spaced and parallel with the centerline of the armature shaft within 0.005 inch (0,13 mm) of true position. All mica must be removed from the edges of the undercut.

- (5) Use triangular scraping tool to remove sharp edges and burrs.
- (6) Remove undercut residue from between commutator bars with a soft bristle brush.

WARNING: WHEN USING COMPRESSED AIR FOR CLEANING OR DRYING, CONTROL PRESSURE TO 29 PSIG (200 KPA) OR LESS. WEAR GOGGLES OR FACE SHIELD TO PROTECT EYES. TAKE PRECAUTIONS TO AVOID INJURY TO OTHER PERSONNEL IN AREA.

CAUTION: MAKE SURE THAT COMPRESSED AIR USED TO CLEAN OR DRY COMPONENTS IS FREE FROM OIL AND WATER. THIS WILL PREVENT CONTAMINATION OF THE COMPONENTS.

- (7) Clean the armature (470) surfaces with compressed air, 29 PSIG (200 kPa) maximum.
- (8) After the commutator is refinished, measure the commutator bar-to-bar and total indicator reading (TIR) run-out in full circumference of the commutator. Support the armature (470) on two "V" blocks.
- (9) REPLACE the armature if the damage cannot be repaired.

REPAIR

24-30-63



Component Maintenance Manual with Illustrated Parts List DC Starter-Generator, 23046 Series I

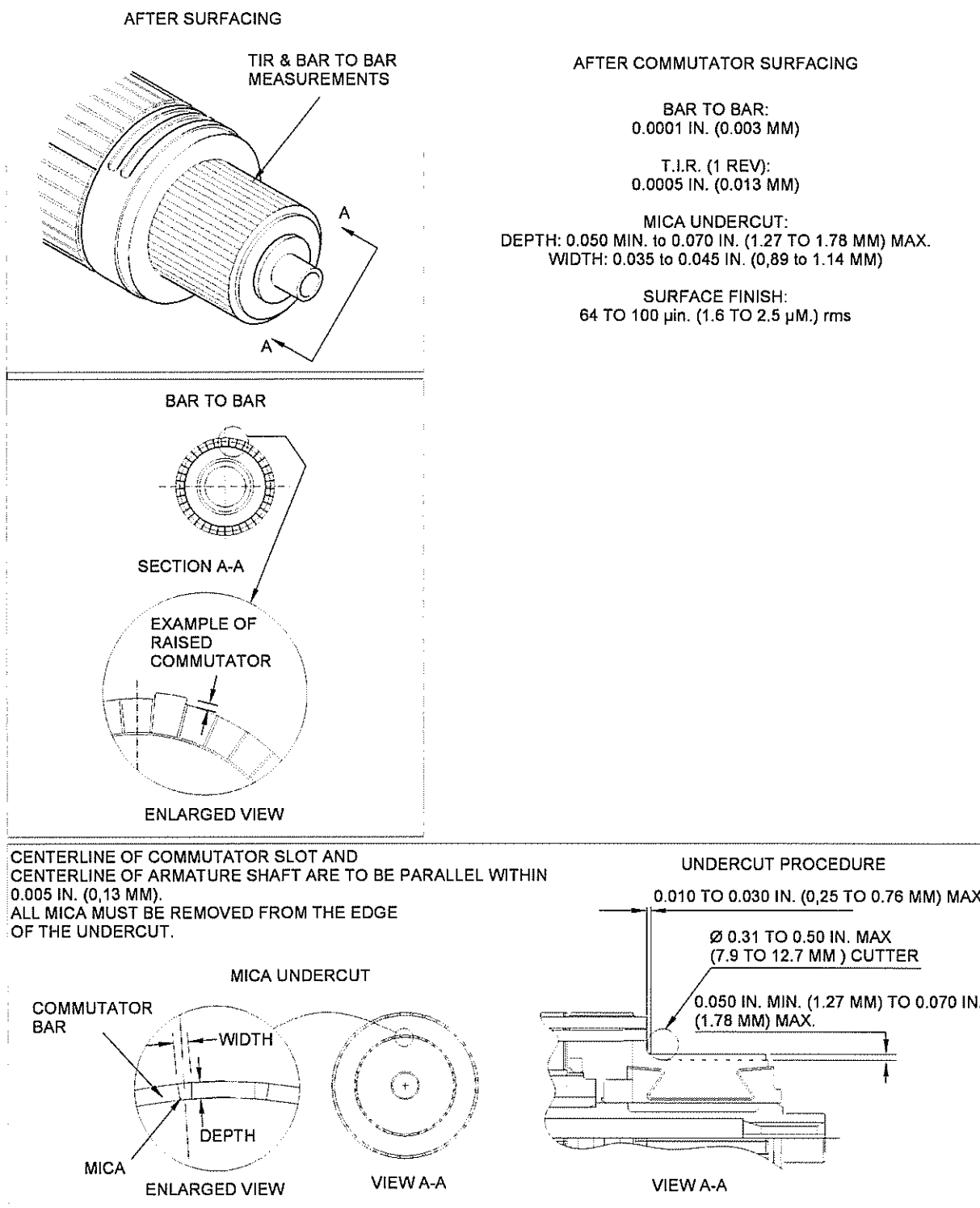


Figure 6004 - Armature Repair

REPAIR

24-30-63

Page 6011
Mar 23/10



Component Maintenance Manual with Illustrated Parts List
DC Starter-Generator, 23046 Series I

9. **Balancing Armature**

NOTE: Unless otherwise specified, numbers in parentheses () refer to item numbers in Figure 10001 of ILLUSTRATED PARTS LIST.

A. **Procedure**

Balance the armature (470) to the requirements listed in the FITS AND CLEARANCES section of this CMM according to the procedure in SPD 1001.

10. **Restoring the Surface Coatings of Parts and Assemblies**

A. **Removal of surface coatings.**

Remove the surface coatings of parts and assemblies that have been damaged using plastic blasting equipment and procedures in accordance with (IAW) SPD 1003.

NOTE: Follow the quality assurance guidelines listed in SPD 1003 for the use of plastic media blasting equipment.

B. **Restoration of surface coatings.**

Restore surface coatings of parts and assemblies with materials and procedures given in Goodrich Corporation SPD 1002.

Clean all parts per CLEANING section of this CMM.

Use Table 6003 to find the correct primer and paint specification when using SPD 1002.

WARNING: CHEMICAL CONVERSION MATERIALS ARE POISONOUS AND CAN CAUSE EXPLOSIONS WHEN MIXED WITH ACIDS, REDUCING AGENTS COMBUSTIBLE AND OXIDIZING MATERIALS. ISOLATED STORAGE OF THESE MATERIALS IS MANDATORY.

WARNING: BEFORE YOU USE CHEMICAL CONVERSION MATERIALS, PUT ON A RESPIRATOR, RUBBER APRON, RUBBER GLOVES AND EYE PROTECTION. THIS WILL PREVENT INJURY FROM SPILLS AND FROM THE FUMES.

WARNING: KEEP CHEMICAL CONVERSION MATERIALS AWAY FROM SOURCES OF HEAT, SPARKS AND FLAME. CHEMICAL CONVERSION MATERIALS ARE FLAMMABLE.



Component Maintenance Manual with Illustrated Parts List
DC Starter-Generator, 23046 Series I

IPL Item Number	Nomenclature	Process	Coating Repair
50	Drive End Pad End Bell	Refinish	Chemical film touch-up IAW MIL-C-5541, Class 3
130	Brush Access Cover	Refinish	Chemical film touch-up IAW MIL-C-5541, Class 3
180	Cover Assembly	Refinish	Chemical film touch-up IAW MIL-C-5541, Class 3
200	Air Inlet Assembly	Refinish/ Paint	Dichromate touch-up IAW SAE ASM-M-3171, Type III. Epoxy primer, yellow. White polyurethane enamel, IAW MIL-PRF-85285, Type 1
240	Fan Cover	Refinish/ Paint	Phosphate IAW TT-C-490. Black enamel IAW TT-E-489G, Class B.
250	Fan	Refinish	Chemical film touch-up IAW MIL-C-5541, Class 3
340	Dampener Backplate	Refinish phosphate	Re-phosphate IAW TT-C-490 or phosphate touch-up
390	Drive End End Bell	Refinish	Chemical film touch-up IAW MIL-C-5541, Class 3
390C	Drive End End Bell	Refinish/ Paint	Chemical film touch-up IAW MIL-C-5541, Class 1. Zinc chromate primer. Black enamel IAW TT-E-489G, Class B
520	Anti-Drive End End Bell	Refinish	Chemical film touch-up IAW MIL-C-5541, Class 3
680	Stator and Housing Assembly	Refinish	Chemical film touch-up IAW MIL-C-5541, Class 3

Table 6003 - Refinishing Procedures

11. Polarizing Output Voltage Field

NOTE: If polarity of output voltage of starter-generator has been reversed, use following procedure to correct output voltage polarity.

A. Turn power OFF and make connections.

- (1) As illustrated in Figure 6005, connect negative power lead of 6 volt battery (or equivalent DC power source) to stud E of terminal block.

REPAIR

24-30-63

Page 6013
Mar 23/10



Component Maintenance Manual with Illustrated Parts List
DC Starter-Generator, 23046 Series I

- (2) Connect positive power lead and bump switch to stud A of terminal block.

CAUTION: NO OTHER POWER SUPPLY IS TO BE CONNECTED TO THE STARTER-GENERATOR. DO NOT POLARIZE FIELD WHILE STARTER-GENERATOR IS IN OPERATION.

- (3) Turn power supply ON for one to two seconds to polarize field.

NOTE: If the polarity of the field has not been corrected, check for incorrect connections or a defective field circuit.

- (4) Turn power OFF and disconnect power leads.

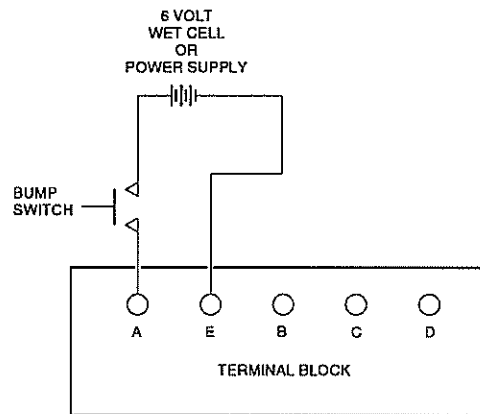


Figure 6005 - Polarization Connection Diagram

12. **Brush Access Cover.** See Figure 6006.

NOTE: Unless otherwise specified, numbers in parentheses () refer to item numbers in Figure 10001 of ILLUSTRATED PARTS LIST.

A. **Replacing brush access cover (130) insulating tape.**

- (1) Removal
 - (a) Remove all signs of insulating tape and clean inside surface of brush access cover (130). Refer to the CLEANING section.
 - (b) Touch up surface coating, if necessary. Refer to Paragraph 10. or details.
- (2) Installation
 - (a) Cut leading edge of insulating tape square with sides of roll.
 - (b) Starting at the side of the brush access cover (130) with holes, line up the side of the roll with the edge of the holes.



Component Maintenance Manual with Illustrated Parts List
DC Starter-Generator, 23046 Series I

- (c) Apply insulating tape to inner surface of brush access cover (130).

NOTE: Edge of tape to be flush to 0.03 inch (0,76 mm) maximum from edges of holes and 0.06 inch (1,5 mm) maximum inside the opposite edge of the band and both ends. Tape can be 0.02 inch (0,5 mm) wider than band on side with no holes.

- (d) Cut off unwanted insulating tape at the brush access cover (130) ends.

B. Brush access cover (130) self-locking blind rivet nut (150) replacement.

NOTE: Use a PlusNut® Fastener Head Crimping Tool for repair procedure.

(1) Removal

- (a) Turn fillister head screw (140) in a COUNTERCLOCKWISE direction until screw is fully disengaged from self locking blind rivet nut (150).
- (b) Remove damaged self-locking blind rivet nut (150) from brush access cover (130) by drilling a 0.25 inch (6,35 mm) hole through self-locking blind rivet nut. The inboard and outboard sides of nut should become loose and fall off.

(2) Rivet Nut Replacement

- (a) Using a PlusNut® Fastener Header Tool, loosen ½ inch (12,7 mm) nut by turning COUNTERCLOCKWISE until stud is fully extended.
- (b) Thread the tool completely into the self-locking blind rivet nut (150) until it stops.
- (c) Tighten the ½ inch (12,7 mm) nut down until hand-tight against washer and tool body.
- (d) Insert fastener header tool into hole in brush access cover (130).

CAUTION: TO AVOID UNREPAIRABLE THREAD DAMAGE, DO NOT OVERTIGHTEN NUT.

- (e) Use a ¾ inch (19 mm) open ended wrench on tool body while holding tool stationary using a ¾ inch (19 mm) wrench on half nut. While holding tool perpendicular to hole, turn tool CLOCKWISE>
- (f) Turn nut until firm resistance is felt. Self-locking blind rivet nut (150) should be physically reformed. If self-locking blind rivet nut (150) is loose, tighten nut until self-locking blind rivet nut is secure. Loosen nut by turning COUNTERCLOCKWISE.
- (g) Remove fastener header tool from self-locking blind rivet nut (150) by turning COUNTERCLOCKWISE.
- (h) TORQUE test self-locking blind rivet nut (150) to 60 lbf.in (6,8 N•m).
- (i) Touch up the chemfilm IAW the instructions in Paragraph 10.

REPAIR

24-30-63

Page 6015
Mar 23/10

Component Maintenance Manual with Illustrated Parts List
DC Starter-Generator, 23046 Series I

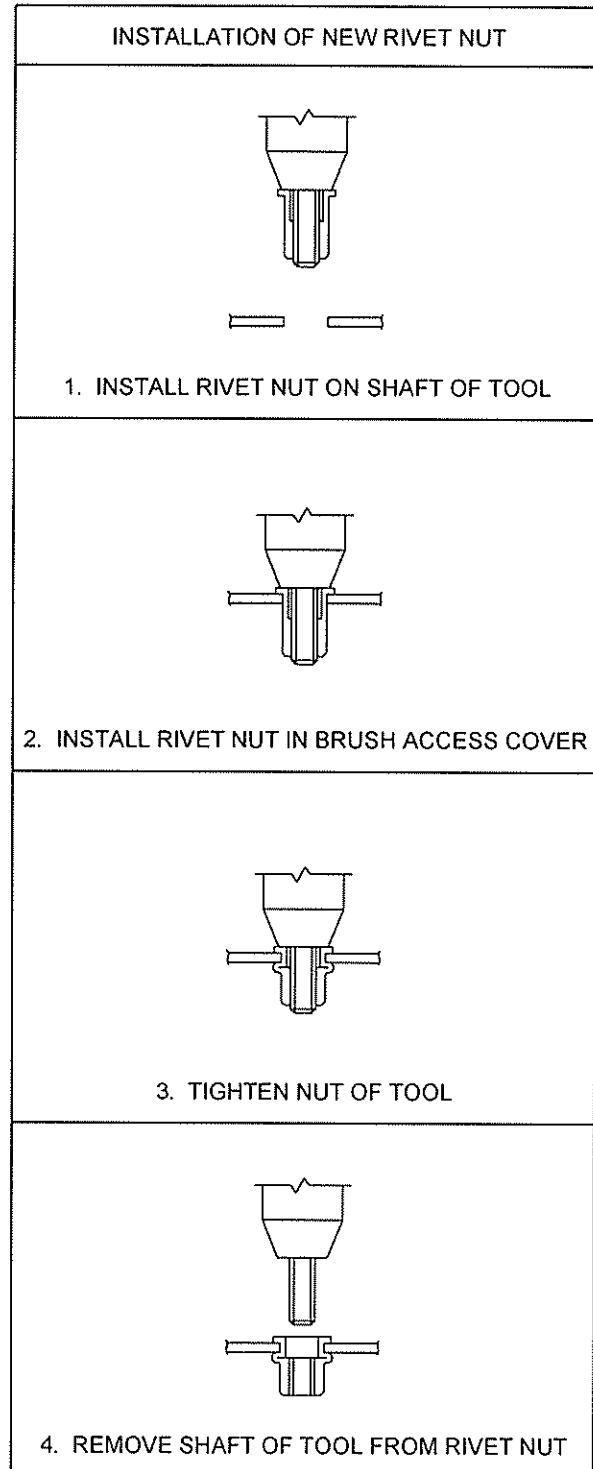
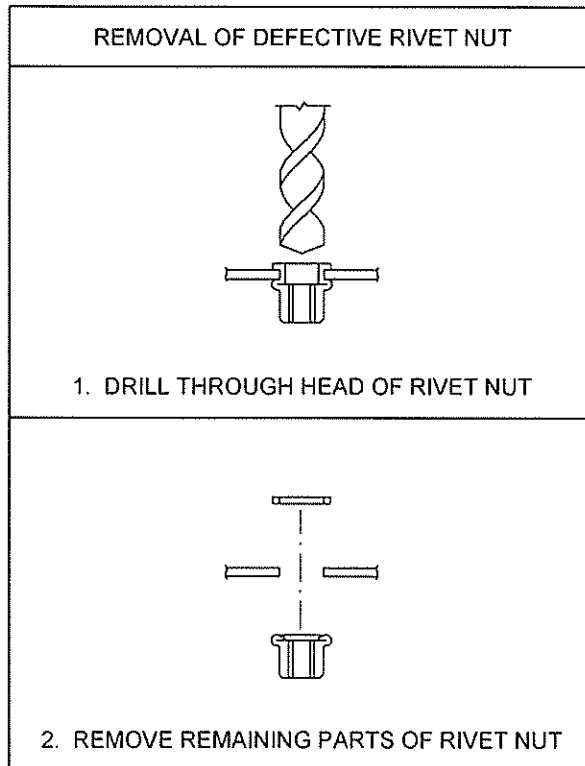
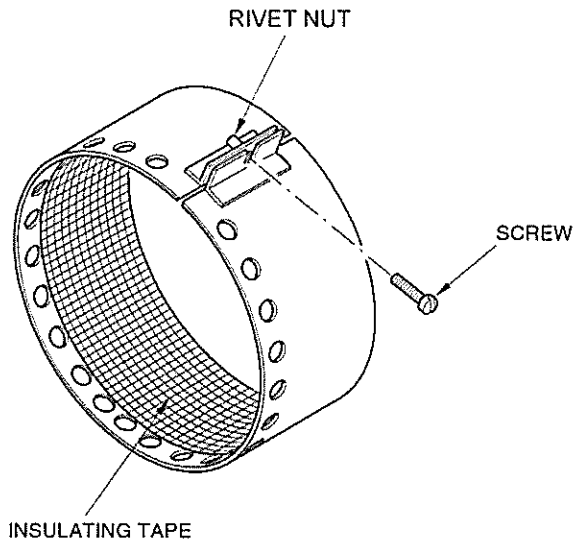


Figure 6006 - Replacement of the Brush Access Cover Rivet Nut and Insulating Tape



Component Maintenance Manual with Illustrated Parts List
DC Starter-Generator, 23046 Series I

13. Terminal Lug Replacement

NOTE: Unless otherwise specified, numbers in parentheses () refer to item numbers in Figure 10001 of ILLUSTRATED PARTS LIST

A. Remove and replace damaged terminal lugs (730 and 780) as needed.

- (1) Cut wire at base of terminal(s). Discard terminal(s).
- (2) Strip 0.25 inch (6,4 mm) insulation from end of electrical wire(s).
- (3) Use a crimping tool to install new terminal(s).

14. Removal and Replacement of the Drive End Pad End Bell Guide Pin

NOTE: Unless otherwise specified, numbers in parentheses () refer to item numbers in Figure 10001 of ILLUSTRATED PARTS LIST.

Use the following repair procedure to remove and replace mounting adapter guide pins (60).

A. Tools

A set of pliers, vice grips or any suitable tool that lets you grip the guide pin (60) and pull it out.

B. Procedure. See Figure 6007.

- (1) Secure the drive end pad end bell (50) in a vice making sure it is secured enough to handle the force required to pull the guide pin (60).
- (2) Using the suitable tool, twist and pull on the guide pin (60) until it releases from the drive end pad end bell (50).

WARNING: PRIMER CAN BE DANGEROUS TO PERSONS. USE ONLY IN AN AREA WITH A GOOD FLOW OF AIR. KEEP AWAY FROM SOURCES OF IGNITION. DO NOT BREATHE THE FUMES. PREVENT SKIN CONTACT. PUT ON PROTECTIVE CLOTHING AND EYE PROTECTION. IF NECESSARY, PUT ON A RESPIRATOR.

- (3) Apply zinc chromate primer to the new guide pin (60) before inserting into the drive end pad end bell (50).
- (4) Insert a new guide pin (60) into the hole in the drive end pad end bell (50).

NOTE: If the guide pin is grooved, make sure the grooves are oriented as shown in enlarged view A of Figure 6007.

- (5) Using an arbor press or light hammer, lightly tap the guide pin (60) into the hole.

NOTE: The finished height of the guide pin (60) is 0.11 to 0.13 inch (2,8 to 3,3 mm)



Component Maintenance Manual with Illustrated Parts List
DC Starter-Generator, 23046 Series I

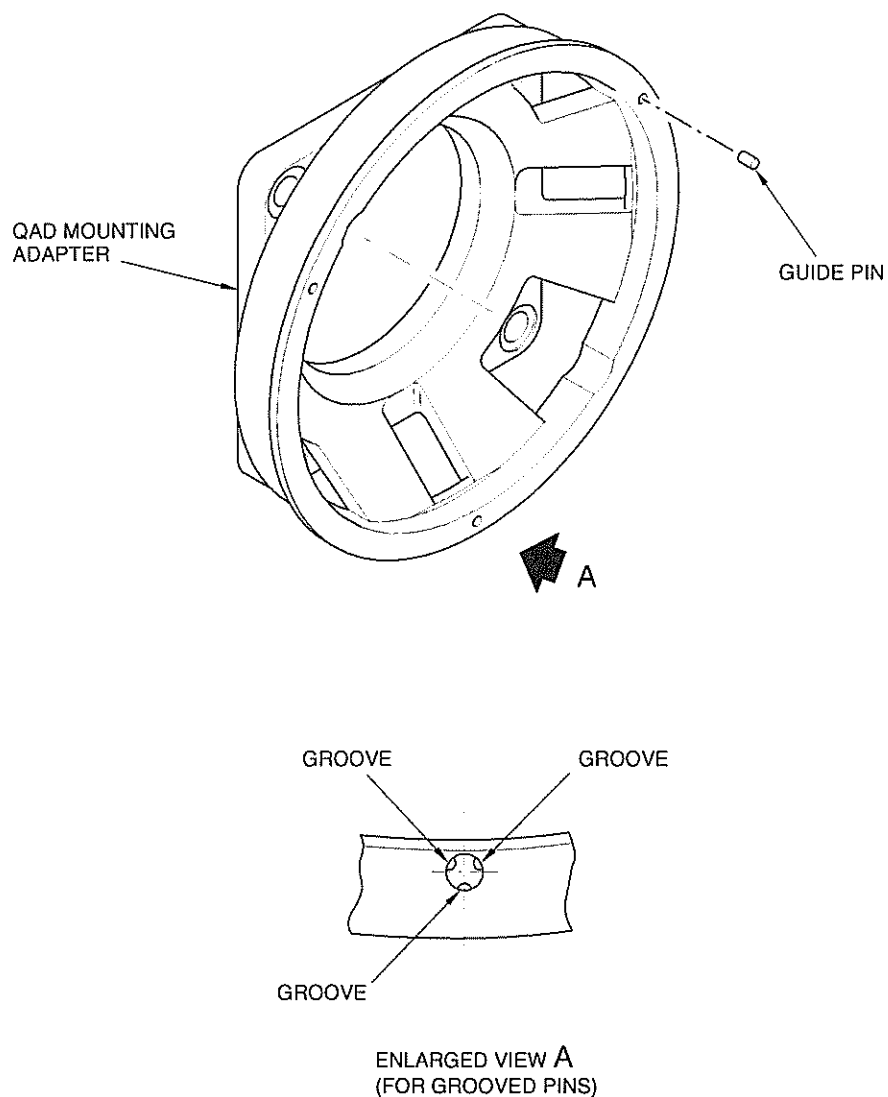


Figure 6007 - Repair of the Mounting Adapter Guide Pin



Component Maintenance Manual with Illustrated Parts List
DC Starter-Generator, 23046 Series I

ASSEMBLY

1. Introduction

This Section gives assembly instructions for 23046 Series I DC Starter-Generators. Assemble unit in a clean work area away from machining or other metal removing operations. Clean all parts per instructions in CLEANING section before final assembly.

2. Required Tools for assembly

In addition to normal shop tools, tools in Table 7001 are needed for starter-generator assembly. The tools listed are not available for purchase from Goodrich Corporation.

NOTE: Equivalent substitutes may be used for the tools listed in Table 7001.

Tool Description	Reference
Arbor Press	Not illustrated
Armature Support	Figure 9007
Drive End Hub Support	Figure 9009
Inner Race Bearing Driver	Figure 9004
Outer Race Bearing Driver	Figure 9005
Leather or Plastic Mallet	Not illustrated
Pliers, External Snap Ring	Not illustrated
Spline Wrench	Figure 9014
Horizontal Stator Support	Figure 9010
Vertical Stator Support	Figure 9011
Wire Hook Tool	Not illustrated
High Potential Tester	Not illustrated
Dampener Plate Driver	Figure 9013

Table 7001 - Assembly Tools



Component Maintenance Manual with Illustrated Parts List
DC Starter-Generator, 23046 Series I

3. Materials Required for Assembly

Table 7002 lists materials required to assemble the starter-generator. Equivalent materials may be used.

WARNING: BEFORE USING ANY OF THE FOLLOWING MATERIALS, BE AWARE OF ALL HANDLING, STORAGE, AND DISPOSAL PRECAUTIONS RECOMMENDED BY THE MANUFACTURER OR SUPPLIER. FAILURE TO COMPLY WITH THE RECOMMENDATIONS MAY RESULT IN SERIOUS INJURY, PHYSICAL DISORDER, OR DEATH.

Item	Description/ Material Specification	Source (CAGE Code)
Acrylic Coating	MIL-I-46058/HumiSeal 1B31, Type AR	Chase Corporation Woodside, NY www.humiseal.com (VOSR97)
Anti-seize Compound	Gasoila, soft set, lead free	GSA Supply on-line www.gsasupplyco.com or Federal Process Co 4620 Richmond Road Beachwood, OH44128
Enamel, Red Insulating	Synthite ER-41	John C. Dolph Company Monmouth NJ www.dophs.com (V72688)
Isopropyl Alcohol	TT-I-735, Grade A See <u>WARNING</u> before using this material. Flash Point: 53° F (12° C), FLAMMABLE Refer to Material Safety Data (MSD) Sheet for the material for additional safety information.	Commercially Available
Sandpaper	180 Grit, non-aluminum oxide	Commercially available
Thread-locking Adhesive	MIL-S-22473/Loctite grade D	Loctite Corporation Aurora, IL www.loctite.com (V7V827)
Thread-locking Primer	MIL-S-22473/Loctite grade N	

Table 7002 - Assembly Materials



Component Maintenance Manual with Illustrated Parts List
DC Starter-Generator, 23046 Series I

4. Assembly Procedure

Unless otherwise specified, numbers in parentheses () refer to the item numbers given in Figure 10001 of the ILLUSTRATED PARTS LIST.

A. **Attach the terminal block (690) to the stator and housing assembly (680).**

NOTE: Before acceptance testing, attach an unserviceable terminal block to the stator and housing assembly. Acceptance testing can damage the internal capacitors of a terminal block (690).

- (1) Set the stator and housing assembly (680) onto a horizontal stator support.

CAUTION: DO NOT BEND THE LARGE STATOR LEADS MORE THAN NECESSARY TO INSTALL THE TERMINAL BLOCK.

- (2) Set the terminal block (690) onto the stator and housing assembly and slide it into position in the large stator leads C, B+ and E-.

NOTE: Pass the leads with terminal lugs between the large stator leads and the terminal block (690).

- (3) Put one terminal lug (780) onto each of terminal studs A and D.
- (4) Put a lock washer (710) and a flat washer (720) onto each of two screws (700).
- (5) Put the two screws into the end holes of the terminal block (690) and tighten the screws to a torque of 22.7 to 35.0 lbf.in. (2,6 to 4,0 N•m)

NOTE: Lockwire is not used on screw (700 or 700A).

B. **Install hardware on terminal block (690).**

- (1) Install one new nut (760) and flat washer (770) each on terminals C, B+ and E-.

NOTE: Do not tighten the nuts (760) on terminals C, B+ and E-. They will be tightened at the time of installation into the aircraft.

- (2) Install one flat washer (750) and one new nut (740) on each terminal A and D.

NOTE: Do not tighten the nuts (740) on terminals A and D. They will be tightened at the time of installation into the aircraft. It is necessary to install the wires from the GCU at that time.

C. **Assemble brush springs (600) to brush holder (610). See Figure 7001.**

CAUTION: DO NOT OVER-WIND BRUSH SPRINGS. OVER-STRESSING BRUSH SPRINGS CAN RESULT IN REDUCED BRUSH SPRING PRESSURE.

- (1) Wind each brush spring (600) $\frac{3}{4}$ of a turn in tension from spring resting position.



Component Maintenance Manual with Illustrated Parts List
DC Starter-Generator, 23046 Series I

CAUTION: CORRECT SEATING OF BRUSH SPRINGS ON BRUSH SPRING SUPPORT IS CRITICAL DURING ASSEMBLY.

- (2) Press brush springs (600) onto brush spring support. Make sure brush springs are properly seated on brush spring support.

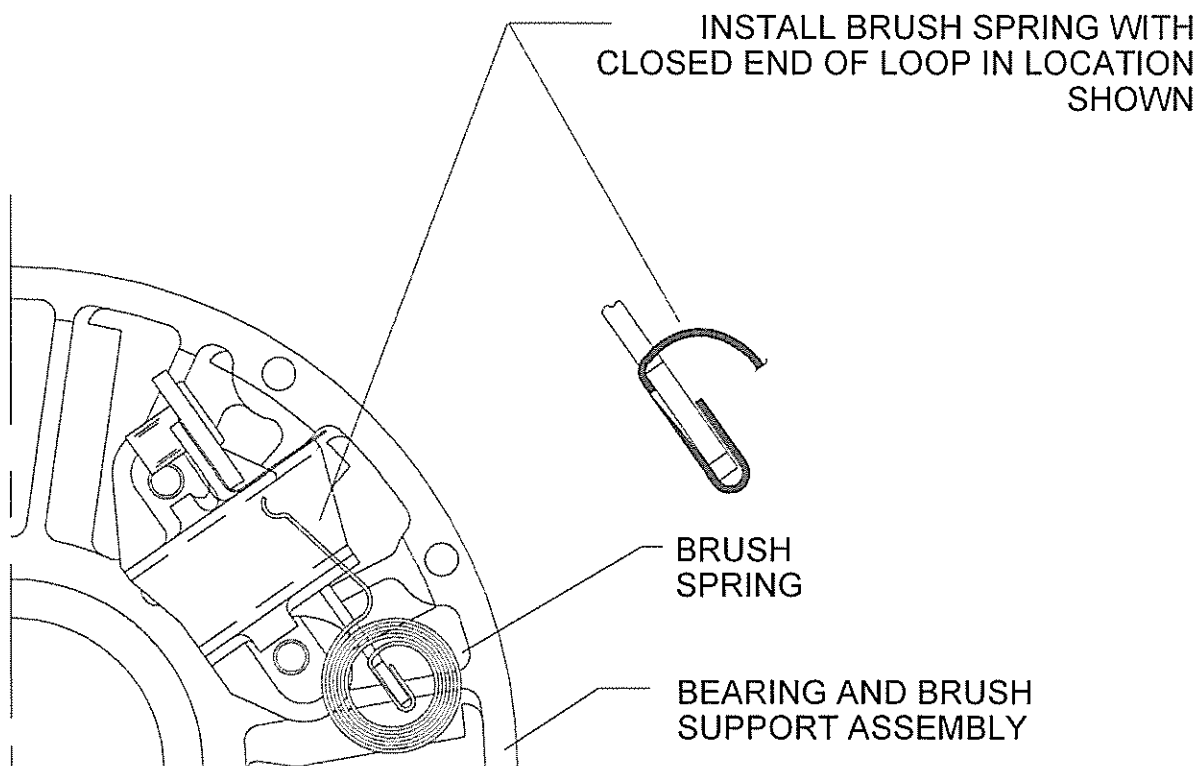


Figure 7001 - Brush Spring Assembly



Component Maintenance Manual with Illustrated Parts List
DC Starter-Generator, 23046 Series I

D. Assembly of bearing and brush support assembly. Reference SPD 1004.
See Figure 7002.

CAUTION: PROPER INSTALLATION OF INSULATION SLEEVES IS CRITICAL. USE EXTREME CARE WHEN PRESSING INSULATION SLEEVES INTO MOUNTING HOLES IN ANTI-DRIVE END END BELL. IF INSULATION SLEEVE IS PINCHED BETWEEN ANTI-DRIVE END END BELL AND BRUSH HOLDER ASSEMBLY DURING ASSEMBLY, CARBON WILL BUILD UP IN AREA WHERE INSULATION SLEEVE IS DAMAGED, RESULTING IN A PATH TO GROUND AND EQUIPMENT FAILURE. DAMAGE IS NOT DETECTABLE DURING HIGH POTENTIAL TESTING AFTER ASSEMBLY BECAUSE BEARING AND BRUSH SUPPORT AND BRUSH HOLDER ASSEMBLIES ARE CLEAN AT TIME OF ASSEMBLY.

- (1) Carefully press one insulation sleeve (640) into each mounting hole in anti-drive end end bell (520).
- (2) If brush holder assembly (-550) has studs (630), proceed as follows:
 - (a) Place an insulation plate (650) on the studs (630) of a brush holder assembly (-550).
 - (b) Carefully pass the studs of the brush holder assembly through two insulation sleeves on the anti-drive end end bell (520).

NOTE: Make sure the brush holder assembly is on the inboard side of the anti-drive end end bell.

- (c) Put a non-metallic washer (590), flat washer (580) and self-locking nut (570) on each stud.

NOTE: Do not tighten the self-locking nuts (570) at this time.

- (d) Repeat Paragraph 4.D.(2)(a) thru Paragraph 4.D.(2)(c) for remaining brush holder locations.
- (3) If brush holder assembly (-550) does not have studs, proceed as follows:
 - (a) Put a flat washer (580) and non-metallic washer (590) on two bolts (560).
 - (b) Insert bolts (560) in mounting holes in anti-drive end end bell (520).
 - (c) Put an insulation plate (650) on bolts (560).



Component Maintenance Manual with Illustrated Parts List
DC Starter-Generator, 23046 Series I

WARNING: THREAD LOCKING PRIMER AND ADHESIVE MAY IRRITATE SENSITIVE SKIN. WASH THOROUGHLY AFTER CONTACT WITH SKIN. REFER TO APPLICABLE MATERIAL SAFETY DATA SHEET FOR ADDITIONAL MATERIAL AND SAFETY INFORMATION.

WARNING: THREAD LOCKING PRIMER AND ADHESIVE IS HIGHLY FLAMMABLE AND TOXIC. DO NOT USE NEAR OPEN FLAMES OR SPARKS. USE IN A WELL-VENTILATED AREA.

- (d) Apply thread locking primer, Grade N, followed by adhesive, Grade D, to the threads of the two bolts (560).

NOTE: Let the thread locking primer, Grade N, dry for a few minutes before applying thread locking adhesive, Grade D.

- (e) Put a brush holder assembly (-550) against the two bolts (560) on the inboard side of the anti-drive end end bell (520).

- (f) Attach the brush holder assembly (-550) loosely until correctly aligned.

NOTE: Do not tighten the bolts (560) at this time.

- (g) Repeat Paragraph 4.D.(3)(a) thru Paragraph 4.D.(3)(f) for remaining brush holder locations.

CAUTION: MAKE SURE THE COMPLETE BRUSH HOLDERS ARE PROPERLY ALIGNED IN THE BRUSH HOLDER ALIGNMENT FIXTURE. IMPROPER ALIGNMENT CAN CAUSE ELECTRICAL FAILURE.

- (4) Adjust all brush holder assemblies (-550) until they are properly aligned. Refer to SPD 1004.
- (5) Tighten bolts (560) to a torque of 25 to 30 lbf.in. (2,8 to 3,4 N•m) or tighten self-locking nuts (570).
- (6) Remove the bearing and brush support assembly (-480) from the brush holder alignment fixture.

WARNING: INSULATING ENAMEL IS TOXIC AND FLAMMABLE. DO NOT BREATHE VAPORS. USE IN A WELL VENTILATED AREA FREE FROM SPARKS, FLAME AND/OR HOT SURFACES. WEAR SPLASH GOGGLES, SOLVENT-RESISTANT GLOVES, AND OTHER PROTECTIVE GEAR. IN CASE OF EYE CONTACT, FLUSH WITH WATER AND SEEK MEDICAL ATTENTION. IN CASE OF SKIN CONTACT, WASH WITH SOAP AND WATER.

- (7) Apply a thick coat of red insulating enamel IAW SPD 1002 to the exposed surfaces of the bearing and brush support assembly.

Component Maintenance Manual with Illustrated Parts List
DC Starter-Generator, 23046 Series I

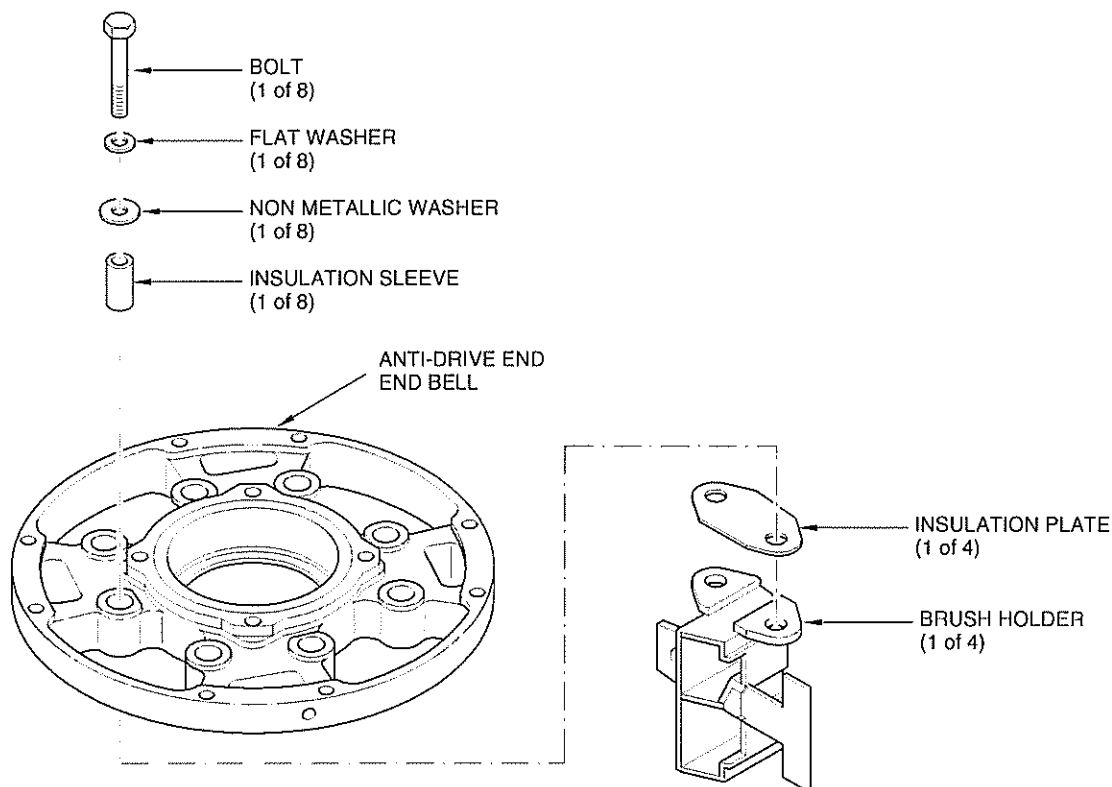


Figure 7002 - Bearing and Brush Support Assembly Detail (with Bolts)

- (8) Perform a dielectric test of the bearing and brush support assembly. Refer to CHECK section.

**E. Press a ball bearing (450) into bearing and brush support assembly (-480).
See Figure 7003.**

- (1) Set a drive end hub support on arbor press table.
- (2) Set bearing and brush support assembly, brush holder side down, on drive end hub support. See Figure 7003.
- (3) Set a ball bearing onto bearing liner of anti-drive end end bell (520).
- (4) Set an outer race bearing driver onto ball bearing.
- (5) Press ball bearing into bearing liner of anti-drive end end bell (520).
- (6) Make sure that ball bearing is fully seated against inner lip of bearing liner.



Component Maintenance Manual with Illustrated Parts List
DC Starter-Generator, 23046 Series I

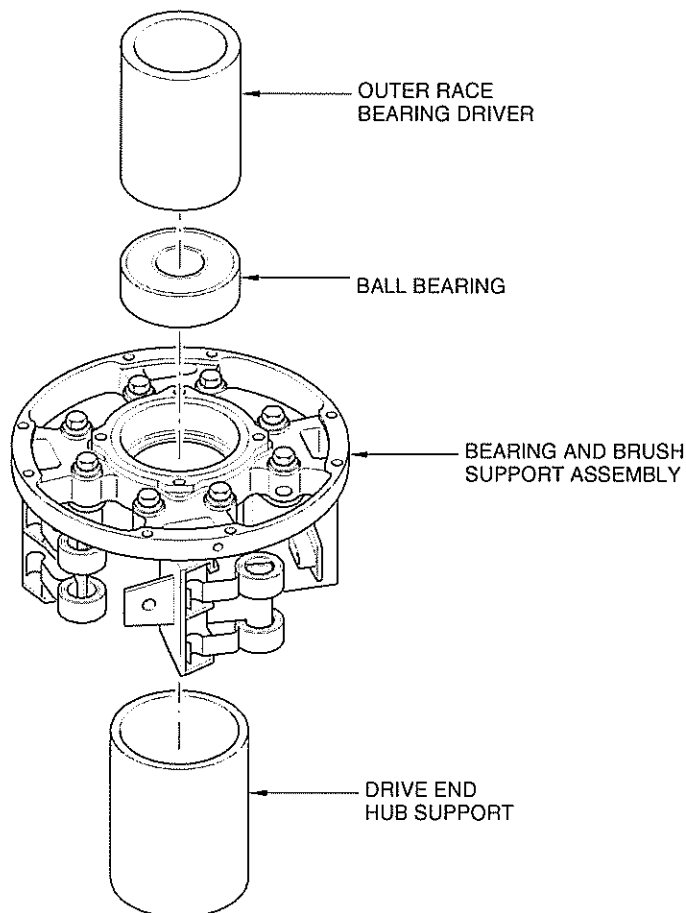


Figure 7003 - Ball Bearing Installation into Bearing and Brush Support Assembly.

- F. Attach bearing retainer (660) to bearing and brush support assembly (-480). See Figure 7004.

WARNING: ANTI-SEIZE COMPOUND IS FLAMMABLE. DO NOT USE NEAR OPEN FLAMES, NEAR WELDING AREAS, OR ON HOT SURFACES.

- (1) Apply anti-seize compound to threads of the four attaching screws (670).
- (2) Attach bearing retainer (660) to bearing and brush support assembly (-480) using attaching screws.
- (3) Tighten the screws (670) to a torque of 9.0 to 12.0 lbf.in. (1.0 to 1.4 N•m).

NOTE: Lockwire is not used on screw (670 or 670A).



Component Maintenance Manual with Illustrated Parts List
DC Starter-Generator, 23046 Series I

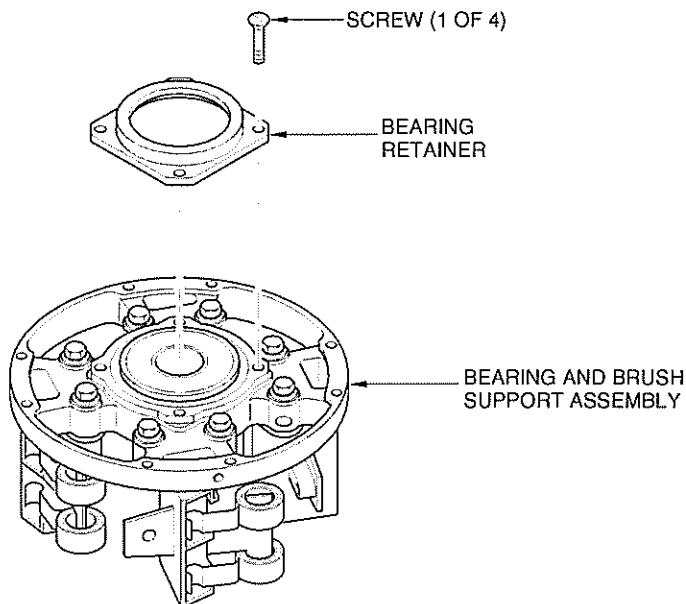


Figure 7004 - Attaching Bearing Retainer

G. Install baffle disc (440) or bearing shield (460) and ball bearing (450) onto drive end of armature (470) shaft. See Figure 7005.

CAUTION: DO NOT TOUCH POLISHED SURFACE OF COMMUTATOR WITH BARE HANDS. SKIN ACIDS AND OILS CAN CONTAMINATE CONDUCTING SURFACES CAUSING CORROSION AND/OR POOR ELECTRICAL CONTACT.

- (1) Put an armature support on arbor press table.
- (2) Put armature, commutator end down, on armature support.
- (3) Put a baffle disc on drive end of armature shaft with concave side facing toward the armature.

NOTE: For models 23046-019 and 23046-020 refer to SB 23046-0XX-24-06. If these models are at MOD D they have a drive end bearing shield (460) assembled instead of the baffle disc (440).

- (4) Set a ball bearing onto armature shaft.
- (5) Set an inner race bearing driver on ball bearing.
- (6) Press ball bearing down onto armature shaft.
- (7) Make sure that ball bearing is fully seated against baffle disc or drive end bearing shield.



Component Maintenance Manual with Illustrated Parts List
DC Starter-Generator, 23046 Series I

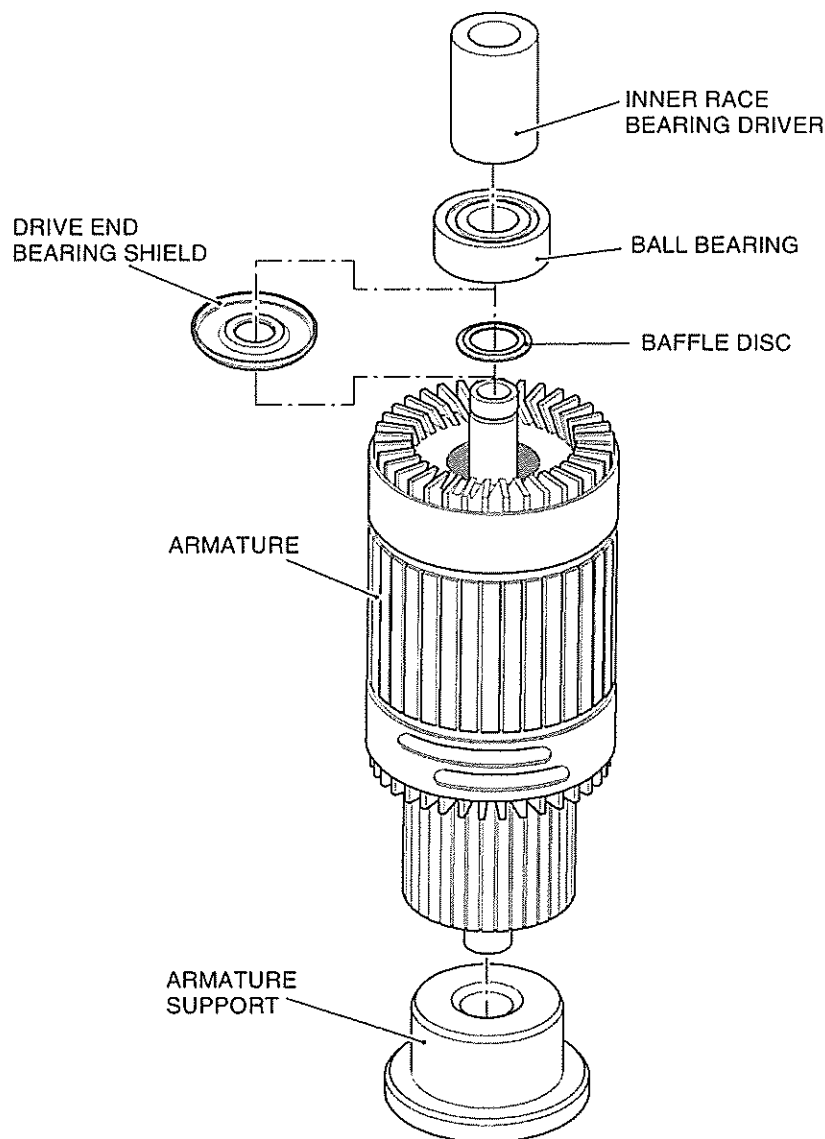


Figure 7005 - Baffle Disc or Bearing Shield and Ball Bearing Installation on Drive End of Armature Shaft.

H. Install baffle disc (440) and retaining ring (430) onto drive end of armature (470) shaft. See Figure 7006.

- (1) Put an outer baffle disc (440) on drive end of armature shaft with concave side facing away from armature stack.
- (2) Put a retaining ring (430) in groove on drive end of armature shaft using external snap ring pliers. Make sure it is fully engaged in groove of armature shaft.

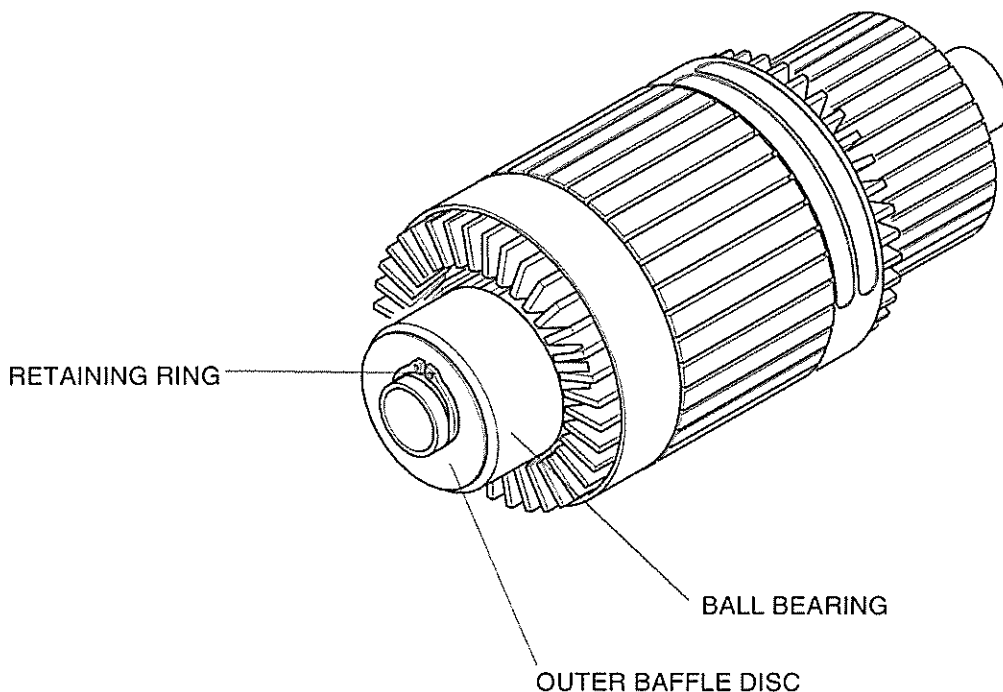


Figure 7006 - Installing Outer Baffle Disc and Retaining Ring

I. Attach bearing and brush support assembly(-480) to armature (470).

CAUTION: DO NOT TOUCH POLISHED SURFACE OF COMMUTATOR WITH BARE HANDS. SKIN ACIDS AND OILS CAN CONTAMINATE CONDUCTING SURFACES CAUSING CORROSION AND/OR POOR ELECTRICAL CONTACT.

- (1) Put armature support on arbor press table.
- (2) Set armature, drive end down, on armature support.
- (3) Set bearing and brush support assembly, brush holder side down, onto armature shaft. Refer to Figure 7007.
- (4) Set an inner race bearing driver onto ball bearing installed in bearing and brush support assembly.
- (5) Press ball bearing in bearing and brush support assembly onto armature shaft.
- (6) Make sure that ball bearing is fully seated against shoulder of armature shaft.



Component Maintenance Manual with Illustrated Parts List
DC Starter-Generator, 23046 Series I

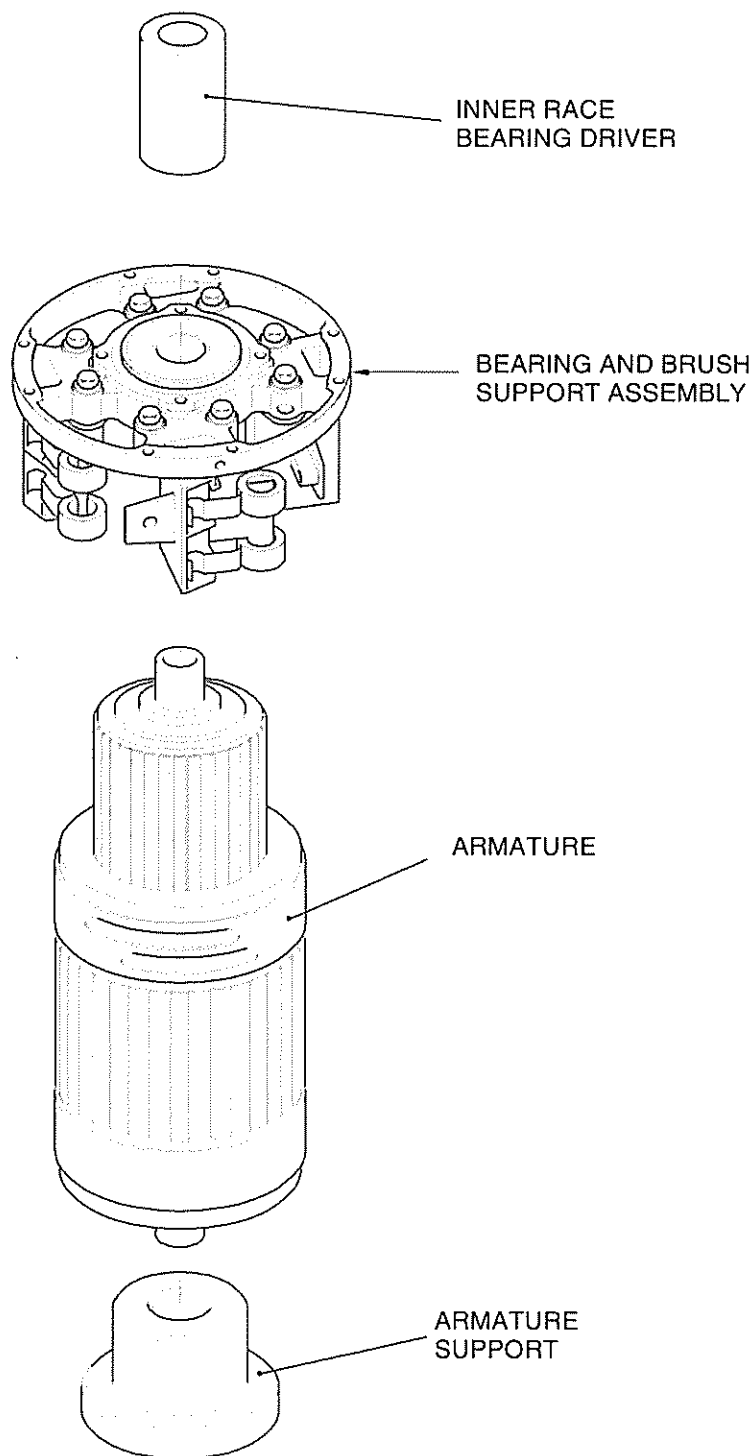


Figure 7007 - Attaching Bearing and Brush Support Assembly and Armature



Component Maintenance Manual with Illustrated Parts List
DC Starter-Generator, 23046 Series I

J. Attach bearing and brush support assembly, (-480), with attached armature (470), to stator and housing assembly (680). See Figure 7008.

- (1) Set a vertical stator support onto a flat work surface.
- (2) Carefully set stator and housing assembly (680), anti-drive end up, on vertical stator support.
- (3) Insert armature (470), with attached bearing and brush support assembly (-480), into stator and housing assembly (680).
- (4) Make sure that brush stator leads are aligned with each brush holder assembly. If necessary, use a probe to move brush stator leads to their correct position.

NOTE: On older models a lead from the stator windings should be connected to the negative brush holder. Keep this lead with the negative stator lead. It will be secured when the brushes are installed.

- (5) Place a lock washer (500) and flat washer (510) on each of the eight attaching screws (490).

WARNING: ANTI-SEIZE COMPOUND IS FLAMMABLE. DO NOT USE NEAR OPEN FLAMES, NEAR WELDING AREAS, OR ON HOT SURFACES.

- (6) Apply anti-seize compound to the threads of eight attaching screws (490).
- (7) Attach bearing and brush support assembly to stator and housing assembly with the eight attaching screws.
- (8) Make sure that mounting surfaces of bearing and brush support assembly and stator and housing assembly are fully seated against each other.
- (9) Tighten the screws (490) to a torque of 13.5 to 18.0 lbf.in, (1,5 to 2.0 N•m).



Component Maintenance Manual with Illustrated Parts List
DC Starter-Generator, 23046 Series I

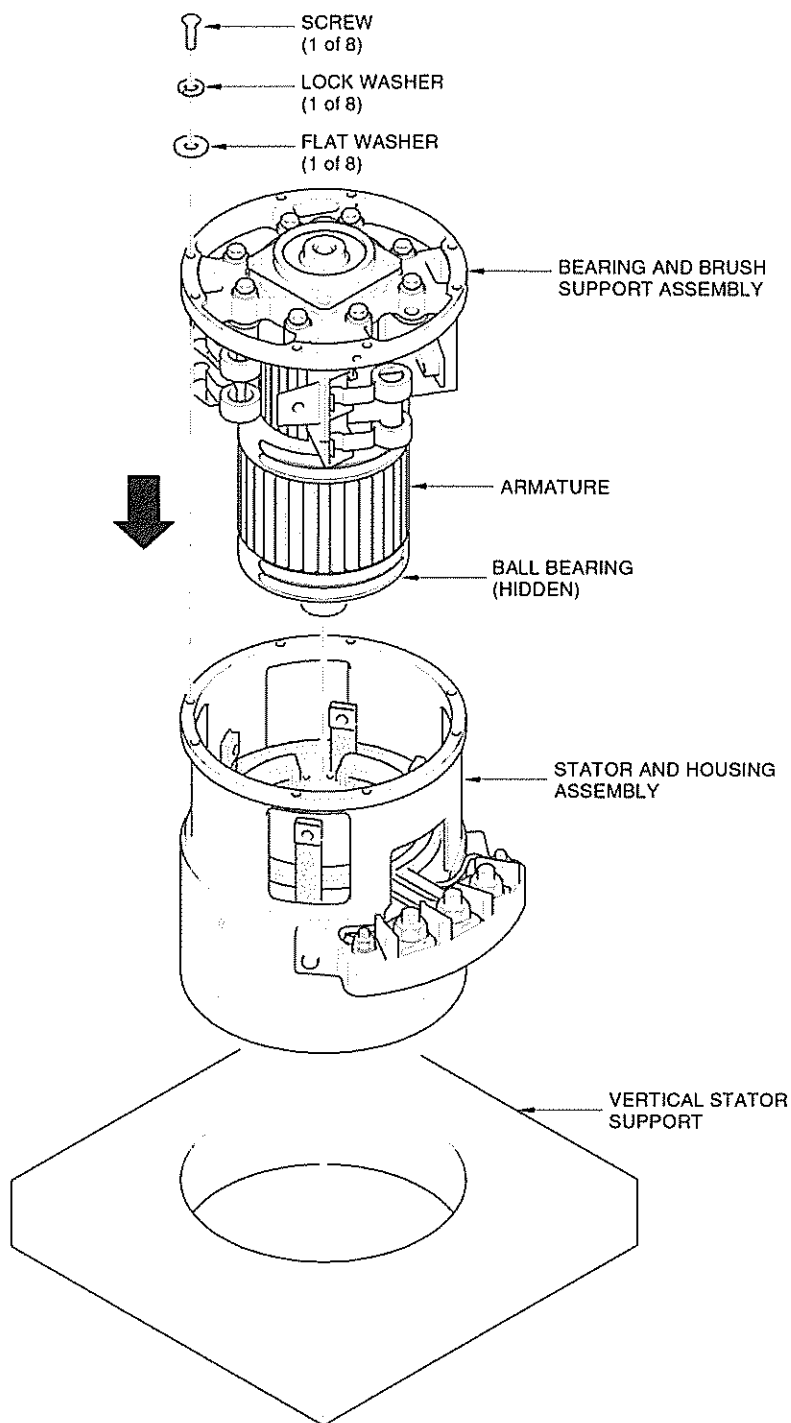


Figure 7008 - Assembling Bearing and Brush Support Assembly with Stator and Housing Assembly

ASSEMBLY

24-30-63

Page 7014

Mar 23/10



Component Maintenance Manual with Illustrated Parts List
DC Starter-Generator, 23046 Series I

K. Attach drive end bearing support assembly (-350) to stator and housing assembly (680).

- (1) Set a vertical stator support onto a flat work surface.
- (2) Place stator and housing assembly, drive end up, on vertical support.
- (3) Position drive end bearing support assembly on stator and housing assembly. Ensure that it is firmly seated against the bearing (450).

NOTE: For model 23046-001 make sure that one flat of the mounting flange is parallel with the top of the starter-generator.

NOTE: For all other models make sure that one dowel hole on the mounting flange is in the 12 o'clock position (looking from the drive end).

WARNING: ANTI-SEIZE COMPOUND IS FLAMMABLE. DO NOT USE NEAR OPEN FLAMES, NEAR WELDING AREAS, OR ON HOT SURFACES.

- (4) Apply anti-seize compound to the threads of 12 attaching flat-head screws (360). (Eight socket-head screws (360B) for Model 23046-001)
- (5) For all units, except 23046-001, install and tighten twelve screws (360) to secure drive end bearing support assembly (-350) to the stator and housing assembly (680). Tighten the screws (360) to a torque of 13.5 to 18.0 lbf.in. (1,5 to 2,0 N•m).
- (6) For unit 23046-001, assemble lock washers (370) and flat washers (380) on eight socket head screws (360B). Install the screws to secure the drive end bearing support assembly (-350) to the stator and housing assembly (680). Tighten the socket head screws (360B) to a torque of 13.5 to 18.0 lbf.in. (1,5 to 2,0 N•m).

L. Install dampener backplate (340) onto armature (470) shaft.

CAUTION: DO NOT USE AN ARBOR PRESS TO FORCE DAMPENER BACK PLATE ONTO ARMATURE SHAFT OR DAMPENER HUB ONTO DRIVE SHAFT. PROPER TIGHTENING OF FAN NUT (260) WILL PRODUCE CORRECT SEATING OF THESE COMPONENTS.

- (1) Set stator and housing assembly (680) on a horizontal stator support.
- (2) Twist dampener backplate (340) onto drive end of armature (470) shaft until it is fully seated.
- (3) Check the runout of the radial surface of the backplate while rotating the armature on its bearings.

Runout must not exceed 0.003 inch (0,0762 mm) T.I.R.



Component Maintenance Manual with Illustrated Parts List
DC Starter-Generator, 23046 Series I

M. Install dampener hub (310) and dampener plate (320) onto drive shaft (290). See Figure 7009.

CAUTION: DO NOT FORCE DAMPENER HUB ONTO DRIVE SHAFT MATING TAPER OR THE HUB CAN FRACTURE.

- (1) If dampener hub (310) was removed from drive shaft (290), twist dampener hub by hand onto drive shaft taper and make sure it is fully seated.
- (2) Insert drive shaft (290) through dampener plate (320) and dampener plate driver. Align plate and hub splines.
- (3) Lightly tap drive end of drive shaft using a leather or plastic mallet to seat dampener plate onto hub. Make certain that spline on plate fully engages the dampener hub spline.

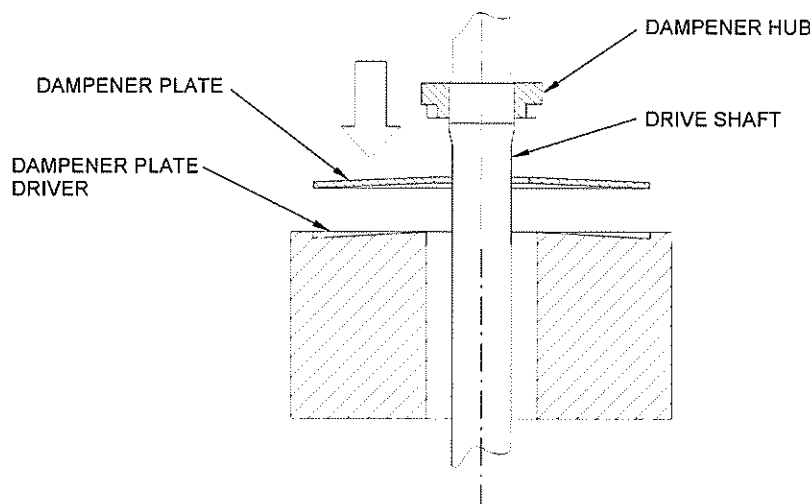


Figure 7009 - Dampener Plate and Driver

N. Insert drive shaft (290) into armature (470) shaft. See Figure 7010.

CAUTION: DO NOT USE FORCE TO ENGAGE DRIVE SHAFT AND ARMATURE SHAFT MATING SPLINES. FAILURE TO ENGAGE THE DRIVE SHAFT SPLINES CORRECTLY CAN DAMAGE THE DRIVE AND ARMATURE SHAFTS.

- (1) Put friction ring (330) in recess of dampener backplate (340).
- (2) Insert drive shaft (290) into drive end of armature (470) shaft.



Component Maintenance Manual with Illustrated Parts List
DC Starter-Generator, 23046 Series I

- (3) Push drive shaft through armature shaft until dampener plate (320) is fully engaged against friction ring (330).
- (4) Turn drive shaft in direction of rotation to make sure that armature shaft and drive shaft splines are correctly engaged.

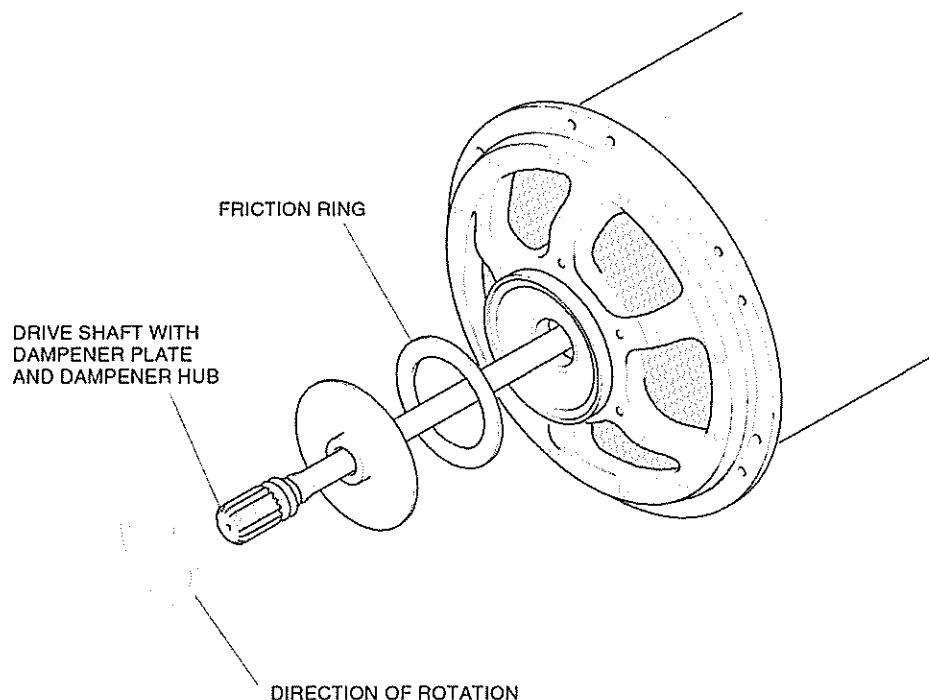


Figure 7010 - Installing Drive Shaft into Armature Shaft

O. Attach fan (250) to drive shaft (290). See Figure 7011.

NOTE: To keep drive shaft from turning, use a spline wrench on the drive spline when tightening self-locking nut onto drive shaft.

- (1) For models 23046-019 and 23046-020 at MOD D, assemble the anti-drive end bearing shield (280) on the flange of the fan (250). Make sure the part identification stamping on the anti-drive end bearing shield faces the fan blades.
- (2) Put fan (250) onto drive shaft (290).
- (3) Attach fan to drive shaft with a flat washer (270) and a self-locking nut (260).
- (4) Tighten the nut to a torque of 100 to 120 lbf.in. (11,3 to 13,6 N•m).



Component Maintenance Manual with Illustrated Parts List
DC Starter-Generator, 23046 Series I

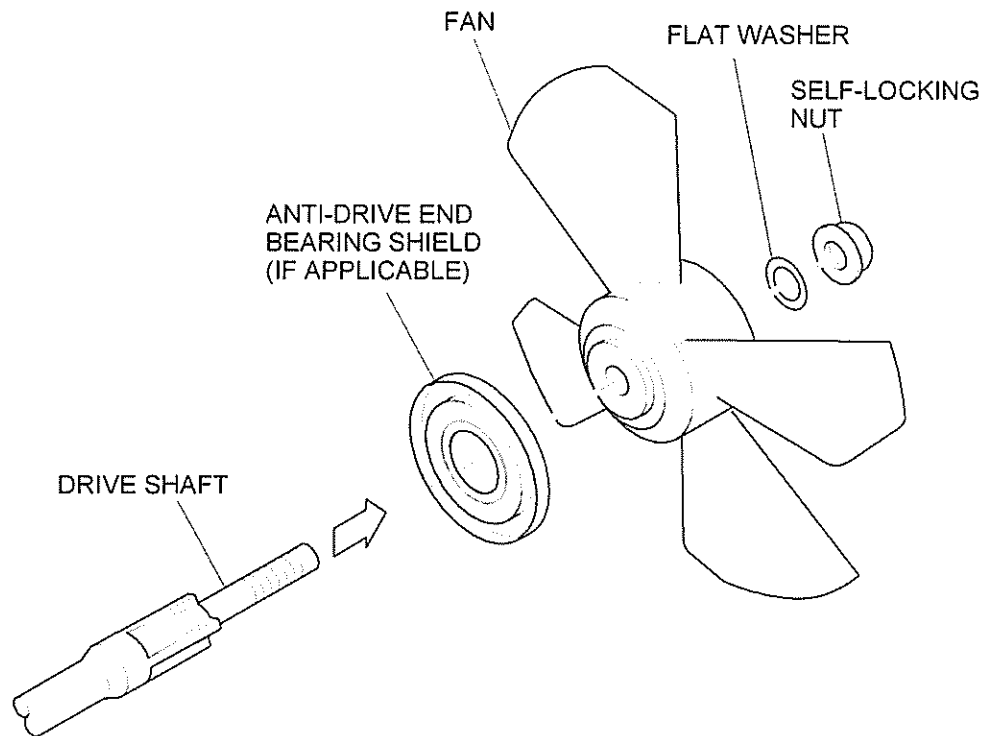


Figure 7011 - Installing Fan

P. Install brushes (160) into starter-generator. See Figure 7012.

NOTE: If new brushes (160) are to be used, identify the brushes with the numbers 1 through 4 as shown in Figure 7011.

CAUTION: RAISE AND LOWER BRUSH SPRINGS SLOWLY. DO NOT LIFT BRUSH SPRINGS MORE THAN NEEDED TO INSERT BRUSH SETS INTO BRUSH HOLDER ASSEMBLY.

- (1) Set starter-generator, anti-drive end up, onto a vertical stator support.
- (2) At each brush holder location, lift one brush spring at a time with a wire hook tool and insert brush set into a brush holder assembly.
- (3) Slowly lower springs on top of brushes. Make sure that brush leads are not caught under brush springs.

CAUTION: FAILURE TO PROPERLY FORM THE BRUSH LEADS CAN RESULT IN BRUSH HANG UP.

- (4) Form the brush leads IAW SPD 1006.

ASSEMBLY

24-30-63

Page 7018
Mar 23/10



Component Maintenance Manual with Illustrated Parts List
DC Starter-Generator, 23046 Series I

- (5) Attach the four brush (160) leads and four braided stator leads to the brush holders (610) with four screws (170). Tighten the screws to a torque of 22.7 to 35.0 lbf.in (2,6 to 4,0 N•m).

NOTE: On older models there is an extra negative brush lead with a terminal lug. Attach this lead immediately under the screw (170) head when attaching brush number 1. Refer to Figure 7012.

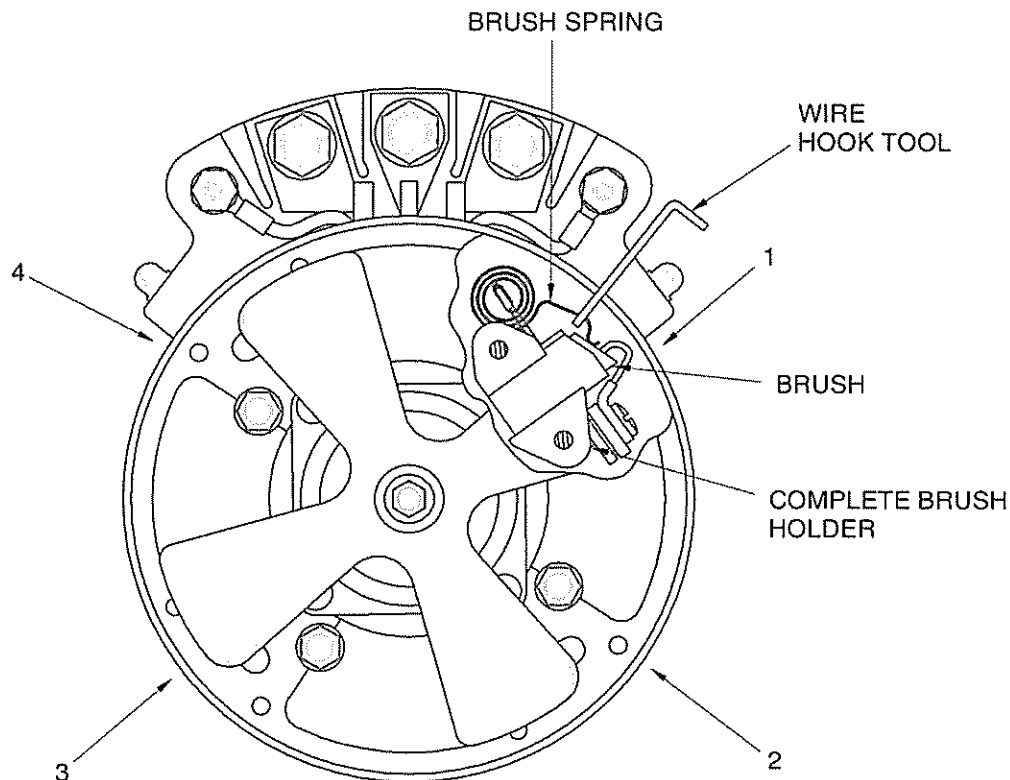


Figure 7012 - Installing Brushes

Q. Coarse seat the new brushes (160).

CAUTION: FAILURE TO FULLY SEAT BRUSHES CAN DECREASE BRUSH LIFE, LOWER STARTER-GENERATOR PERFORMANCE, OR CAUSE STARTER-GENERATOR FAILURE.

- (1) Coarse seat the brushes (160) IAW SPD 1006.

NOTE: The sandpaper should first touch leading edge (short side) of brush (160) when armature (470) is rotated in its normal direction.

- (2) Remove sandpaper IAW SPD 1006 when done with brush seating.



Component Maintenance Manual with Illustrated Parts List
DC Starter-Generator, 23046 Series I

R. Final run-in.

CAUTION: FAILURE TO FULLY RUN-IN THE BRUSHES CAN DECREASE BRUSH LIFE, LOWER STARTER-GENERATOR PERFORMANCE, OR CAUSE STARTER-GENERATOR FAILURE.

CAUTION: DO NOT LET THE STARTER-GENERATOR HANG UNSUPPORTED DURING INSTALLATION ONTO AND REMOVAL FROM THE DRIVE STAND. TOO MUCH LOAD ON THE DRIVE SHAFT SHEAR SECTION CAN DAMAGE THE UNIT.

- (1) Run-in the brushes (160) IAW SPD 1006.

NOTE: A correctly run-in brush (160) has a smooth and semi-gloss contact surface.

S. Attach commutation viewing adapter to starter-generator.

- (1) Put commutation viewing adapter over openings in stator and housing assembly (680).

CAUTION: THE JOINT FORMED BY THE BRUSH ACCESS COVER MUST NOT BE POSITIONED OVER AN ACCESS WINDOW IN THE HOUSING. THE BRUSH SHUNT WIRES CAN SHORT TO THIS AREA.

- (2) Center brackets of commutation viewing adapter between two brush access openings in stator and housing assembly (680).
- (3) Tighten screw (140) into blind rivet nut attached to commutation viewing adapter.

NOTE: Lockwire is not used on screw (140).

T. Attach cover assembly (180) to bearing and brush support assembly (-480). (All models except 23046-019 and 23046-020).

- (1) Set starter-generator, anti-drive end up, onto a vertical stator support.
- (2) Position cover assembly (180) on brush support assembly (-480).

WARNING: ANTI-SEIZE COMPOUND IS FLAMMABLE. DO NOT USE NEAR OPEN FLAMES, NEAR WELDING AREAS, OR ON HOT SURFACES.

- (3) Apply anti-seize compound to the threads of four screws (190).
- (4) Install and tighten screws (190) to secure cover assembly (180) to the bearing and brush support assembly (-480). Tighten the screws to a torque of 7.7 to 10.3 lbf.in. (0,87 to 1,16 N•m).

U. Attach air inlet assembly (200) with accompanying fan cover (240) to starter-generator. (Models 23046-019 and 23046-020 only).

- (1) If air inlet assembly (200) and fan cover (240) have been disassembled, re-assemble as follows:



Component Maintenance Manual with Illustrated Parts List
DC Starter-Generator, 23046 Series I

- (a) Place a washer (230) on each of the two screws (220).

WARNING: ANTI-SEIZE COMPOUND IS FLAMMABLE. DO NOT USE NEAR OPEN FLAMES, NEAR WELDING AREAS, OR ON HOT SURFACES.

- (b) Apply anti-seize compound to the threads of two screws (220).
(c) Place the fan cover (240) in the air inlet assembly (200).
(d) Attach the two halves of the air inlet assembly with the two screws (220) and washers (230).
(e) Tighten the two screws (220) to a torque of 22.7 to 35.0 lbf.in (2,6 to 4,0 N•m).

NOTE: Lockwire is not used on screw (220).

- (2) Place starter-generator, anti-drive end up, onto a vertical stator support.
(3) Place air inlet assembly (200) in position on anti-drive end of generator.
(4) Bring rim-clenching clamp (210) into position and tighten self-locking nut onto T-bolt. Tighten the self-locking nut to the torque etched on the clamp.

V. Attach identification plate (70), information plate (90), instruction decal (110) and FAA-PMA label (-120) to stator and housing assembly (680).

CAUTION: DO NOT STAMP INFORMATION DIRECTLY ONTO ANY PART OF STATOR AND HOUSING ASSEMBLY.

- (1) Identification plate (70) replacement, if required.

NOTE: When re-identifying a starter-generator, refer to ILLUSTRATED PARTS LIST section for applicable modification status.

- (a) Transfer information from old identification plate to replacement identification plate (70).
(b) Position replacement identification plate (70) on stator and housing assembly (680) in place of the old identification plate and secure with four drive screws (80).

WARNING: KEEP ACRYLIC COATING AWAY FROM SOURCES OF IGNITION. CONFORMAL COATING IS FLAMMABLE.

WARNING: DO NOT GET ACRYLIC COATING ON YOUR SKIN AND DO NOT BREATHE THE FUMES. ACRYLIC COATING IS A POISONOUS MATERIAL.

- (c) Coat replacement identification plate (70) with acrylic coating and let the coating air dry.



Component Maintenance Manual with Illustrated Parts List
DC Starter-Generator, 23046 Series I

- (2) Information plate (90) replacement, if required.
 - (a) Transfer information from old information plate to replacement information plate (90).
 - (b) Position replacement information plate (90) on stator and housing assembly (680) in place of the old information plate and secure with two drive screws (100).

WARNING: KEEP ACRYLIC COATING AWAY FROM SOURCES OF IGNITION. CONFORMAL COATING IS FLAMMABLE.

WARNING: DO NOT GET ACRYLIC COATING ON YOUR SKIN AND DO NOT BREATHE THE FUMES. ACRYLIC COATING IS A POISONOUS MATERIAL.

- (c) Coat replacement information plate (90) with acrylic coating and let the coating air dry.
- (3) If instruction decal (110) or FAA-PMA label (-120) is damaged or unreadable:
 - (a) Remove instruction decal (110) or FAA-PMA label (-120) from stator and housing assembly (680).

WARNING: ISOPROPYL ALCOHOL IS DANGEROUS TO PERSONS. USE ONLY IN AN AREA WITH A GOOD FLOW OF AIR. KEEP AWAY FROM SOURCES OF HEAT, SPARKS AND FLAME. DO NOT BREATHE THE FUMES. PREVENT SKIN CONTACT. PUT ON PROTECTIVE CLOTHING AND EYE PROTECTION.

- (b) Clean surface of stator and housing assembly (680) with cloth moistened with isopropyl alcohol.
 - (c) Apply new instruction decal (110) to stator and housing assembly (680) next to terminal block.
 - (d) Apply new FAA-PMA label (-120) to stator and housing assembly (680) next to identification plate (70).

NOTE: Make sure decals are oriented to read in the same direction as the identification plate.

5. Performance Acceptance Test on Starter-Generator

CAUTION: ACCEPTANCE TESTING CAN DAMAGE CAPACITORS IN THE STARTER-GENERATOR.

A. Attach a dummy terminal block, refer to DISASSEMBLY section and ASSEMBLY section for details.

NOTE: Alternatively, remove the screw that secures the terminal block to the housing that is adjacent to terminal D. Cover the terminal lug attached to the terminal block with electrical tape.

ASSEMBLY

24-30-63



Component Maintenance Manual with Illustrated Parts List
DC Starter-Generator, 23046 Series I

- B. Perform acceptance test procedure on the assembled generator as detailed in the TESTING AND FAULT ISOLATION section.**

6. Final Assembly After Acceptance Testing

NOTE: Unless otherwise specified, numbers in parentheses () refer to item numbers given in Figure 10001 of the ILLUSTRATED PARTS LIST.

NOTE: After starter-generator has successfully completed and passed acceptance testing, proceed with following final assembly instructions.

- A. Attach a serviceable terminal block (690) to stator and housing assembly (680) or replace the screw (700) that secures the terminal block to the housing.**

NOTE: A dummy terminal block is used during acceptance testing, or the terminal block securing screw was removed.

- (1) Remove dummy terminal block according to instructions in Paragraph 4.R. of the DISASSEMBLY section.
- (2) Attach a terminal block according to instructions in Paragraph 4.A. of this section.

- B. Remove commutation viewing adapter.**

Unthread screw (140) from blind rivet nut that secures commutation viewing adapter to starter-generator.

- C. Attach brush access cover (130).**

- (1) Position brush access cover (130) around stator and housing assembly (680) with air bleed holes towards terminal block.

CAUTION: TO PREVENT ELECTRICAL SHORTS. DO NOT POSITION BRACKETS DIRECTLY OVER OPENING IN STATOR AND HOUSING ASSEMBLY.

- (2) Thread screw (140) into blind rivet nut (150). Tighten screw to a torque of 25.0 to 35.0 lbf.in. (2,8 to 4,0 N•m).

- D. If present, attach QAD mounting kit (10) to starter-generator.**

NOTE: When a starter-generator has been removed from service, the QAD mounting kit usually stays on the aircraft.

- (1) Engage guide pins (-60) of end bell, drive end pad (50) to pin holes in drive end bearing support assembly (-350).
- (2) Be sure that guide pins are correctly located and that pilot diameters of drive end pad and drive end end bell assembly are fully engaged.
- (3) While holding pad firmly in place, put rim-clenching clamp (20) over flanges of pad and drive end end bell assembly.



Component Maintenance Manual with Illustrated Parts List
DC Starter-Generator, 23046 Series I

- (4) Tighten self-locking nut (-30) onto T-bolt (-40) to lock QAD mounting kit to starter-generator. Tighten the self-locking nut to the torque etched on the clamp.

E. Prepare starter-generator for storage/shipment.

Refer to STORAGE section for storage/shipment instructions.



Component Maintenance Manual with Illustrated Parts List
DC Starter-Generator, 23046 Series I

FITS AND CLEARANCES

1. Torque Values

NOTE: Unless otherwise specified, numbers in parentheses () refer to item numbers in Figure 10001 of ILLUSTRATED PARTS LIST.

Table 8001 gives information necessary to tighten fasteners to specified torque. Values shown do not include frictional torque caused by self-locking devices or rundown resistance. Frictional torque values must be added to the specified torque.

Nuts, bolts, and screws not listed in Table 8001 are to be tightened in accordance with Goodrich Corporation methods and techniques or practices acceptable to your administration as specified in FAR PART 43.

Nomenclature and IPL Number	Torque Acceptance Limits
Screw (140)	25.0 to 35.0 lbf.in. (2,8 to 4,0 N•m)
Screw (170)	22.7 to 35.0 lbf.in. (2,6 to 4,0 N•m)
Screw, Binding Head (190)	7.7 to 10.3 lbf.in. (0,87 to 1,16 N•m)
Screw (220)	22.7 to 35.0 lbf.in. (2,6 to 4,0 N•m)
Nut, Self-locking (260)	100 to 120 lbf.in. (11,3 to 13,6 N•m)
Screw, Flat Head (360)	13.5 to 18.0 lbf.in. (1,5 to 2,0 N•m)
Screw (490)	13.5 to 18.0 lbf.in. (1,5 to 2,0 N•m)
Bolt (560)	25 to 30 lbf.in. (2,8 to 3,4 N•m)
Screw (670)	9.0 to 12.0 lbf.in. (1,0 to 1,4 N•m)
Screw (700)	22.7 to 35.0 lbf.in. (2,6 to 4,0 N•m)

Table 8001 - Torque Values



Component Maintenance Manual with Illustrated Parts List
DC Starter-Generator, 23046 Series I

2. Component Acceptance Limits

Refer to Table 8002 for acceptance limits applicable to starter-generator components and subassemblies. Paragraph designations in the CHECK column refer to applicable inspection procedures found in the CHECK section, except where noted.

NOTE: Unless otherwise specified, Item numbers in parentheses () refer to items found in Figure 10001 of the ILLUSTRATED PARTS LIST.

CHECK	Nomenclature and IPL Number	Inspect for...	Acceptance Limits
8.B.	Drive End Pad End Bell (Mounting Adapter) (50)	Pilot flange diameter P/N 23046-1660 and 23046-1662	4.122 to 4.120 inch (104,70 to 104,65 mm)
		P/N 23046-1790	3.185 to 3.184 inch (80,90 to 80,87 mm)
		P/N 23046-2090	3.185 to 3.183 inch (80,90 to 80,85 mm)
		P/N 23046-1761	3.249 to 3.247 inch (82,52 to 82,47 mm)
		Guide pin height	0.11 to 0.13 inch (2,8 to 3,3 mm)
8.K.	Drive Shaft (290)	16-tooth spline, diameter over 0.096 inch (2,438 mm) gage pins	0.921 inch (23,393 mm) min.
		12-tooth spline, diameter over 0.1094 inch (2,779 mm) gage pins	0.757 inch (19,228 mm) min.
8.L.(3)	Dampener Plate (320)	Thickness	0.038 inch (0,965 mm) min.
		Internal spline diameter between 0.090 inch (2,28 mm) diameter gage pins	0.6547 inch (16,629 mm) max.
8.M.	Friction Ring (330)	Thickness	0.060 inch (1,524 mm) min.
8.O	Drive End Bearing Support Assembly (-350)	Bearing liner diameter.	1.8501 to 1.8504 inch (46,993 to 47,000 mm). See Figure 8001.

Table 8002 - Acceptance Limits



Component Maintenance Manual with Illustrated Parts List
DC Starter-Generator, 23046 Series I

CHECK	Nomenclature and IPL Number	Inspect for...	Acceptance Limits
8.R.	Armature (470)	Commutator bar to bar (Before refinishing)	Bar to Bar. 0.0008 inch (0,020 mm) max.
		Bearing journal diameter (A,C) See Figure 5011.	0.7872 to 0.7875 inch (19,995 to 20,003 mm). See Figure 8002.
		Commutator outside diameter (B)	2.060 inch (52,197 mm) min.
		Mica undercut (before recut)	Depth: 0.045 inch (1,14 mm) min. to 0.070 inch (1,78 mm) max. Width: 0.035 inch (0,89 mm) to 0.045 inch (1,14 mm)
		Mica undercut (after recut)	Depth: 0.050 inch (1,27 mm) min. to 0.070 inch (1,78 mm) max. Width: 0.035 inch (0,89 mm) to 0.045 inch (1,14 mm)
		Commutator bar to bar acceptance test values	Bar to Bar: 0.0002 inch (0,005 mm) max. T.I.R: 0.0008 inch (0,020 mm) max.
		Commutator bar to bar and concentricity (After refinishing)	Bar to Bar: 0.0001 inch (0,0025 mm) max T.I.R: 0.0005 inch (0,0127 mm) max.
		Balance	5 grain inch (8,2 gram mm) max. See SPD 1001
8.S.	Bearing and brush support assembly (-480)	Bearing liner diameter.	1.8501 to 1.8504 inch (46,993 to 47,000 mm) See Figure 8001.
		Brush spring force P/N 23033-1120	Position 'A' 2.0 lb (0,9 kg) min. Position 'B' 3.85 lb (1,7 kg) max.
		P/N 23064-1350	Position 'A' 2.7 lb (1,2 kg) min. Position 'B' 3.6 lb (1,6 kg) max.
		P/N 23033-1121	Position 'A' 2.62 lb (1,2 kg) min. Position 'B' 3.4 lb (1,5 kg) max.

Table 8002 - Acceptance Limits (Continued)

FITS AND CLEARANCES



Component Maintenance Manual with Illustrated Parts List
DC Starter-Generator, 23046 Series I

CHECK	Nomenclature and IPL Number	Inspect for...	Acceptance Limits
8.V.	Terminal block (690)	Capacitance values B+ to E- C to Ground B+ to Ground	1.6 to 3.0 μ F with a value of 100 VDC/<1 μ A. 1.6 to 3.0 μ F with a value of 100VDC/<1 μ A. 0.0176 to 0.033 μ F with a value of 100 VDC/<1 μ A.

Table 8002 - Acceptance Limits (Continued)

3. Bearing Liner and Journal Limits

A. Bearing liners. See Figure 8001.

During overhaul/service inspection, diameter of bearing liner is considered to be acceptable if not more than 25% of circumference is above the maximum diameter limit. See Figure 8001. When the starter-generator is to be repaired and returned to service on a "continued time" basis, the maximum diameter limit of bearing liner diameter is acceptable if up to 25% of the circumference is not more than 0.0001 inch (0,0025 mm) above maximum diameter limit.

NOTE: Acceptance criteria for used liners, not new or repaired liners.

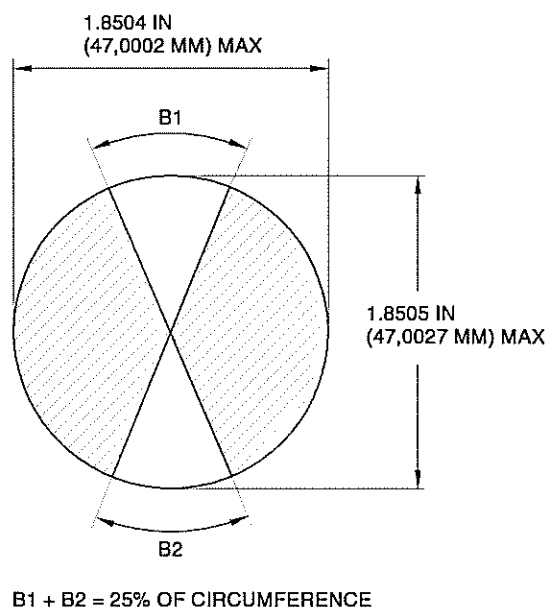


Figure 8001 - Measuring Bearing Liner Diameters



Component Maintenance Manual with Illustrated Parts List
DC Starter-Generator, 23046 Series I

B. Bearing journals. See Figure 8002.

During overhaul/service inspection, diameter of bearing journal is considered to be acceptable if not more than 25% of circumference is below the minimum diameter limit. See Figure 8002. When the starter-generator is to be repaired and returned to service on a "continued time" basis, the minimum diameter limit of bearing journal diameter is acceptable if up to 25% of the circumference is not more than 0.0001 inch (0,0025 mm) below minimum diameter limit.

NOTE: Acceptance criteria for used journals, not new or repaired journals.

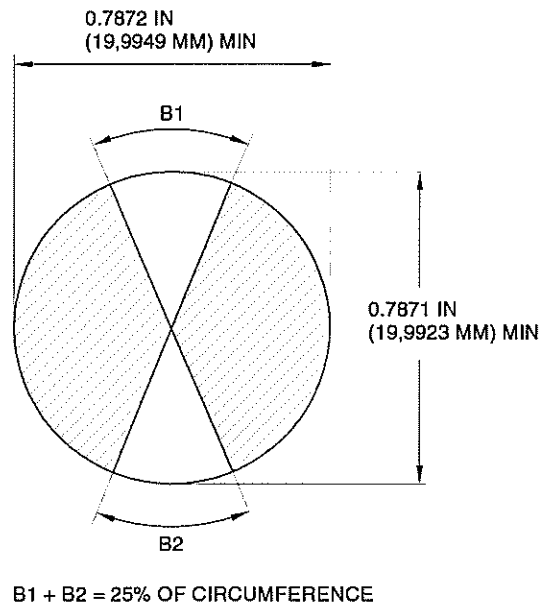


Figure 8002 - Measuring Armature Bearing Journal Diameters



Component Maintenance Manual with Illustrated Parts List
DC Starter-Generator, 23046 Series I

THIS PAGE INTENTIONALLY LEFT BLANK



Component Maintenance Manual with Illustrated Parts List
DC Starter-Generator, 23046 Series I

SPECIAL TOOLS, FIXTURES, AND EQUIPMENT

1. Introduction

Table 9001 lists the approved special tools, fixtures, and equipment necessary for use during repair and overhaul of 23046 Series I starter generators.

Reference Standard Practice Document (SPD) 1004 for the 23046 Series I Brush Holder Alignment Fixture.

Item	Figure No.
Armature Shaft Adapter, Drive End	Figure 9016
Armature Shaft Adapter, Anti-drive End	Figure 9002
Commutation Viewing Adapters	Figure 9003
Inner Race Bearing Driver	Figure 9004
Outer Race Bearing Driver	Figure 9005
Commutator Turning Fixture	Figure 9006
Armature Support	Figure 9007
Bearing and Brush Support Assembly Support	Figure 9008
Drive End Hub Support	Figure 9009
Horizontal Stator Support	Figure 9010
Vertical Stator Support	Figure 9011
Dampener Hub Driver	Figure 9012
Dampener Plate Driver	Figure 9013
Spline Wrench	Figure 9014
Anti-Drive End Bearing Hub Support	Figure 9015
PlusNut® Fastener Header	Figure 9016

Table 9001 - Special Tools, Fixtures, and Equipment

2. Illustrated Description of Tools

The drawings in this section provide material and fabrication instructions for the tools listed in Table 9001. Where dimensions and fabrication instructions are provided, the tool or fixture is approved for local manufacture.

The tools and fixtures cannot be procured from Goodrich Corporation. Equivalent tooling and fixture requirements meeting with Goodrich Corporation specifications can be used.



Component Maintenance Manual with Illustrated Parts List
DC Starter-Generator, 23046 Series I

A. Armature shaft adapters. See Figure 9001 and Figure 9002.

Tolerances on decimals 0.XX $\pm$ 0.01 inch (0,25 mm), 0.XXX $\pm$ 0.005 inch (0,127 mm).
Machined ends to be parallel to within 0.001 inch (0,025 mm) and square with
sidewalls within $\pm$ 0.05 degree. Materials: Brass.

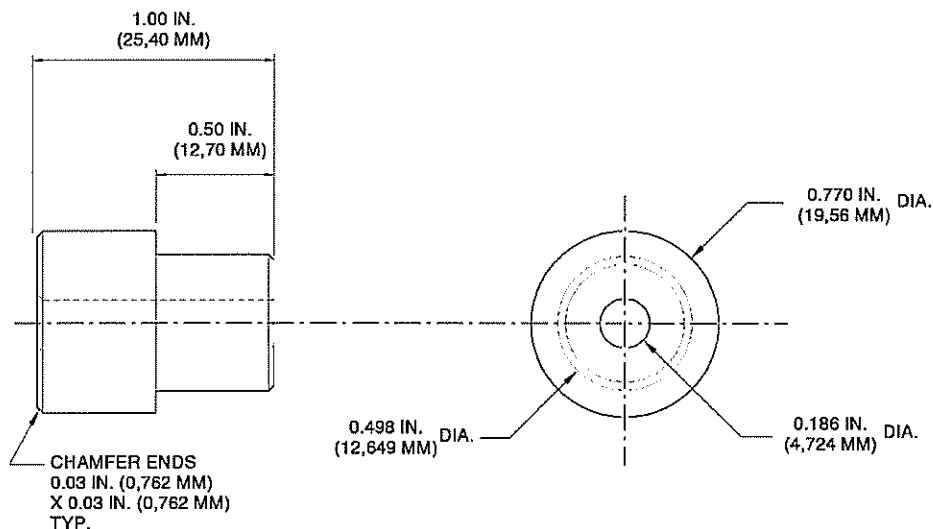


Figure 9001 - Armature Shaft Adapter, Drive End

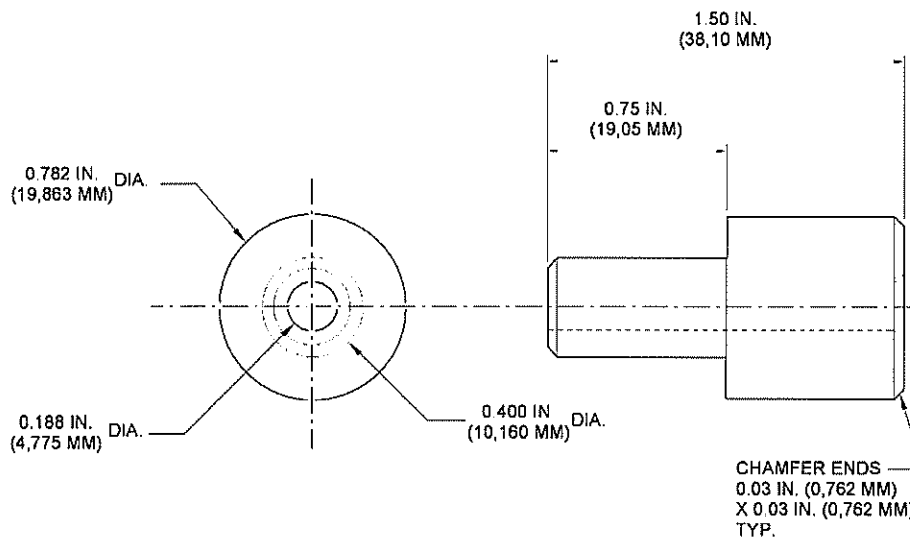


Figure 9002 - Armature Shaft Adapter, Anti-Drive End



Component Maintenance Manual with Illustrated Parts List
DC Starter-Generator, 23046 Series I

B. Commutation viewing adapters. See Figure 9003.

This adapter is constructed using a used brush access cover. Two designs are shown. Design one uses four pieces of acrylic, 2.00 inch (50,8 mm) high by 1.75 inch (44,5 mm) wide by 0.63 inch (1,60 mm) thick. Design two uses one piece of acrylic, 19.00 inch (482,6 mm) long by 2.30 inch (58,4 mm) wide by 0.063 inch (1,60 mm) thick. Construction of these designs is as follows:

(1) Design One

- (a) Cut four holes in a brush access cover. Each hole should be positioned above one of the four brush holder assemblies to allow for viewing the brushes as they make contact with the commutator during test.
- (b) Cut four pieces of clear acrylic to be larger than the brush access holes in the stator and housing assembly.
- (c) Apply adhesive on the inside surface of the viewing adapter to the frame of the cut out viewing windows.
- (d) Attach the four pieces of acrylic to the viewing windows on the inner surface of the viewing adapter using four pop rivets.

(2) Design Two

- (a) Remove brackets from a used brush access cover.
- (b) Attach brackets to a piece of clear acrylic.

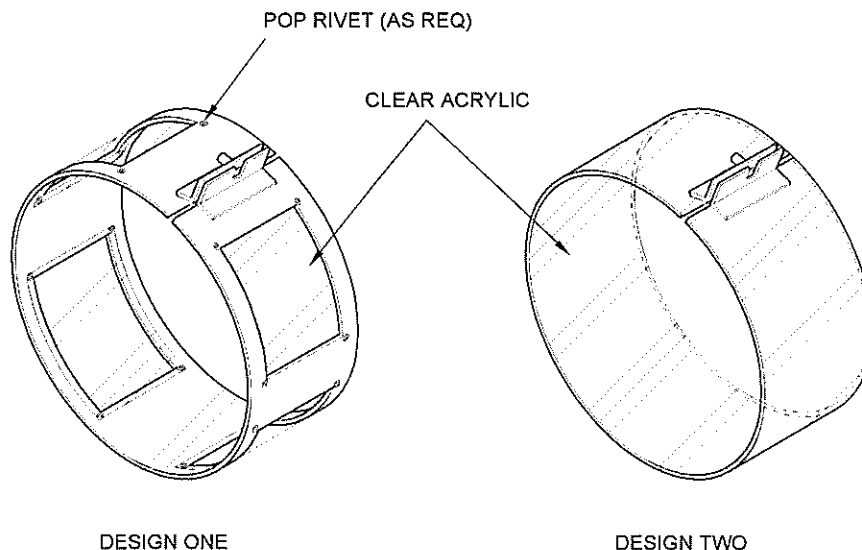


Figure 9003 - Commutation Viewing Adapters



Component Maintenance Manual with Illustrated Parts List
DC Starter-Generator, 23046 Series I

C. Bearing drivers. See Figure 9004 and Figure 9005

Tolerances on decimals 0.XX $\pm$ 0.01 inch (0,25 mm), 0.XXX $\pm$ 0.005 inch (0,127 mm).
Machined ends to be parallel to within 0.001 inch (0,025 mm) and square with
sidewalls within $\pm$ 0.05 degree. Harden steel to Rockwell Rc 55-60. Material: 1040
Steel, 2.00 inch (50,8 mm) diameter stock size.

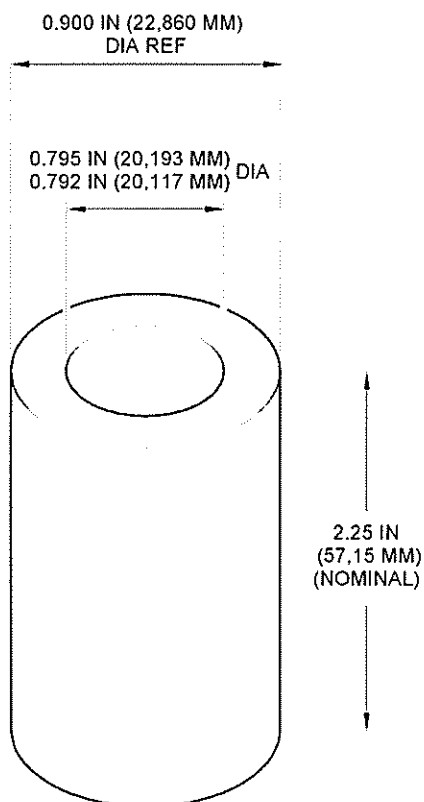


Figure 9004 - Inner Race Bearing Driver



Component Maintenance Manual with Illustrated Parts List
DC Starter-Generator, 23046 Series I

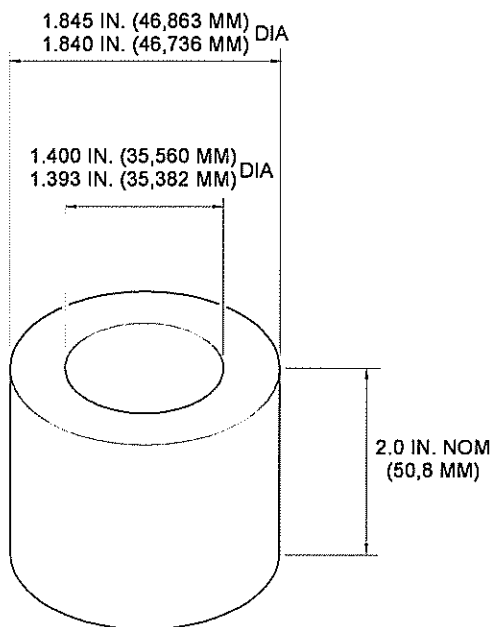


Figure 9005 - Outer Race Bearing Driver



Component Maintenance Manual with Illustrated Parts List
DC Starter-Generator, 23046 Series I

D. Commutator turning fixture. See Figure 9006.

Tolerances on decimals 0.XX ± 0.01 inch (0,25 mm), 0.XXX ± 0.005 inch (0,127 mm). Machined ends to be parallel to within 0.001 inch (0,025 mm) and square with sidewalls within ± 0.05 degree. Harden steel to Rockwell Rc 35-40. Material: 1040 Steel, Stock size: 2.50 inch (63,5 mm) diameter.

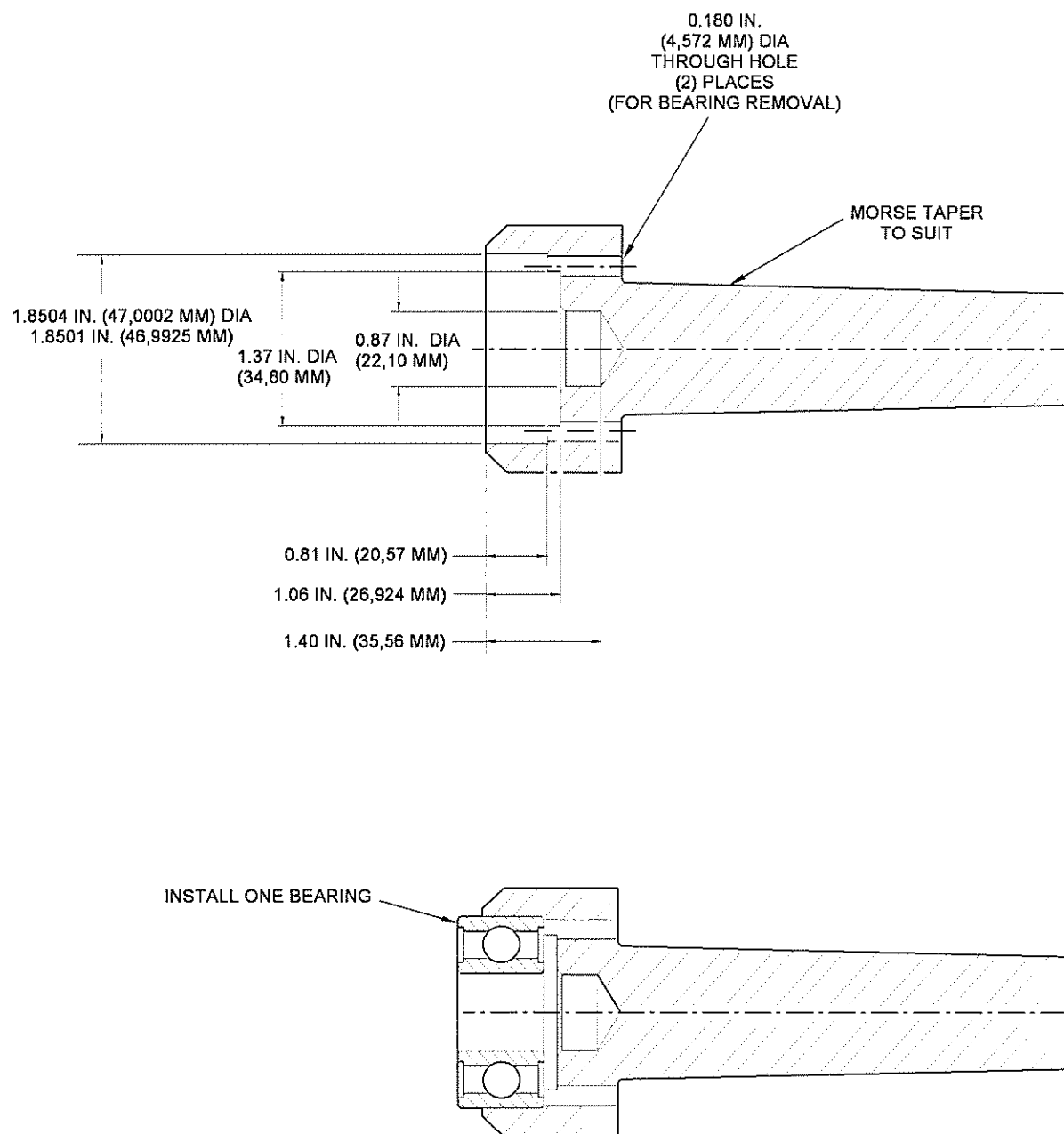


Figure 9006 - Commutator Turning Fixture



Component Maintenance Manual with Illustrated Parts List
DC Starter-Generator, 23046 Series I

E. Armature support. See Figure 9007.

Material 0-1 tool steel, Rc 35-40. Tolerances on decimal 0.XX ± 0.01 inch (0,25 mm) 0.XXX ± 0.005 inch (0,127 mm). Machined ends to be parallel to within 0.001 inch (0,025 mm) and square with bore within ± 0.05 degree.

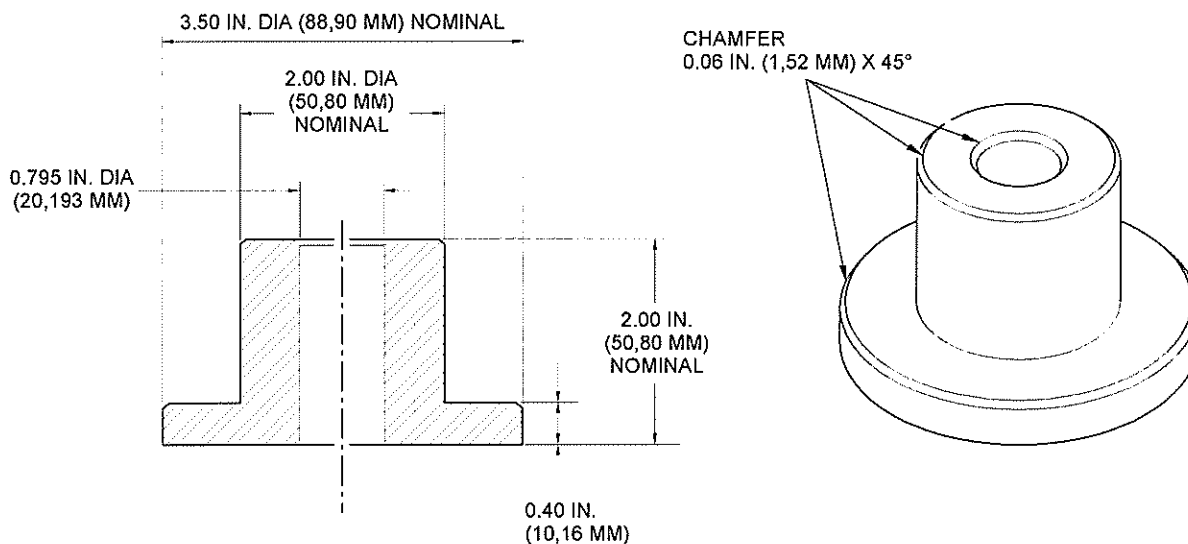


Figure 9007 - Armature Support



Component Maintenance Manual with Illustrated Parts List
DC Starter-Generator, 23046 Series I

F. Bearing and brush support assembly support. See Figure 9008.

Tolerances on decimals 0.XX $\pm$ 0.01 inch (0,25 mm), 0.XXX $\pm$ 0.005 inch (0,127 mm).
Tolerance on angles $\pm$ 5 degree. Material: 01 tool steel or equivalent.

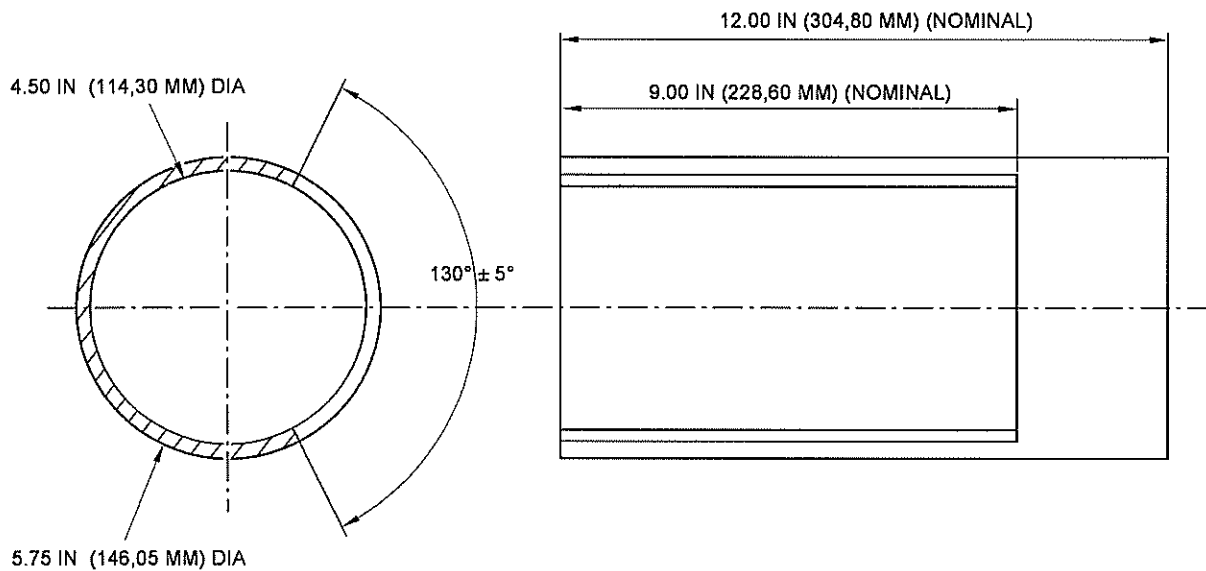


Figure 9008 - Bearing and Brush Support Assembly Support



Component Maintenance Manual with Illustrated Parts List
DC Starter-Generator, 23046 Series I

G. Drive end hub support. See Figure 9009.

Tolerances on decimals 0.XX $\pm$ 0.01 inch (0,25 mm), 0.XXX $\pm$ 0.005 inch (0,127 mm).
Machined ends to be parallel to within 0.001 inch (0,025 mm) and square with
sidewalls within $\pm$ 0.05 degree. Material is 1040 Steel hardened to Rc 35-40, Stock
size: 2.00 inch (50,8 mm) diameter.

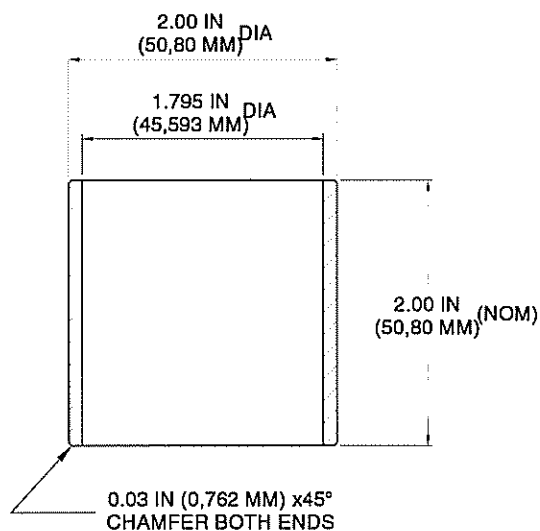


Figure 9009 - Drive End Hub Support



Component Maintenance Manual with Illustrated Parts List
DC Starter-Generator, 23046 Series I

H. Horizontal stator support. See Figure 9010.

All dimensions are nominal. The material is a suitable hardwood.

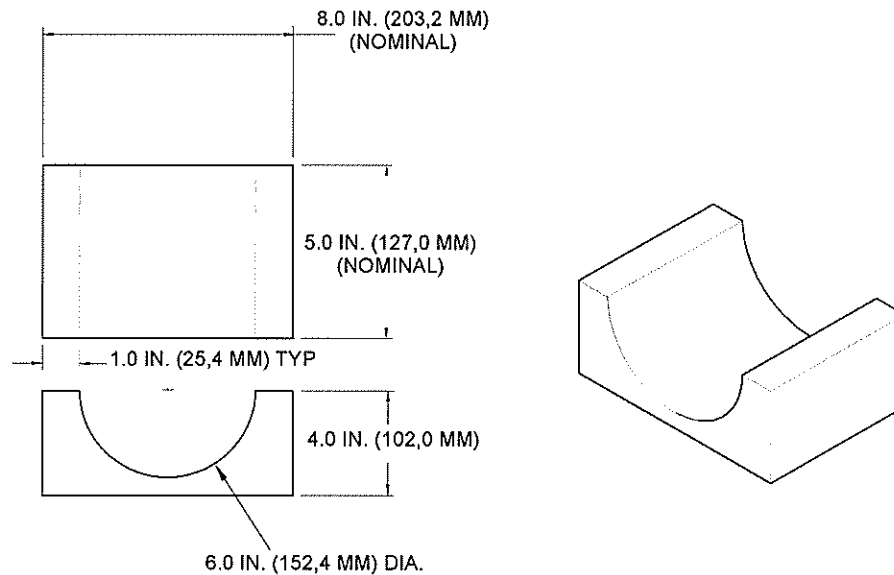


Figure 9010 - Horizontal Stator Support



Component Maintenance Manual with Illustrated Parts List
DC Starter-Generator, 23046 Series I

I. Vertical stator support. See Figure 9011.

All dimensions are nominal. The material is a suitable hardwood.

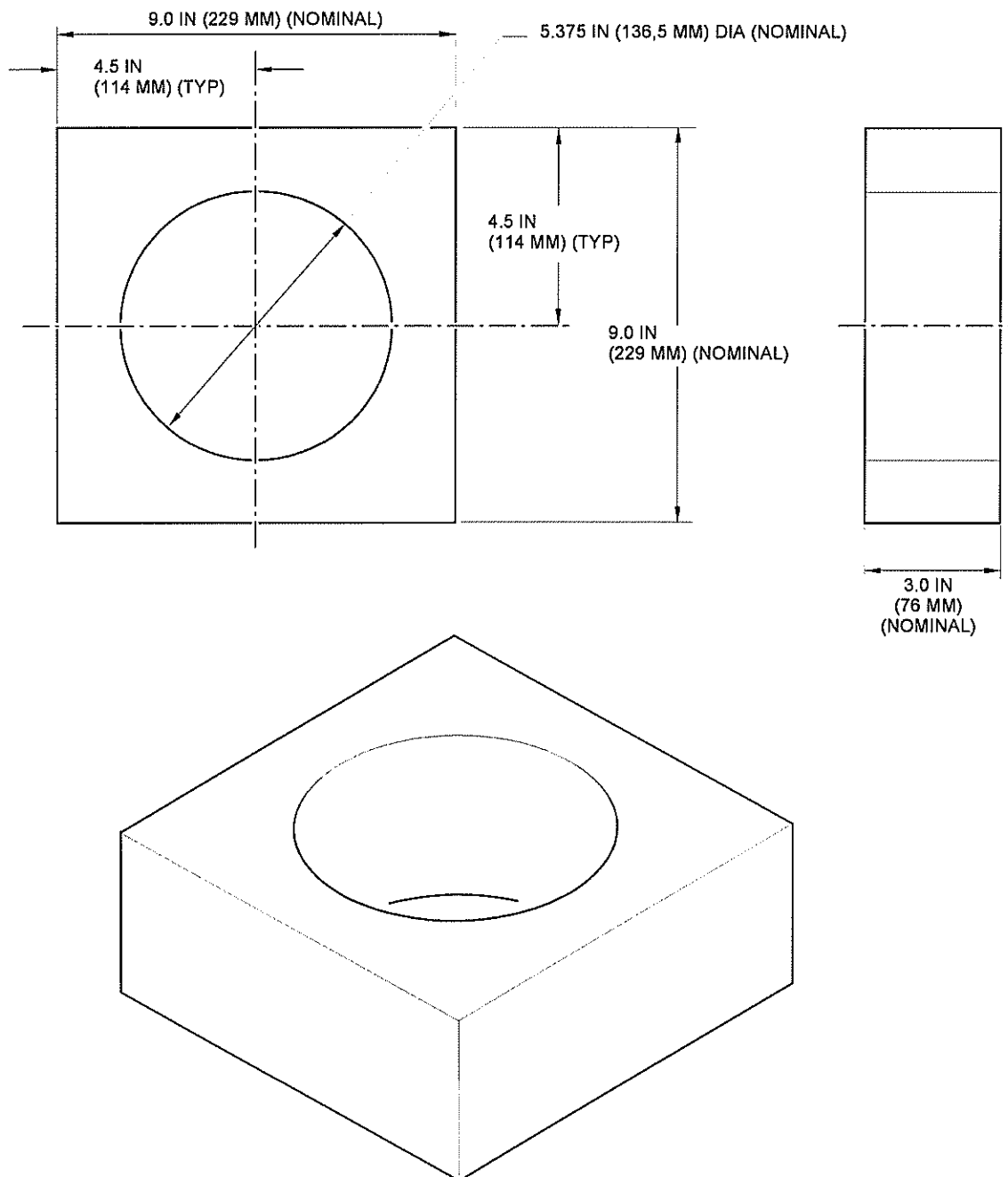


Figure 9011 - Vertical Stator Support



Component Maintenance Manual with Illustrated Parts List
DC Starter-Generator, 23046 Series I

J. Dampener hub driver. See Figure 9012.

Tolerances on decimals 0.XX $\pm$ 0.01 inch (0,25 mm), 0.XXX $\pm$ 0.005 inch (0,127 mm).
Material: 1040 Steel.

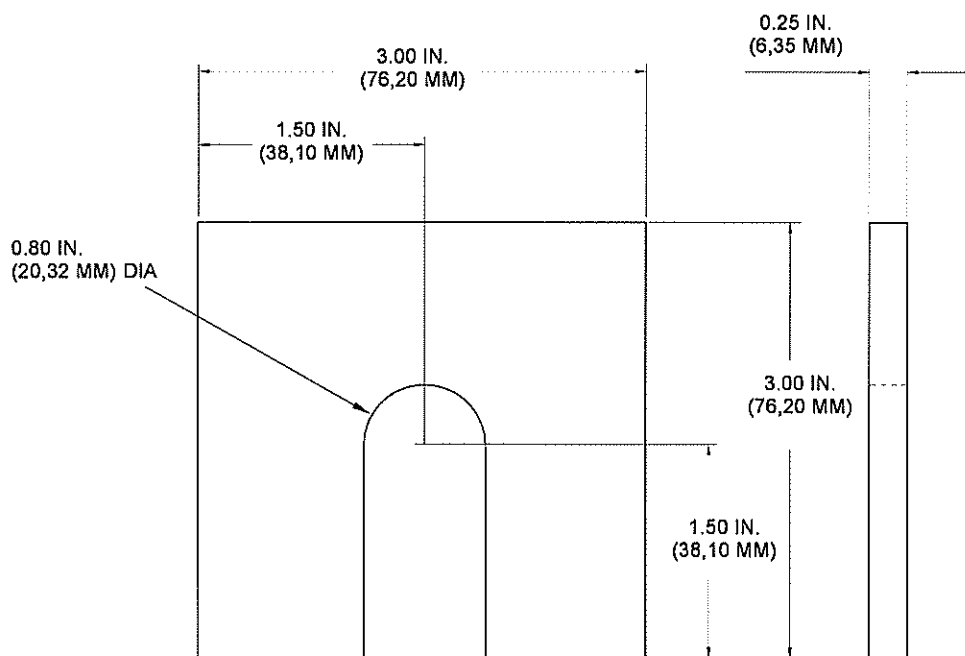


Figure 9012 - Dampener Hub Driver



Component Maintenance Manual with Illustrated Parts List
DC Starter-Generator, 23046 Series I

K. Dampener plate driver. See Figure 9013.

Tolerances on decimals 0.XX $\pm$ 0.01 inch (0,25 mm), 0.XXX $\pm$ 0.005 inch (0,127 mm).
Tolerance on angles $\pm$ 1 degree. Material: 01 tool steel or equivalent.

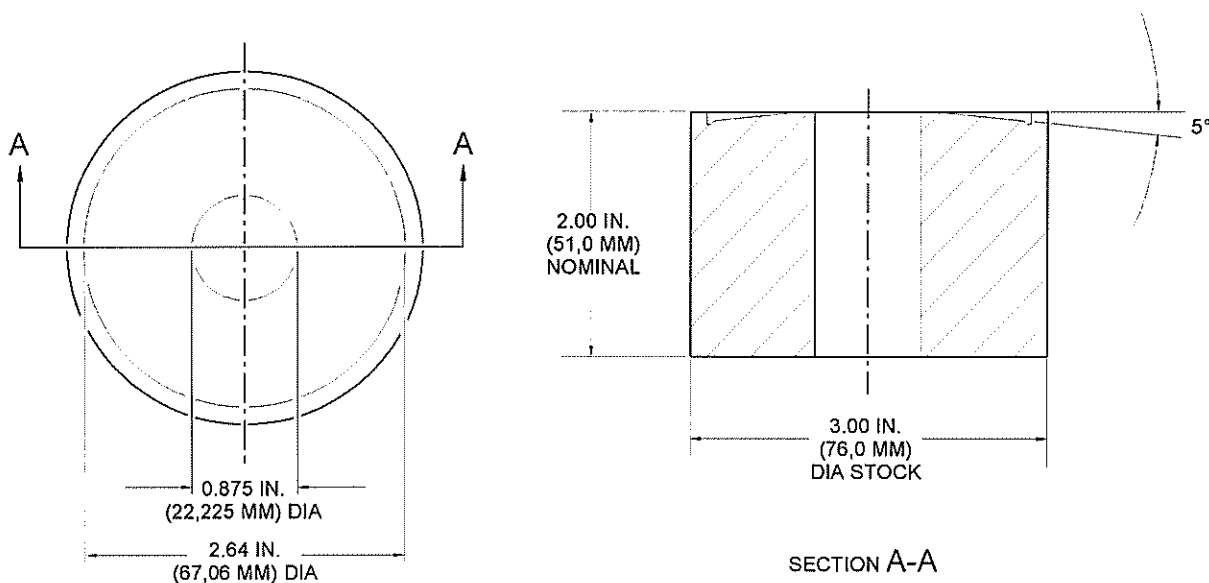


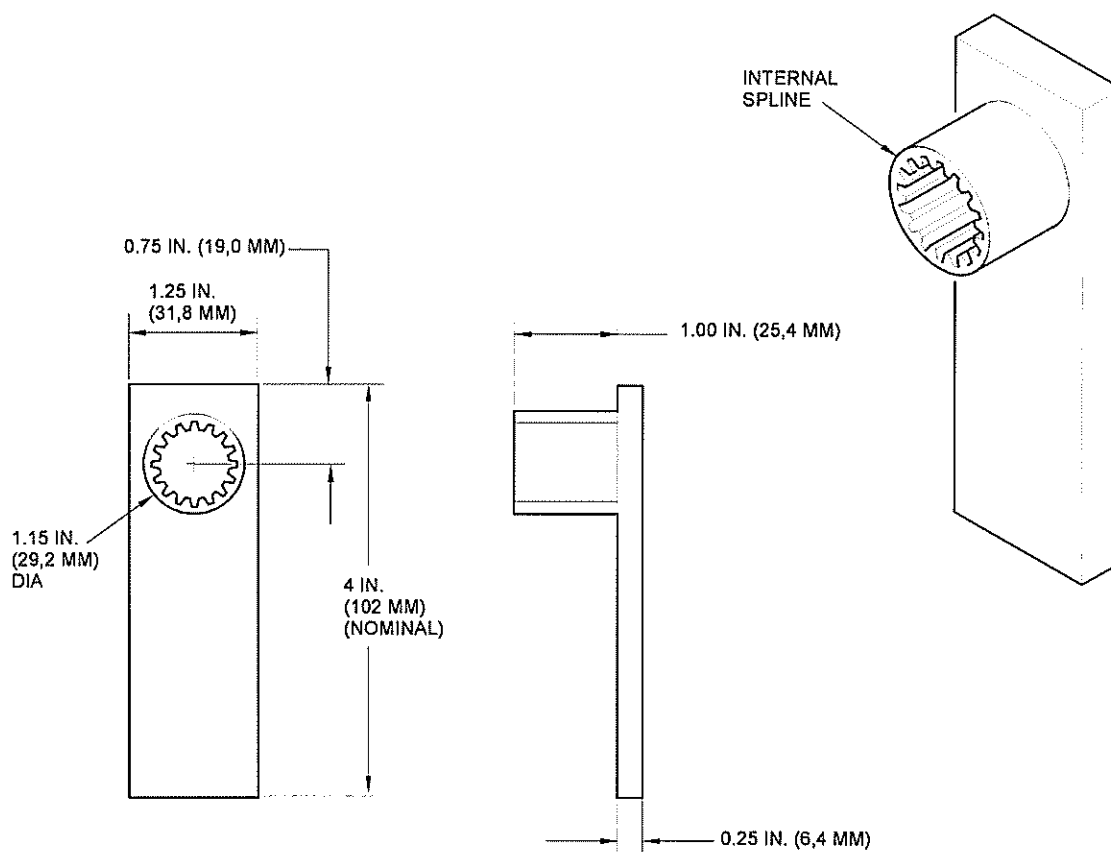
Figure 9013 - Dampener Plate Driver



Component Maintenance Manual with Illustrated Parts List
DC Starter-Generator, 23046 Series I

L. Spline wrench. See Figure 9014.

Tolerances on decimals 0.XX $\pm$ 0.01 inch (0,25 mm), 0.XXX $\pm$ 0.005 inch (0,127 mm).
Machined ends to be parallel to within 0.001 inch (0,025 mm) and square with
sidewalls within $\pm$ 0.05 degree. Angles $\pm$ 2 degree. Material is 1040 Steel hardened
to Rockwell C 55 to 60.



12 TOOTH SPLINE

INTERNAL INVOLUTE SPLINE DATA		
FILLET ROOT SIDE FIT	INCH	MM
NUMBER OF TEETH	12	
DIAMETRAL PITCH	20/40	
PRESSURE ANGLE	30°	
PITCH DIAMETER	0.6000	15,240
MINOR DIAMETER (MIN)	0.5550	14,097
MAJOR DIAMETER (MIN)	0.6500	16,510
CHORD SPACE (MIN)	0.0730	1,853
PIN DIAMETER	0.0720	1,829
BETWEEN THREE 0.0720 IN (1,829 MM) PINS (MIN)	0.5095	12,941

16 TOOTH SPLINE

INTERNAL INVOLUTE SPLINE DATA		
FILLET ROOT SIDE FIT	INCH	MM
NUMBER OF TEETH	16	
DIAMETRAL PITCH	20/40	
PRESSURE ANGLE	30°	
PITCH DIAMETER	0.8000	20,320
MINOR DIAMETER (MIN)	0.7270	18,590
MAJOR DIAMETER (MIN)	0.8640	21,946
CHORD SPACE (MIN)	0.0728	1,849
PIN DIAMETER	0.0720	1,829
BETWEEN THREE 0.0720 IN (1,829 MM) PINS (MIN)	0.7096	18,024

Figure 9014 - Spline Wrench



Component Maintenance Manual with Illustrated Parts List
DC Starter-Generator, 23046 Series I

M. Anti-drive end bearing hub support. See Figure 9015.

Tolerances on decimals $0.XX \pm 0.01$ inch (0,25 mm), $0.XXX \pm 0.005$ inch (0,127 mm).
Machined ends to be parallel to within 0.001 inch (0,025 mm) and square with
sidewalls within ± 0.05 degree. Material is 1040 Steel hardened to Rc 35-40, Stock
size: 2.50 inch (63,5 mm) diameter.

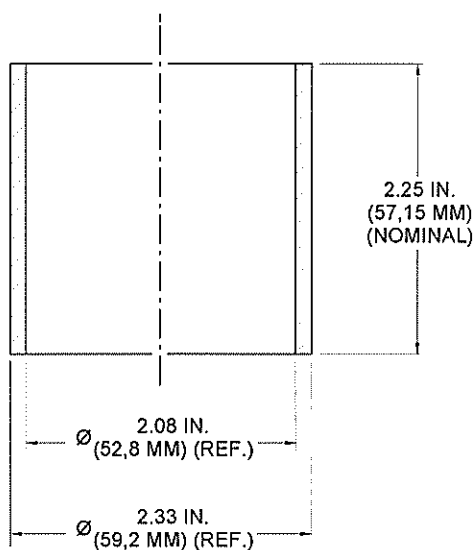


Figure 9015 - Anti-Drive End Bearing Hub Support



Component Maintenance Manual with Illustrated Parts List
DC Starter-Generator, 23046 Series I

N. PlusNut® fastener header. See Figure 9016.

Part Number: C1000-1032

Vendor Cage Code: V 0ZVN9

Vendor details: Omni Fasteners Inc.
DBA Fastener Distributor
909 Towpath Road
Broadview Heights, OH
Ph: 440-838-1800
Fax: 440-838-6200

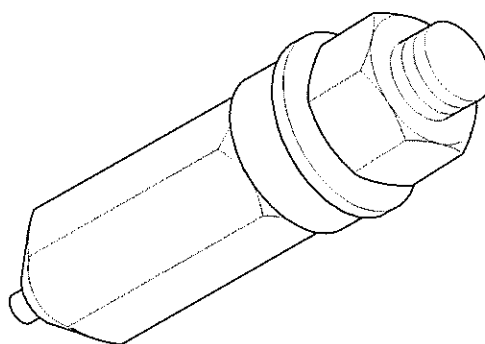


Figure 9016 - PlusNut® Fastener Header



Component Maintenance Manual with Illustrated Parts List
DC Starter-Generator, 23046 Series I

SPECIAL PROCEDURES

1. Introduction

This section is unassigned.



Component Maintenance Manual with Illustrated Parts List
DC Starter-Generator, 23046 Series I

THIS PAGE INTENTIONALLY LEFT BLANK



Component Maintenance Manual with Illustrated Parts List
DC Starter-Generator, 23046 Series I

REMOVAL

1. Introduction

This section is unassigned.



Component Maintenance Manual with Illustrated Parts List
DC Starter-Generator, 23046 Series I

THIS PAGE INTENTIONALLY LEFT BLANK

REMOVAL

24-30-63

Page 12002
Mar 23/10



Component Maintenance Manual with Illustrated Parts List
DC Starter-Generator, 23046 Series I

INSTALLATION

1. Introduction

This section is unassigned.



Component Maintenance Manual with Illustrated Parts List
DC Starter-Generator, 23046 Series I

THIS PAGE INTENTIONALLY LEFT BLANK



Component Maintenance Manual with Illustrated Parts List
DC Starter-Generator, 23046 Series I

SERVICING

1. Introduction

This section is unassigned.



Component Maintenance Manual with Illustrated Parts List
DC Starter-Generator, 23046 Series I

THIS PAGE INTENTIONALLY LEFT BLANK



Component Maintenance Manual with Illustrated Parts List
DC Starter-Generator, 23046 Series I

STORAGE

1. Storage

Materials required to pack the starter-generator for storage or shipment are listed in Table 15001.

Description	Specification	Quantity
Bag, waterproof, vapor-proof	Commercially available. Bag must totally enclose instructions for installation and then be sealed.	1
Bag, waterproof, vapor-proof	Kraft-foil or suitable equivalent. Bag must totally enclose generator and then be sealed.	1
Box - domestic class, single wall, corrugated cardboard.	PPP-B-636 or suitable equivalent. Box must be large enough to totally enclose and restrain bagged and cushioned generator.	1
Box - WC5 overseas shipping container.	PPP-B-636 or suitable equivalent. Box must be large enough to completely enclose domestic class packaging.	1
Cardboard Tubing	Commercially available	AR
Chemically Neutral Protective Paper	MIL-B-121A, Grade A, Type II	AR
Grease, Lubricating	MIL-PRF-81322	AR
Packing Material - shock absorbing foam rubber, styrofoam, bubble wrap, or expanded foam. (Goodrich Corporation's recommended method: 3 inch (76,2 mm) thick minimum expanded foam surrounding machine on all sides).	Commercially available.	AR
Polyethylene (Plastic) Wrap	Commercially available	AR
Tag	Commercially available	1 (Domestic) 2 (Internat'l)
Tape - waterproof, pressure sensitive.	Commercially available	AR

Table 15001 - Packing Material

STORAGE

24-30-63

Page 15001
Mar 23/10



Component Maintenance Manual with Illustrated Parts List
DC Starter-Generator, 23046 Series I

A. General information.

CAUTION: IF MACHINE STORAGE TIME IS MORE THAN 24 MONTHS WITH NO USE, IT IS RECOMMENDED THAT ITS BEARINGS BE REPLACED. IT MUST THEN BE RE-TESTED ACCORDING TO TESTING AND FAULT ISOLATION SECTION BEFORE BEING PLACED INTO SERVICE.

- (1) Unit must have successfully completed tests specified in TESTING AND FAULT ISOLATION section of this manual before preparing unit for shipment or storage.
- (2) Record following information to tag(s):
 - Model Number
 - Serial Number
 - Mod Status
 - Test Date (PASSED)
 - Packing Date
- (3) Use packaging materials as specified in Table 15001.

B. Documentation.

Include all applicable documentation with unit:

- Testing Records
- Repair Reports
- Final Inspection/Check Records
- Packing List and Certificate of Conformance

C. Domestic packaging.

NOTE: Unless otherwise specified, numbers in parentheses () refer to item numbers in Figure 10001 of ILLUSTRATED PARTS LIST.

- (1) For models 23046-019 and 23046-020, place O-ring (300) into a small plastic bag with instructions to user to lubricate and install O-ring onto drive end of drive shaft before installation of starter-generator into the aircraft. Attach plastic bag to machine.
- (2) Apply lubricating grease to drive spline according to MIL-PRF-81322 and wrap it in protective paper according to MIL-B-121A, Grade A, Type II. Secure with waterproof tape.
- (3) Put a section of cardboard tubing over drive shaft to prevent drive shaft from puncturing shipping container.

STORAGE

24-30-63

Page 15002
Mar 23/10



Component Maintenance Manual with Illustrated Parts List
DC Starter-Generator, 23046 Series I

- (4) Place unit into a waterproof and vapor-proof heat-seal bag and partially heat-seal it while forcing as much air as possible from bag.
- (5) Insert vacuum tube into bag to draw remaining air out of bag.
- (6) Withdraw vacuum tube quickly and complete heat-sealing of bag.
- (7) Using machine's original shipping container if possible, place bagged machine into box surrounded with a minimum of 3 inch (76,2 mm) thick shock absorbing, cushioning material on all six sides.
- (8) Securely and completely seal all flapped openings of box with tape.
- (9) Tape tag to exterior surface of box. Make sure all information is visible.

D. Overseas shipment on surface vessels.

- (1) Accomplish domestic packaging of machine in accordance with Paragraph 1.C..
- (2) Place domestically packaged starter-generator into WC5 shipping container.
- (3) Securely and completely, seal all flapped openings of shipping container with tape.
- (4) Tape tag to exterior surface of shipping container. Make sure all information is visible.



Component Maintenance Manual with Illustrated Parts List
DC Starter-Generator, 23046 Series I

THIS PAGE INTENTIONALLY LEFT BLANK



Component Maintenance Manual with Illustrated Parts List
DC Starter-Generator, 23046 Series I

REWORK

1. Introduction

This section is unassigned.

REWORK

24-30-63

Page 16001
Mar 23/10



Component Maintenance Manual with Illustrated Parts List
DC Starter-Generator, 23046 Series I

THIS PAGE INTENTIONALLY LEFT BLANK



Component Maintenance Manual with Illustrated Parts List
DC Starter-Generator, 23046 Series I

ILLUSTRATED PARTS LIST

1. Introduction

A. General

- (1) This section provides a listing of assemblies and detail parts for the 23046 Series I DC Starter-Generator.
- (2) All replacement parts are manufactured or source-controlled by Goodrich Corporation with the exception of those parts given an MS, AN, NAS, ANSI, ASME or ISO, etc. part number. See Paragraph 2.B. below.

CAUTION: ANY USE OF PARTS, MATERIALS OR PROCEDURES NOT AUTHORIZED BY GOODRICH CORPORATION FOR MAINTENANCE OR OVERHAUL OF UNIT(S) CAN AFFECT CONTINUED AIRWORTHINESS OR INVALIDATE CERTIFICATION.

To order authorized Goodrich Corporation parts, contact your regional Goodrich Corporation Customer Service Center.

- (3) The Illustrated Parts List (IPL) is divided into the parts that follow:
 - Vendor List (when applicable)
 - Numerical List (when applicable)
 - Detailed Parts List

B. Vendor List (when applicable)

- (1) The Vendor List is included in the IPL when parts are not supplied by Goodrich Corporation.
- (2) The list gives all supplier codes used in this manual and the names and addresses of the suppliers.

C. Numerical Index (when applicable)

- (1) A Numerical Index is included when there are more than five pages of Detailed Parts List.
- (2) The Numerical Index gives, in alpha-numerical sequence, all part numbers given in the Detailed Parts List. An Airline Stock Number column is given for airline use. The Total Req'd column gives the total quantity required for each part number at the figure and item entry.

D. Detailed Parts List

- (1) The Detailed Parts List includes parts lists and related illustrations.



Component Maintenance Manual with Illustrated Parts List
DC Starter-Generator, 23046 Series I

- (2) Each list is in disassembly sequence but that attaching parts are given immediately after the parts they attach. These are headed (ATTACHING PARTS).
- (3) More data on the Detailed Parts List is given in Paragraph 2.

2. Arrangement of Parts List

A. Figure and Item Number

The figure number indicates the figure where the part is illustrated. Each item number corresponds to a part illustrated in the applicable figure. Several item numbers may have a letter following a number (e.g. 10 A). This indicates that the part has the same function and location as the base number (e.g. 10), but can be different in form and material. An item number that is preceded by a dash (e.g. -30) is not illustrated in the applicable figure.

B. Part Number

The part numbers listed in this column are the only authorized parts for replacement and overhaul of the starter-generator. The list consists mainly of Goodrich Corporation part numbers but does include Military Standard (MS), Army Navy (AN), National Aerospace Standard (NAS) and industry standard (ANSI, ASME, ISO, etc.) part numbers.

C. Nomenclature

The proper name and type of each part is provided in this column. The description of each item is indented by columns to indicate the relationship to the next higher assembly (NHA). The number of indentures and bullets depicts the relationship of the item to the associated next higher assembly as follows:

- 1 2 3 4 5
Assembly or Installation Descriptive Title
 . Assembly
 . . Attaching Parts for Assembly
 . . . Detail Parts for Assembly
 . . . Subassembly
 Attaching Parts for Subassembly
 Detail Parts for Subassembly



Component Maintenance Manual with Illustrated Parts List
DC Starter-Generator, 23046 Series I

The interchangeability relationship between parts is identified in the NOMENCLATURE column of the parts list. A list of the terms used to show interchangeability and their definition is as follows:

Term	Parts List Abbreviation	Definition
Alternate	ALT	One of the two part numbers can be used. If the part number in the Nomenclature column is NOT specified as 'ALT', it can not be used as an alternative. The information is for reference only or as a way to help identify the part.
Modification	MOD	Modification (Mod) status details information about the effectivity of parts in regard to upgrades and modifications. Alpha variants (A through Y) (except I O Q S X and Z) are assigned to existing model numbers when necessary. Includes details of SB applicability.
Superseded	SUPSD BY	Part number listed is obsolete and must be replaced by new part listed at the next overhaul or repair. The superseded part is not to be installed or reinstalled.
Supersedes	SUPSDS	The part number listed must replace the obsolete part number at the next overhaul or repair.
Replaced by	REPLD BY	Part number listed is discontinued and should be replaced by the new part number. The original part may be used until current stock is depleted and then must be replaced by the new part listed.
Replaces	REPLS	The part listed replaces and is interchangeable with the item number shown in the notation.
Attaching Parts		Will be listed in the same column as the assembly they attach and the listing of parts will be followed by _____ * _____ in the same column.

D. Effect Code

This column establishes part relationships with starter-generator models, which are essentially the same, but have minor variations. These starter-generator models are the end items on the detailed parts list and are assigned reference letters such as A, B, C, D, etc. Subassemblies or detail parts, which are not common to all configurations, but are associated with one or more of the coded end items, carry the letter or letters assigned to the end item(s) with which they are associated. When parts are used on all models, the column is left blank. Effectivity codes are redefined for each major subassembly figure.



Component Maintenance Manual with Illustrated Parts List
DC Starter-Generator, 23046 Series I

E. Units Per Assembly

The numbers listed in this column indicate the quantity of parts used per assembly at the location shown and are not necessarily the total quantity per unit. For bulk items, the abbreviation AR is used to indicate the part quantity is "as required". The abbreviation NP indicates non-procurable and refers to items which are not procurable and may not be ordered. The abbreviation RF indicates that the item is listed for reference only.

3. Vendors List

In the case of an item supplied by a vendor and not listed under the prime manufacturer's part number, a vendor CAGE code is prefixed by the capital letter V appearing in the nomenclature column. This CAGE code number designates the original manufacturer of non-Goodrich Corporation components, and is in accordance with Cataloging Hand Books H4-1, H4-2, and H4-3. CAGE codes are listed in the NOMENCLATURE column, except for the government codes below, which are not listed:

V80205	National Aircraft Standard (prefix NAS)
V81349	Military Specifications (prefix M)
V88044	Army/Navy Standard (prefix AN)
V96906	Military Standard (prefix MS)

The Customer Support division in your region is to be contacted for parts dispatch.